

Shor's Algorithm

From the Postulates of Quantum Mechanics
to Factoring and the Elliptic-Curve Discrete Logarithm

A complete derivation: prerequisites, circuits, proofs, and resource analysis

How to read this document

What this is

This is a self-contained derivation of Shor’s algorithm — the 1994 quantum algorithm that factors integers and computes discrete logarithms in polynomial time — built from the ground up. *Every* fact is proved or explicitly attributed. Nothing is asserted because a textbook says so.

The word “self-contained” is meant literally. We assume linear algebra over $\mathbb{C}$ at the level of a first course (vector spaces, bases, matrices, eigenvalues) and elementary number theory (divisibility, primes, modular arithmetic). We assume *no* quantum mechanics: the formalism is built from its four postulates in [Chapter 3](#), in exactly the amount needed to give mathematical meaning — *semantics* — to the circuits that follow, and no more. We assume no elliptic-curve theory: the group law is derived from the chord–tangent construction in [Chapter 16](#).

Two things distinguish this treatment from a typical exposition.

First, **the semantics are made explicit**. A quantum circuit diagram is a picture; what it *denotes* is a unitary operator on a finite-dimensional Hilbert space, and what a measurement *means* is a specific probability distribution together with a specific state update. [Chapters 3](#) and [4](#) make that correspondence precise, so that when [Chapter 8](#) computes an amplitude, the computation is a theorem about vectors rather than a manipulation of suggestive notation.

Second, **the cost is taken seriously**. It is common to write “ $O(n^3)$ gates, see [refs]” and move on. [Chapters 12](#) and [13](#) instead build the reversible modular arithmetic explicitly — adders, comparators, modular multipliers, the controlled- U tower — and count. The conclusion, which reorients one’s entire intuition about the algorithm, is that the quantum Fourier transform is roughly 0.003% of the circuit. Shor’s algorithm is arithmetic wearing a Fourier hat.

The claim, stated once

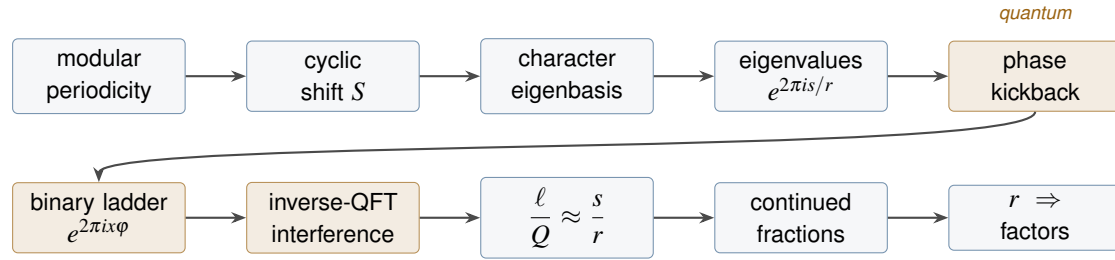
Shor’s theorem

There is a randomised quantum algorithm which, given a composite n -bit integer N , outputs a non-trivial factor of N using $\Theta(n)$ qubits and $\tilde{O}(n^3)$ elementary gates, with success probability bounded below by a constant. The same machinery computes discrete logarithms in any finite cyclic group whose group operation admits an efficient reversible circuit — in particular in $\mathbb{F}_p^\times$ and in the group of points of an elliptic curve over $\mathbb{F}_p$.

Consequently RSA, finite-field Diffie–Hellman, DSA, ECDH and ECDSA are all broken by one machine. The best known classical factoring algorithm, the general number field sieve, runs in $\exp(O(n^{1/3} \log^{2/3} n))$ time. The gap is super-polynomial.

The chain of ideas

The entire algorithm is one chain of nine links. Each arrow is a theorem, and the point of this document is to prove each of them.



In words. Multiplication by a modulo N , restricted to the orbit of 1, is the cyclic shift on $\mathbb{Z}/r\mathbb{Z}$ where r is the order of a . The shift is diagonalised — by the spectral theorem, and uniquely — by the characters of $\mathbb{Z}/r\mathbb{Z}$, whose eigenvalues are the r -th roots of unity $e^{2\pi i s/r}$. A controlled application of a unitary re-attaches its eigenvalue to the control qubit as a *relative* phase, which is observable. A ladder of controlled powers writes that phase across a whole register as $e^{2\pi i x s/r}$. The inverse Fourier transform converts phase into probability, concentrating the measurement outcome ℓ near Qs/r . Continued fractions promote the approximation ℓ/Q to the exact fraction s/r , whose denominator is r . And r yields the factors.

#	Link	Established by
0	factoring $\leftarrow$ order finding	Lemma 9.1.1 and Theorem 9.2.1
1	periodicity = cyclic shift	Proposition 5.5.3
2	shift $\Rightarrow$ roots of unity	Theorem 5.3.1
3	eigenvectors are characters	Theorem 5.3.1, Corollary 5.5.5
4	the input $ 1\rangle$ is free	Theorem 10.1.1
5	eigenvalue $\rightarrow$ relative phase	Lemma 7.2.1
6	relative phase $\rightarrow$ ladder	Theorem 7.4.1
7	ladder $\rightarrow$ Fejér peak	Theorem 8.3.2, Theorem 8.3.4
8	peak $\rightarrow$ exact fraction	Theorem 1.4.3, Corollary 11.1.1
9	$r \rightarrow$ factors	Corollary 10.5.2

The fork for elliptic curves

Links 5–7 are untouched. Three things change, and [Chapters 15](#) to [17](#) treat them: the hiding function becomes $f(k, \ell) = [k]P + [\ell]Q = [k + m\ell]P$, hiding the subgroup $L = \langle (-m, 1) \rangle \leq \mathbb{Z}/r\mathbb{Z} \times \mathbb{Z}/r\mathbb{Z}$; the transform becomes $\text{QFT} \otimes \text{QFT}$, because the group is two-dimensional and its dual factors; and the post-processing becomes one modular inversion $m \equiv y_2 y_1^{-1}$, with no continued fractions, because r is public.

One sentence for the fork

Factoring is the one-dimensional hidden subgroup problem, where the unknown is the *size* of the hidden subgroup; the ECDLP is the two-dimensional one, where the size is known and the unknown is the *slope*. That is why factoring needs continued fractions and the ECDLP needs one modular inverse.

Structure

Part	Topic	Contents
I	Foundations	Number theory and continued fractions; Hilbert spaces and the spectral theorem in Dirac notation; the four postulates of quantum mechanics; the circuit model, reversible computation, and what a gate costs.
II	The Fourier machinery	The cyclic shift and why the Fourier basis is forced; the QFT with its product form, circuit and approximate version; phase kickback; quantum phase estimation with its exact output distribution and precision bounds.
III	Factoring	The classical reduction; order finding as phase estimation; continued-fraction post-processing; explicit reversible modular arithmetic; the controlled- U tower; a complete worked example at $N = 21$.
IV	Discrete logarithms	The DLP as a two-dimensional hidden subgroup problem; elliptic curves and their group law; the reversible point-addition oracle down to Toffoli gates.
V	Cost and context	Resource estimates for RSA and ECC; formal pre/post-condition specifications of every subroutine; the abelian hidden subgroup problem and where the mechanism stops.

Conventions used throughout

Standing conventions

(C1) All Hilbert spaces are finite-dimensional over $\mathbb{C}$. Hence completeness is automatic, operators are matrices, and every normal operator is diagonalisable.

(C2) Phases are normalised into $[0, 1)$ inside $e^{2\pi i(\cdot)}$, so that a phase's binary expansion is what an algorithm returns.

(C3) Registers are **big-endian**: for $x = x_1x_2\cdots x_t \in \{0, 1\}^t$ we write $x = \sum_{s=1}^t x_s 2^{t-s}$, so x_1 is the most significant bit and sits on the *top* wire of every circuit diagram.

(C4) Circuits are read left to right in time but composed right to left as operators: a diagram showing G_1 then G_2 denotes the operator G_2G_1 .

(C5) Two symbols are overloaded and never occur in the same formula. $Q = 2^t$ is the size of the counting register in Parts II–III; $Q = [m]P$ is the public point in Part IV, where register sizes are written 2^{n+1} instead.

The single most common implementation bug

Mismatching the bit order between the ladder of controlled powers and the QFT. It fails *silently*: the circuit runs and returns plausible but wrong output. Qiskit uses the opposite (little-endian) convention from this document. Fix the convention once, and check it end to end on a known basis state before trusting any output. See [Section 3.8](#).

Notation

Symbol	Meaning	Symbol	Meaning
N	modulus to be factored	$\mathcal{H}$	a Hilbert space
n	$= \lceil \log_2 N \rceil$, work-register size	$\mathcal{L}(\mathcal{H})$	linear maps $\mathcal{H} \rightarrow \mathcal{H}$
a	base, $\gcd(a, N) = 1$	$\mathbb{I}$	identity operator
$r = \text{ord}_N(a)$	order of a modulo N	$A^\dagger$	adjoint (conjugate transpose)
t	counting-register size	$ \psi\rangle, \langle\psi $	ket and bra
$Q = 2^t$	counting-register dimension	$ \psi\rangle\langle\psi $	rank-one projector
ω_M	$= e^{2\pi i/M}$, principal M -th root	$\otimes$	tensor product
φ	an eigenphase, $U u\rangle = e^{2\pi i\varphi} u\rangle$	H, X, Y, Z	Hadamard, Paulis
δ_y	$= \varphi - y/Q$, detuning	$P(\theta)$	$\text{diag}(1, e^{i\theta})$
$ u_s\rangle$	eigenvector of U_a	R_k	$P(2\pi/2^k)$
U_a	$ y\rangle \mapsto ay \bmod N\rangle$	$c-U$	controlled- U
S	cyclic shift $ k\rangle \mapsto k+1\rangle$	CNOT, TOF	CNOT, Toffoli
$\phi(\cdot)$	Euler totient	$v_2(\cdot)$	2-adic valuation
$E(\mathbb{F}_p), \mathcal{O}$	curve points, point at infinity	$[k]P$	k -fold sum $P + \dots + P$
m	the discrete logarithm, $Q = [m]P$	λ	chord/tangent slope

The Fourier convention is fixed once: for $Q \geq 2$ and indices in $\mathbb{Z}/Q\mathbb{Z}$,

$$\text{QFT}_Q |x\rangle = \frac{1}{\sqrt{Q}} \sum_{y=0}^{Q-1} \omega_Q^{xy} |y\rangle, \quad \text{QFT}_Q^\dagger |x\rangle = \frac{1}{\sqrt{Q}} \sum_{y=0}^{Q-1} \omega_Q^{-xy} |y\rangle.$$

Binary fractions are written $0.a_1a_2\cdots a_m := \sum_{s \geq 1} a_s 2^{-s}$.

One statement that will not appear

The phrase “it tries all answers at once” is not used anywhere in this document, and [Section 20.3](#) explains precisely why it is wrong. An unmeasured uniform superposition carries zero information — measuring it returns a uniformly random string, exactly what a classical coin gives. The algorithm’s entire content is the *interference* that is arranged *before* measurement.

Contents

How to read this document	1
I Foundations	10
1 Number-theoretic prerequisites	11
1.1 Groups, orders, and the multiplicative group modulo N	11
1.2 The Euclidean algorithm and Bézout	12
1.3 The Chinese remainder theorem	12
1.4 Continued fractions and Legendre's theorem	13
1.5 Two facts about the totient	14
2 Linear algebra in Dirac notation	15
2.1 Hilbert spaces and the inner product	15
2.2 Kets, bras, and why the notation is shaped that way	15
2.3 Orthonormal bases and the completeness relation	16
2.4 Operators, outer products, and the adjoint	16
2.5 The spectral theorem	17
2.6 Functions of an operator	18
2.7 Tensor products	19
2.8 Trace, partial trace, and density operators	20
3 The postulates of quantum mechanics	21
3.1 Postulate 1: states	21
3.2 Postulate 2: evolution	21
3.3 Where unitarity comes from: time evolution derived	22
3.3.1 The evolution operator and why it must be unitary	22
3.3.2 The infinitesimal generator, and the Schrödinger equation	22
3.3.3 Observables, commutators, and the uncertainty relation	23
3.4 Postulate 3: measurement	23
3.5 Postulate 4: composite systems	24
3.6 Measuring one part of a composite system	25
3.7 Global phase, relative phase, and the qubit	25
3.8 Registers, and how integers are stored in them	26
3.9 Two things quantum states cannot do	27
3.10 Interference, precisely	27
4 The circuit model, gates, and what a gate costs	29
4.1 Semantics of a circuit diagram	29
4.2 The gate zoo	30
4.2.1 One-qubit gates	30
4.2.2 Two- and three-qubit gates	31
4.2.3 Controlling a gate makes a global phase local	31

4.3	Reversible classical computation	32
4.3.1	Two-bit reversible gates are not universal	32
4.3.2	Computing a classical function reversibly	32
4.4	Universality and the cost model	33
4.4.1	Clifford+ T and Gottesman–Knill	33
4.4.2	The Toffoli gate in Clifford+ T	34
4.4.3	The temporary logical-AND: Toffolis in pairs cost $4 T$	34
4.5	What to count: two cost regimes	35
II	The Fourier Machinery	36
5	Where the Fourier basis comes from	37
5.1	Simultaneous diagonalisation	37
5.2	The cyclic shift and the circulants	38
5.3	Diagonalising the shift: roots of unity are forced	39
5.3.1	The eigenvectors are the characters	39
5.4	What a Fourier coefficient computes	40
5.5	The bridge: modular multiplication <i>is</i> the shift	41
6	The quantum Fourier transform	43
6.1	Definition and first properties	43
6.1.1	The QFT diagonalises addition	43
6.2	The product form	44
6.3	The circuit	44
6.4	The approximate QFT	46
7	Phase kickback and the interferometer picture	47
7.1	The Mach–Zehnder interferometer as a one-qubit circuit	47
7.2	Simulating the phase shifter with a controlled gate	48
7.3	When kickback fails	49
7.4	The kickback ladder	49
7.5	From two paths to Q paths	50
8	Quantum phase estimation	51
8.1	The problem and the circuit	51
8.2	The exact case	52
8.3	The general case: the Fejér kernel	52
8.3.1	Three regimes	53
8.4	Precision: how many qubits for how much accuracy	54
8.5	Input that is not an eigenvector	55
8.6	Iterative phase estimation and the semiclassical QFT	56
8.7	Resource summary	57
III	Factoring	58
9	The classical reduction: factoring to order finding	59

9.1	Non-trivial square roots of 1 split N	59
9.2	A random base usually works	59
9.3	The complete classical algorithm	60
9.4	The pipeline, with the quantum part isolated	61
9.5	Two running examples	61
10	Order finding is phase estimation	63
10.1	The free input	63
10.2	The order-finding circuit	63
10.3	The state before the inverse QFT, in two bases	64
10.4	The state through the circuit	65
10.5	Correctness and cost	65
10.6	The original formulation, and why it is the same circuit	66
11	Post-processing: from ℓ to r	67
11.1	Legendre's theorem and the register-size rule	67
11.2	The recovery procedure	68
11.3	What is actually recovered	69
11.4	Two related questions, answered precisely	69
12	Reversible modular arithmetic	70
12.1	Addition: the carries are the cost	70
12.1.1	The CDKM ripple-carry adder	71
12.2	Subtraction, comparison, and controlled addition	72
12.3	Modular addition	72
12.4	Modular multiplication and exponentiation	73
12.5	The Fourier (Draper) adder, and why it loses	74
12.6	Reducing the constant factor: windowed arithmetic	75
12.7	Resource summary for the order-finding circuit	75
13	The controlled-U tower	77
13.1	The control-propagation principle	77
13.2	Level 4: $c\text{-}U_a$ from two multipliers and a swap	78
13.3	Level 3: the controlled multiplier	78
13.4	Level 2: the doubly-controlled modular adder	79
13.5	Level 1: doubly-controlled to singly-controlled	80
13.6	The complete tower, and what it costs	80
13.7	Three things that bite	81
13.8	Two ways to make the control cheaper still	81
13.8.1	One control qubit instead of t	81
13.8.2	Fold the control into the address	82
14	Worked example: $N = 21, a = 2$	83
14.1	Setup	83
14.2	One branch, by hand	83
14.3	The full measured distribution	84
14.4	Reconstruction, branch by branch	84

14.4.1	Following $\ell = 171$ through the post-processing	84
14.5	Success probability, exactly	85
14.6	Finishing the factorisation	85
14.7	Contrast with $N = 15$	85
14.8	What the published demonstrations show	86
IV	Discrete Logarithms and Elliptic Curves	87
15	The discrete logarithm as a hidden subgroup problem	88
15.1	Setup	88
15.2	The measured relation, derived	89
15.3	Why two QFTs	90
15.4	Factoring and DLP side by side	90
15.5	The finite-register version and post-processing	90
15.6	Is the finite-field discrete logarithm the same algorithm?	91
16	Elliptic curves and their group law	92
16.1	The curve and its group	92
16.2	Deriving the group law	92
16.3	Scalar multiplication, order, and register size	93
16.4	The running curve	93
16.5	The algorithm and its circuit	94
17	The reversible elliptic-curve oracle	96
17.1	The cost of each field operation	96
17.2	Modular addition and doubling	97
17.3	Modular multiplication, squaring, and inversion	97
17.3.1	Kaliski's binary extended GCD, made reversible	97
17.4	Reversible controlled point addition	98
17.5	Exceptional cases	100
17.6	Worked example over $\mathbb{F}_5$, end to end	100
17.7	Resources	101
V	Cost, Specification, and Context	102
18	Resource estimates and cryptographic impact	103
18.1	Logical resources for factoring	103
18.1.1	Simulated counts (Häner–Roetteler–Svore, minimal width)	103
18.2	Twenty years of constant factors	104
18.3	Elliptic curves versus RSA	105
18.4	Variants that change the picture	106
18.5	Cryptographic impact and migration	106
19	Formal specifications of every subroutine	108
19.1	The top-level specification	108
19.2	The quantum core	109

19.2.1	Block-level assertions	109
19.3	Arithmetic blocks	110
19.4	Elliptic-curve blocks	111
19.5	The whole algorithm as one judgement	112
20	The abelian hidden subgroup problem, and where it stops	114
20.1	One mechanism, several broken cryptosystems	114
20.2	What each ingredient contributes — and what it does not	115
20.3	Six common misstatements, and precise replacements	116
20.4	Analogies: which to use, and exactly where each breaks	116
20.5	Fourteen review questions	117
20.6	The mechanism in one line	117
	Appendices	119
A	Sums of roots of unity and elementary inequalities	119
B	The continued-fraction algorithm	120
B.1	The algorithm	120
B.2	The convergent recursion	120
B.3	Worked example: 171/512	120
B.4	Cost	121
C	Gate and cost reference	122
C.1	Everything, in one place	122
C.2	Gate usage in Shor’s algorithm, layer by layer	122
C.3	Reference numbers	123
D	Verification log	124
D.1	What was checked	125
D.2	Circuit identities	125
D.3	Reproducing the numbers	127
	References	128

Part I

Foundations

Chapter 1

Number-theoretic prerequisites

Everything in this chapter is classical mathematics, most of it eighteenth- and nineteenth-century. It is collected here because Shor's algorithm is a *classical* algorithm with one quantum subroutine, and the classical part is where the correctness argument lives: the reduction of factoring to order finding, the proof that a random base works, and the continued-fraction step that turns an approximate measurement into an exact rational number. None of it involves quantum mechanics.

1.1 Groups, orders, and the multiplicative group modulo N

Definition 1.1.1 (Group). A *group* is a set G with an associative operation $\cdot$, an identity e , and inverses. It is *abelian* if the operation commutes. A group is *cyclic* if $G = \langle g \rangle := \{g^k : k \in \mathbb{Z}\}$ for some *generator* g . The *order* of G , written $|G|$, is its cardinality; the *order of an element* g , written $\text{ord}(g)$, is the least $k \geq 1$ with $g^k = e$ (which exists whenever G is finite).

Lemma 1.1.2 (Order divides the exponent set). *Let g have order r in a group. Then $g^k = e$ if and only if $r \mid k$. Consequently, in a finite group, $\text{ord}(g)$ divides $|G|$.*

Proof. If $r \mid k$, write $k = rq$ and $g^k = (g^r)^q = e$. Conversely, divide with remainder: $k = rq + s$ with $0 \leq s < r$. Then $e = g^k = (g^r)^q g^s = g^s$, and minimality of r forces $s = 0$. The last claim is Lagrange's theorem applied to $\langle g \rangle$, whose order is r : the cosets of $\langle g \rangle$ partition G into $|G|/r$ blocks of equal size. $\square$

Definition 1.1.3 (The ring $\mathbb{Z}/N\mathbb{Z}$ and its units). For $N \geq 2$ let $\mathbb{Z}/N\mathbb{Z} = \{0, 1, \dots, N-1\}$ with addition and multiplication modulo N . An element a is a *unit* if $ab \equiv 1 \pmod{N}$ for some b ; the units form a group under multiplication, written $(\mathbb{Z}/N\mathbb{Z})^\times$.

Proposition 1.1.4 (Units are exactly the coprime residues). *$a \in \mathbb{Z}/N\mathbb{Z}$ is a unit if and only if $\gcd(a, N) = 1$. Hence $|(\mathbb{Z}/N\mathbb{Z})^\times| = \phi(N)$, where ϕ is Euler's totient function.*

Proof. $(\Rightarrow)$ If $ab \equiv 1 \pmod{N}$ then $ab - 1 = kN$, so any common divisor of a and N divides 1. $(\Leftarrow)$ If $\gcd(a, N) = 1$, Bézout's identity (Theorem 1.2.1) supplies integers u, v with $au + Nv = 1$, whence $au \equiv 1 \pmod{N}$ and $b = u \pmod{N}$ works. $\square$

Definition 1.1.5 (Order modulo N). For $\gcd(a, N) = 1$, the *order of a modulo N* is

$$r = \text{ord}_N(a) := \min\{k \geq 1 : a^k \equiv 1 \pmod{N}\}.$$

It is finite because $(\mathbb{Z}/N\mathbb{Z})^\times$ is a finite group, and by Lemma 1.1.2 it divides $\phi(N)$ (Euler's theorem).

Example 1.1.6. $\text{ord}_{21}(2) = 6$, since $2^1, \dots, 2^6 \equiv 2, 4, 8, 16, 11, 1 \pmod{21}$. Note that $\phi(21) = 12$ and indeed $6 \mid 12$. By contrast $\text{ord}_{15}(2) = 4$ and $\text{ord}_{15}(7) = 4$.

Why the order is the quantity of interest

The map $k \mapsto a^k \pmod{N}$ is periodic with least period exactly $r = \text{ord}_N(a)$. Finding a hidden period is the one thing a Fourier transform is built to do. Chapter 9 shows that knowing r for a random a suffices, with probability at least $\frac{1}{2}$, to split N .

1.2 The Euclidean algorithm and Bézout

Theorem 1.2.1 (Bézout). *For integers a, b not both zero there exist integers u, v with $au + bv = \gcd(a, b)$, computable together with $\gcd(a, b)$ in $O(\log^2 \max(|a|, |b|))$ bit operations by the extended Euclidean algorithm.*

Proof. Let d be the least positive element of $I := \{au + bv : u, v \in \mathbb{Z}\}$, which is non-empty and contains positive elements. Division with remainder gives $a = qd + s$ with $0 \leq s < d$; then $s = a - qd \in I$, and minimality forces $s = 0$, so $d \mid a$, and symmetrically $d \mid b$. Thus d is a common divisor. Conversely any common divisor of a, b divides every element of I , hence divides d ; so $d = \gcd(a, b)$. The Euclidean algorithm computes it by the recursion $\gcd(a, b) = \gcd(b, a \bmod b)$, which halves the larger argument every two steps, giving $O(\log \max(|a|, |b|))$ iterations of $O(\log \max(|a|, |b|))$ -bit arithmetic. Back-substituting through the recursion produces u, v . $\square$

Lemma 1.2.2 (Gauss). *If $d \mid xy$ and $\gcd(d, x) = 1$ then $d \mid y$.*

Proof. By Theorem 1.2.1 write $du + xv = 1$; multiplying by y gives $y = duy + (xy)v$. Both terms are divisible by d — the first visibly, the second because $d \mid xy$. Hence $d \mid y$. $\square$

1.3 The Chinese remainder theorem

Theorem 1.3.1 (CRT). *Let $N = N_1 N_2 \cdots N_m$ with the N_j pairwise coprime. Then the map*

$$\mathbb{Z}/N\mathbb{Z} \longrightarrow \mathbb{Z}/N_1\mathbb{Z} \times \cdots \times \mathbb{Z}/N_m\mathbb{Z}, \quad x \longmapsto (x \bmod N_1, \dots, x \bmod N_m)$$

is an isomorphism of rings, and it restricts to a group isomorphism $(\mathbb{Z}/N\mathbb{Z})^\times \cong (\mathbb{Z}/N_1\mathbb{Z})^\times \times \cdots \times (\mathbb{Z}/N_m\mathbb{Z})^\times$.

Proof. The map is a ring homomorphism because reduction modulo each N_j is. It is injective: if $x \equiv 0 \pmod{N_j}$ for every j , then each $N_j \mid x$, and since the N_j are pairwise coprime, repeated application of Lemma 1.2.2 gives $N = \prod N_j \mid x$, i.e. $x = 0$ in $\mathbb{Z}/N\mathbb{Z}$. Both sides have N elements, so injectivity gives bijectivity. A ring isomorphism carries units to units bijectively, which is the second statement. $\square$

Corollary 1.3.2 (Orders combine by lcm). *With the notation of Theorem 1.3.1, if $a \in (\mathbb{Z}/N\mathbb{Z})^\times$ has components a_j and $r_j = \text{ord}_{N_j}(a_j)$, then $\text{ord}_N(a) = \text{lcm}(r_1, \dots, r_m)$.*

Proof. $a^k \equiv 1 \pmod{N}$ iff $a_j^k \equiv 1 \pmod{N_j}$ for every j (by injectivity), iff $r_j \mid k$ for every j (Lemma 1.1.2), iff $\text{lcm}_j r_j \mid k$. Now apply Lemma 1.1.2 again. $\square$

Theorem 1.3.3 (Cyclicity of $(\mathbb{Z}/p^\alpha\mathbb{Z})^\times$ for odd p). *For an odd prime p and $\alpha \geq 1$, the group $(\mathbb{Z}/p^\alpha\mathbb{Z})^\times$ is cyclic of order $p^{\alpha-1}(p-1)$, which is even.*

Proof sketch. For $\alpha = 1$: a finite subgroup of the multiplicative group of a field is cyclic. Indeed, if $G \leq \mathbb{F}_p^\times$ has order g and $d \mid g$, then the elements of order dividing d are roots of $X^d - 1$, of which there are at most d in a field; a standard counting argument with $\sum_{d \mid g} \phi(d) = g$ then forces an element of order exactly g . For $\alpha > 1$ one lifts a generator g modulo p to a generator modulo p^α : replacing g by $g + p$ if necessary ensures $g^{p-1} \not\equiv 1 \pmod{p^2}$, and then induction on α using the binomial expansion of $(1 + cp^k)^p$ shows $\text{ord}_{p^\alpha}(g) = p^{\alpha-1}(p-1)$. The order is even because p is odd, so $p-1$ is even. Full details are in any algebraic number theory text. $\square$

Remark 1.3.4. The hypotheses matter for Theorem 9.2.1. The group $(\mathbb{Z}/2^\alpha\mathbb{Z})^\times$ is *not* cyclic for $\alpha \geq 3$ (it is $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2^{\alpha-2}\mathbb{Z}$), which is exactly why Shor's algorithm strips factors of 2 classically before the quantum part is invoked.

1.4 Continued fractions and Legendre's theorem

The quantum subroutine returns a rational number ℓ/Q which is *close to* s/r . This section supplies the classical machinery that recovers s/r *exactly*, which is what makes the algorithm exact rather than approximate.

Definition 1.4.1 (Continued fraction). For $x \in \mathbb{R}$ define $a_0 = \lfloor x \rfloor$, $x_0 = x - a_0$, and recursively, while $x_k \neq 0$, $a_{k+1} = \lfloor 1/x_k \rfloor$ and $x_{k+1} = 1/x_k - a_{k+1}$. Then

$$x = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{\ddots}}}$$

The expansion terminates precisely when x is rational. The k -th convergent is $p_k/q_k := [a_0; a_1, \dots, a_k]$.

Proposition 1.4.2 (The convergent recursion). With $p_{-1} = 1$, $q_{-1} = 0$, $p_0 = a_0$, $q_0 = 1$, the convergents satisfy

$$p_k = a_k p_{k-1} + p_{k-2}, \quad q_k = a_k q_{k-1} + q_{k-2}, \quad k \geq 1,$$

and moreover

$$p_k q_{k-1} - p_{k-1} q_k = (-1)^{k-1}. \quad (1.1)$$

In particular $\gcd(p_k, q_k) = 1$: every convergent is automatically in lowest terms. The q_k are strictly increasing for $k \geq 1$, and $q_k \geq F_{k+1}$, the $(k+1)$ -st Fibonacci number, so the number of convergents of a rational p/q is $O(\log q)$.

Proof. The recursion is proved by induction on k , the inductive step being the observation that $[a_0; a_1, \dots, a_k]$ is obtained from $[a_0; a_1, \dots, a_{k-1}]$ by replacing a_{k-1} with $a_{k-1} + 1/a_k$; substituting into the (already established) formula for p_{k-1}/q_{k-1} and clearing denominators gives the stated expressions. For Equation (1.1), compute

$$p_k q_{k-1} - p_{k-1} q_k = (a_k p_{k-1} + p_{k-2}) q_{k-1} - p_{k-1} (a_k q_{k-1} + q_{k-2}) = -(p_{k-1} q_{k-2} - p_{k-2} q_{k-1}),$$

so the quantity alternates in sign; and at $k = 0$ the initial conditions give $p_0 q_{-1} - p_{-1} q_0 = a_0 \cdot 0 - 1 \cdot 1 = -1 = (-1)^{-1}$, which anchors the alternation. Any common divisor of p_k, q_k divides the left side of Equation (1.1), hence divides 1. Finally $a_k \geq 1$ for $k \geq 1$ gives $q_k \geq q_{k-1} + q_{k-2}$, which is the Fibonacci recursion, and F_k grows exponentially. $\square$

Theorem 1.4.3 (Legendre). Let $x \in \mathbb{R}$ and let p/q be a rational in lowest terms with $q \geq 1$ and

$$\left| x - \frac{p}{q} \right| < \frac{1}{2q^2}. \quad (1.2)$$

Then p/q is a convergent of the continued-fraction expansion of x .

Proof. Write $x - p/q = \varepsilon \theta / (2q^2)$ with $\text{sgn}(\varepsilon) = \pm 1$ and $0 \leq \theta < 1$. Expand p/q as a finite continued fraction $[a_0; a_1, \dots, a_k]$; by adjusting the last term ($[\dots, a_k] = [\dots, a_k - 1, 1]$ when $a_k > 1$) we may choose the parity of k so that $(-1)^{k-1} = \text{sgn}(\varepsilon)$. Now define a real number η by the equation

$$x = \frac{\eta p_k + p_{k-1}}{\eta q_k + q_{k-1}},$$

where $p_k/q_k = p/q$; solving for η and using Equation (1.1),

$$\eta = -\frac{q_{k-1}}{q_k} + \frac{(-1)^{k-1}}{q_k^2 (x - p_k/q_k)} = -\frac{q_{k-1}}{q_k} + \frac{2}{\theta}.$$

Since $\theta < 1$ we get $2/\theta > 2$, and since $0 \leq q_{k-1}/q_k \leq 1$ we conclude $\eta > 1$. Therefore η has a continued-fraction expansion $\eta = [a_{k+1}; a_{k+2}, \dots]$ with all terms ≥ 1 , and substituting it into the displayed formula for x shows that

$$x = [a_0; a_1, \dots, a_k, a_{k+1}, a_{k+2}, \dots],$$

whose k -th convergent is exactly $p_k/q_k = p/q$. $\square$

Remark 1.4.4 (Sharpness). The constant $\frac{1}{2}$ cannot be replaced by 1: with $x = 0.4$ and $p/q = 1/3$ one has $|x - p/q| = 1/15 < 1/9 = 1/q^2$, yet the convergents of $2/5$ are $0/1, 1/2, 2/5$ — and $1/3$ is not among them. This is why [Corollary 11.1.1](#) demands $Q \geq N^2$ and not merely $Q \geq N$: the factor of 2 in [Equation \(1.2\)](#) is what forces a counting register of $2n + O(1)$ qubits rather than $n + O(1)$.

Algorithm 1.4.5 (Recovering a rational). Given integers ℓ, Q with $0 \leq \ell < Q$, and a bound N : run the continued-fraction expansion of ℓ/Q , computing convergents p_k/q_k by [Proposition 1.4.2](#), and return the last convergent with $0 < q_k < N$. This costs $O(\log^3 Q)$ bit operations.

Cost. By [Proposition 1.4.2](#) there are $O(\log Q)$ convergents; each step is one division and two multiply-adds on $O(\log Q)$ -bit integers, at $O(\log^2 Q)$ bit operations apiece with schoolbook arithmetic. $\square$

1.5 Two facts about the totient

Proposition 1.5.1 (Density of coprime residues). For s drawn uniformly from $\{0, 1, \dots, r-1\}$,

$$\Pr[\gcd(s, r) = 1] = \frac{\phi(r)}{r} = \prod_{p|r} \left(1 - \frac{1}{p}\right) = \Omega\left(\frac{1}{\log \log r}\right).$$

Proof. The first equality is the definition of ϕ ; the product formula follows from multiplicativity of ϕ and $\phi(p^\alpha) = p^\alpha - p^{\alpha-1}$. The lower bound is a standard consequence of Mertens' third theorem, $\prod_{p \leq z} (1 - 1/p) \sim e^{-\gamma} / \log z$: the worst case for $r \leq R$ is r a product of all small primes, and then $\log \log r$ appears. $\square$

Proposition 1.5.2 (Two runs and the lcm). Let s_1, s_2 be independent and uniform in $\{0, \dots, r-1\}$ and put $r_i = r / \gcd(s_i, r)$. If $\gcd(s_1, s_2) = 1$ then $\text{lcm}(r_1, r_2) = r$, and

$$\Pr[\gcd(s_1, s_2) = 1] \gtrsim 1 - \sum_{p \text{ prime}} \frac{1}{p^2} = 1 - 0.4522\dots > 0.54.$$

Proof. Let $p^e \parallel r$ (exactly divides). If $p^e \nmid r_1$ and $p^e \nmid r_2$ then p divides both $\gcd(s_1, r)$ and $\gcd(s_2, r)$, hence $p \mid \gcd(s_1, s_2) = 1$, absurd. So p^e divides one of r_1, r_2 , and since both divide r , $\text{lcm}(r_1, r_2) = r$. For the probability, a union bound over primes gives $\Pr[\gcd(s_1, s_2) > 1] \leq \sum_p \Pr[p \mid s_1] \Pr[p \mid s_2]$, and $\Pr[p \mid s_i] = \lceil r/p \rceil / r \rightarrow 1/p$; the prime zeta value $\sum_p p^{-2} = 0.45224\dots$ gives the bound. $\square$

Where each fact is used

[Lemma 1.1.2](#) and [Proposition 1.1.4](#) define the problem. [Theorems 1.3.1](#) and [1.3.3](#) are the engine of [Theorem 9.2.1](#) (a random base succeeds with probability $\geq \frac{1}{2}$). [Theorem 1.4.3](#) is the engine of [Corollary 11.1.1](#) (why the counting register has $2n + O(1)$ qubits). [Propositions 1.5.1](#) and [1.5.2](#) quantify how often a single quantum run returns r outright rather than a proper divisor of it.

Chapter 2

Linear algebra in Dirac notation

Nothing in this chapter is new mathematics. What is new is the notation, which is engineered so that the objects one manipulates in quantum computing — states, operators, measurement outcomes — read off the page in the order in which one thinks about them. The technical heart is the spectral theorem (Theorem 2.5.1): phase estimation is, in one sentence, an algorithm that reads off an eigenvalue, and the spectral theorem is what guarantees the eigenvalues are there to be read.

2.1 Hilbert spaces and the inner product

Definition 2.1.1 (Inner product). Let V be a complex vector space. An *inner product* on V is a map $V \times V \rightarrow \mathbb{C}$, written $(u, v) \mapsto \langle u|v \rangle$, such that for all $u, v, w \in V$ and $\lambda \in \mathbb{C}$:

(I1) **Linearity in the second argument:** $\langle u|\lambda v + w \rangle = \lambda \langle u|v \rangle + \langle u|w \rangle$;

(I2) **Conjugate symmetry:** $\langle u|v \rangle = \langle v|u \rangle^*$;

(I3) **Positive definiteness:** $\langle v|v \rangle \geq 0$, with equality iff $v = 0$.

Conditions (I1) and (I2) together force *conjugate-linearity* in the first slot: $\langle \lambda u + w|v \rangle = \lambda^* \langle u|v \rangle + \langle w|v \rangle$. Note that (I2) makes $\langle v|v \rangle$ real, so (I3) is meaningful.

Definition 2.1.2 (Norm, Hilbert space). The induced norm is $\|v\| := \sqrt{\langle v|v \rangle}$. A *Hilbert space* is an inner-product space complete for the metric $d(u, v) = \|u - v\|$. Vectors are *orthogonal* when $\langle u|v \rangle = 0$, and a vector is a *unit vector* when $\|v\| = 1$.

Remark 2.1.3 (Why completeness never bothers us). Completeness is a genuine requirement in general — continuous functions on $[0, 1]$ with the L^2 inner product form an incomplete inner-product space. But *every finite-dimensional* complex inner-product space is automatically complete. By convention (C1) all our spaces are finite-dimensional, so “Hilbert space” and “inner-product space” coincide here. This is a simplification we are entitled to, not one we are sweeping under the rug: it is what makes it legitimate to identify operators with matrices throughout. Concretely, every space we meet is $\mathbb{C}^d$ with $\langle u|v \rangle = \sum_{k=1}^d u_k^* v_k$.

2.2 Kets, bras, and why the notation is shaped that way

Notation 2.2.1 (Dirac notation). A vector of $\mathcal{H}$ is written $|\psi\rangle$ and called a *ket*. The symbol inside the bracket is a *label* and carries no mathematical content of its own: $|\psi\rangle$, $|0\rangle$, $|x\rangle$ are all just names for vectors. In particular $|0\rangle$ is a specific *unit vector* and is *not* the zero vector; the zero vector is written 0 , the one exception to the rule that vectors get brackets. Two abbreviations inside the bracket are standard: $|Av\rangle := A|v\rangle$ and $|\lambda v\rangle := \lambda|v\rangle$.

Definition 2.2.2 (Bra). For $|\phi\rangle \in \mathcal{H}$, the *bra* $\langle\phi|$ is the linear functional $\mathcal{H} \rightarrow \mathbb{C}$, $|\psi\rangle \mapsto \langle\phi|\psi\rangle$.

Because the inner product is conjugate-linear in its first slot, the map $|\phi\rangle \mapsto \langle\phi|$ is itself conjugate-linear: the bra of $\lambda|\phi\rangle$ is $\lambda^* \langle\phi|$.

The most common hand-computation error

Forgetting to conjugate when passing from a ket to a bra. Every sign error in a hand-derived amplitude is, with high probability, this.

Proposition 2.2.3 (Riesz: every bra comes from a ket). *The map $|\phi\rangle \mapsto \langle\phi|$ is a conjugate-linear bijection from $\mathcal{H}$ onto the dual space $\mathcal{H}^*$ of linear functionals $\mathcal{H} \rightarrow \mathbb{C}$.*

Proof. Injectivity. If $\langle\phi| = \langle\phi'|$ then $\langle\phi - \phi'|\psi\rangle = 0$ for all $|\psi\rangle$; take $|\psi\rangle = |\phi\rangle - |\phi'\rangle$ and apply (I3). *Surjectivity.* Fix an orthonormal basis $\{|e_k\rangle\}_{k=1}^d$ (Definition 2.3.1; nothing there uses this proposition, so there is no circularity) and let $f \in \mathcal{H}^*$. Put $|\phi\rangle := \sum_k f(|e_k\rangle)^* |e_k\rangle$; then $\langle\phi|$ and f agree on the basis, hence everywhere. $\square$

This is the finite-dimensional Riesz representation theorem. It justifies treating $\langle\phi|$ as “the same information as” $|\phi\rangle$, and it is why $\langle\phi|\psi\rangle$ can be read either as “the inner product of $|\phi\rangle$ and $|\psi\rangle$ ” or as “the functional $\langle\phi|$ applied to the vector $|\psi\rangle$ ”. The two readings are the point of the notation.

Notation 2.2.4 (Columns, rows, and the dagger). Once an ordered orthonormal basis is fixed, $|\psi\rangle$ is a column vector of coordinates and $\langle\psi|$ is the conjugate-transposed row vector, so $\langle\phi|\psi\rangle$ is literally a $1 \times d$ times $d \times 1$ matrix product. Accordingly $|\psi\rangle^\dagger := \langle\psi|$ and $\langle\psi|^\dagger := |\psi\rangle$: the dagger always means “conjugate transpose”.

2.3 Orthonormal bases and the completeness relation

Definition 2.3.1 (Orthonormal basis). A basis $\{|e_1\rangle, \dots, |e_d\rangle\}$ is *orthonormal* if $\langle e_j | e_k \rangle = \delta_{jk}$. Every finite-dimensional inner-product space has one (apply Gram–Schmidt to any basis).

Lemma 2.3.2 (Coordinates are inner products). *If $\{|e_k\rangle\}$ is orthonormal and $|\psi\rangle = \sum_k c_k |e_k\rangle$, then $c_k = \langle e_k | \psi \rangle$; consequently*

$$|\psi\rangle = \sum_k |e_k\rangle \langle e_k | \psi \rangle, \quad \|\psi\|^2 = \sum_k |\langle e_k | \psi \rangle|^2. \quad (2.1)$$

Proof. $\langle e_j | \psi \rangle = \sum_k c_k \langle e_j | e_k \rangle = \sum_k c_k \delta_{jk} = c_j$. The norm identity follows by expanding $\langle\psi|\psi\rangle = \sum_{j,k} c_j^* c_k \langle e_j | e_k \rangle$. $\square$

Equation (2.1) is small but it is the seed of everything: the second identity will become the statement that measurement probabilities sum to one, and the first will become the completeness relation.

Lemma 2.3.3 (Cauchy–Schwarz). *$|\langle u | v \rangle| \leq \|u\| \|v\|$, with equality iff $|u\rangle, |v\rangle$ are linearly dependent. In particular, if $|u\rangle, |v\rangle$ are unit vectors with $\langle u | v \rangle = 1$, then $|u\rangle = |v\rangle$.*

Proof. If $|u\rangle = 0$ both sides vanish. Otherwise put $\lambda := \langle u | v \rangle / \langle u | u \rangle$ and $|w\rangle := |v\rangle - \lambda |u\rangle$, so $\langle u | w \rangle = 0$. Then $0 \leq \langle w | w \rangle = \langle v | v \rangle - |\langle u | v \rangle|^2 / \langle u | u \rangle$, which rearranges to the inequality; equality forces $|w\rangle = 0$ by (I3). For the last claim, equality holds, so $|v\rangle = \lambda |u\rangle$ with $\lambda = \langle u | v \rangle = 1$. $\square$

Definition 2.3.4 (Computational basis). The distinguished orthonormal basis of $\mathbb{C}^2$ used to encode one classical bit is written $\{|0\rangle, |1\rangle\}$: $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. The names 0 and 1 are a convention: any two orthogonal unit vectors would do, and choosing which pair to call “computational” is exactly the act of deciding how classical information is stored.

2.4 Operators, outer products, and the adjoint

Definition 2.4.1 (Outer product). For $|u\rangle, |v\rangle \in \mathcal{H}$, the *outer product* $|u\rangle\langle v|$ is the operator $(|u\rangle\langle v|) |w\rangle := \langle v | w \rangle |u\rangle$. In coordinates it is the column-times-row matrix product, a $d \times d$ matrix.

Proposition 2.4.2 (Completeness relation and matrix elements). *Let $\{|e_k\rangle\}$ be an orthonormal basis of $\mathcal{H}$. Then*

$$\sum_k |e_k\rangle\langle e_k| = \mathbb{I}, \quad (2.2)$$

and every $A \in \mathcal{L}(\mathcal{H})$ satisfies $A = \sum_{j,k} \langle e_j | A | e_k \rangle |e_j\rangle\langle e_k|$, i.e. $A_{jk} = \langle e_j | A | e_k \rangle$.

Proof. Applying the left side of Equation (2.2) to $|\psi\rangle$ gives $\sum_k |e_k\rangle \langle e_k| \psi\rangle = |\psi\rangle$ by Lemma 2.3.2. For the second claim, insert Equation (2.2) on both sides of A : $A = \mathbb{1}A\mathbb{1} = \sum_{j,k} |e_j\rangle \langle e_j| A |e_k\rangle \langle e_k|$. $\square$

A technique, not an identity

Inserting $\mathbb{1}$ in the form Equation (2.2) at a convenient place is the workhorse computation of the whole subject. It is worth recognising it as a *technique*. Almost every amplitude computed in Chapters 6 and 8 is an instance of it.

Definition 2.4.3 (Adjoint). The *adjoint* $A^\dagger$ of $A \in \mathcal{L}(\mathcal{H})$ is the unique operator with $\langle u|Av\rangle = \langle A^\dagger u|v\rangle$ for all $|u\rangle, |v\rangle$. In matrix terms $A^\dagger = (A^T)^*$.

Existence and uniqueness follow from Proposition 2.2.3: for fixed $|u\rangle$ the map $|v\rangle \mapsto \langle u|Av\rangle$ is a functional, hence of the form $\langle w|v\rangle$ for a unique $|w\rangle$, and one sets $A^\dagger |u\rangle := |w\rangle$. The rules

$$(A^\dagger)^\dagger = A, \quad (AB)^\dagger = B^\dagger A^\dagger, \quad (\lambda A)^\dagger = \lambda^* A^\dagger, \quad (|u\rangle\langle v|)^\dagger = |v\rangle\langle u| \quad (2.3)$$

follow directly. A mnemonic covering all of them: *daggering reverses the order of everything and conjugates every scalar*.

Definition 2.4.4 (The operator zoo). Let $A \in \mathcal{L}(\mathcal{H})$. Then A is *Hermitian* if $A^\dagger = A$; *unitary* if $A^\dagger A = AA^\dagger = \mathbb{1}$; *normal* if $A^\dagger A = AA^\dagger$; a *projector* if $A^\dagger = A$ and $A^2 = A$; *positive semidefinite* ($A \geq 0$) if $\langle v|A|v\rangle \geq 0$ for all $|v\rangle$.

Hermitian, unitary and projector operators are visibly normal. Positive operators are too, because over $\mathbb{C}$ positivity forces Hermiticity: if $\langle v|A|v\rangle \in \mathbb{R}$ for all $|v\rangle$ then $\langle v|A - A^\dagger|v\rangle = 0$ for all $|v\rangle$, and an operator B with $\langle v|B|v\rangle = 0$ for all $|v\rangle$ is 0 (expand $\langle u+v|B|u+v\rangle$ and $\langle u+iv|B|u+iv\rangle$ to get $\langle u|B|v\rangle = 0$).

Remark 2.4.5. In finite dimension $A^\dagger A = \mathbb{1}$ already implies $AA^\dagger = \mathbb{1}$ (a left inverse of a square matrix is two-sided), so either condition alone defines unitarity. In infinite dimension this fails — the distinction between isometries and unitaries is real — which is one more reason convention (C1) matters.

Proposition 2.4.6 (Unitaries are exactly the symmetry maps). For $U \in \mathcal{L}(\mathcal{H})$ the following are equivalent: (i) U is unitary; (ii) $\langle Uu|Uv\rangle = \langle u|v\rangle$ for all $|u\rangle, |v\rangle$; (iii) U maps some orthonormal basis to an orthonormal basis; (iv) U maps every orthonormal basis to an orthonormal basis. In particular $\|U|v\rangle\| = \||v\rangle\|$.

Proof. (i) $\Rightarrow$ (ii): $\langle Uu|Uv\rangle = \langle u|U^\dagger U|v\rangle = \langle u|v\rangle$. (ii) $\Rightarrow$ (iv) $\Rightarrow$ (iii) are immediate. (iii) $\Rightarrow$ (i): if $\{|e_k\rangle\}$ and $\{U|e_k\rangle\}$ are both orthonormal bases then $\langle e_j|U^\dagger U|e_k\rangle = \langle Ue_j|Ue_k\rangle = \delta_{jk} = \langle e_j|\mathbb{1}|e_k\rangle$ for all j, k , so $U^\dagger U = \mathbb{1}$ by Proposition 2.4.2. $\square$

Statement (iv) is the reason unitary operators, and nothing else, are allowed as the elementary steps of a quantum computation: they are precisely the maps that turn a legitimate description of a state into another legitimate description of a state (Postulate 2).

2.5 The spectral theorem

Theorem 2.5.1 (Spectral theorem). Let $A \in \mathcal{L}(\mathcal{H})$ with $\dim \mathcal{H} = d < \infty$. Then A is normal if and only if there is an orthonormal basis $\{|a_1\rangle, \dots, |a_d\rangle\}$ of $\mathcal{H}$ consisting of eigenvectors of A . In that case, writing $A|a_k\rangle = \lambda_k|a_k\rangle$,

$$A = \sum_{k=1}^d \lambda_k |a_k\rangle\langle a_k|. \quad (2.4)$$

Grouping equal eigenvalues, let $\sigma(A)$ be the set of distinct eigenvalues and P_λ the orthogonal projector onto $\ker(A - \lambda\mathbb{1})$. Then $A = \sum_{\lambda \in \sigma(A)} \lambda P_\lambda$, with $P_\lambda P_\mu = \delta_{\lambda\mu} P_\lambda$ and $\sum_\lambda P_\lambda = \mathbb{1}$.

Proof. ($\Leftarrow$) If Equation (2.4) holds then $A^\dagger = \sum_k \lambda_k^* |a_k\rangle\langle a_k|$ by Equation (2.3), and both $A^\dagger A$ and $AA^\dagger$ equal $\sum_k |\lambda_k|^2 |a_k\rangle\langle a_k|$ by orthonormality; so A is normal.

($\Rightarrow$) *Step 1 (Schur).* Every A is upper triangular in some orthonormal basis, by induction on d . For $d = 1$ there is nothing to prove. For $d > 1$: the characteristic polynomial of A has a root $\lambda \in \mathbb{C}$ (fundamental theorem of algebra), so there is a unit eigenvector $|v\rangle$. Extend $|v\rangle$ to an orthonormal basis; in it the first column of A is $(\lambda, 0, \dots, 0)^T$, so $A = \begin{pmatrix} \lambda & * \\ 0 & A' \end{pmatrix}$ where A' is the matrix of $|w\rangle \mapsto \Pi A |w\rangle$ on $W = \{|w\rangle : \langle v|w\rangle = 0\}$, with Π the orthogonal projector onto W (the projection is needed because W need not be A -invariant). Apply the induction hypothesis to A' .

Step 2 (normal + triangular $\Rightarrow$ diagonal). Let T be the resulting upper triangular matrix; T is normal, since normality is preserved under unitary change of basis. Compare the $(1,1)$ entries of $TT^\dagger$ and $T^\dagger T$: $(TT^\dagger)_{11} = \sum_j |T_{1j}|^2$ while $(T^\dagger T)_{11} = \sum_j |T_{j1}|^2 = |T_{11}|^2$, because T is upper triangular. Equality forces $T_{1j} = 0$ for $j > 1$. Deleting the first row and column leaves a smaller normal upper triangular matrix; induct. Hence T is diagonal, i.e. the basis consists of eigenvectors, and Proposition 2.4.2 gives Equation (2.4). The final statements follow by collecting terms with a common eigenvalue. $\square$

Corollary 2.5.2 (Eigenvalues, by type). *Let A be normal.*

- (a) *If A is Hermitian, all eigenvalues are real; conversely a normal operator with real eigenvalues is Hermitian.*
- (b) *If A is unitary, all eigenvalues have modulus 1, hence are of the form $e^{2\pi i\theta}$ with $\theta \in [0, 1)$.*
- (c) *If A is a projector, its eigenvalues lie in $\{0, 1\}$.*
- (d) *Eigenvectors belonging to distinct eigenvalues are orthogonal.*

Proof. (a) If $A|a\rangle = \lambda|a\rangle$ with $\| |a\rangle \| = 1$ then $\lambda = \langle a|A|a\rangle = \langle a|A^\dagger|a\rangle = \overline{\langle a|A|a\rangle} = \lambda^*$. The converse is Equation (2.4). (b) $\| |a\rangle \| = \| A|a\rangle \| = |\lambda| \| |a\rangle \|$ by Proposition 2.4.6. (c) $\lambda|a\rangle = A|a\rangle = A^2|a\rangle = \lambda^2|a\rangle$. (d) From Equation (2.4), $A^\dagger|a\rangle = \lambda^*|a\rangle$; hence, with $A|b\rangle = \mu|b\rangle$, $\mu\langle a|b\rangle = \langle a|Ab\rangle = \langle A^\dagger a|b\rangle = \langle \lambda^* a|b\rangle = \lambda^*\langle a|b\rangle$, so $(\lambda - \mu)\langle a|b\rangle = 0$. $\square$

The single most important corollary

Corollary 2.5.2(b) is the exact statement that phase estimation starts from: a unitary U and one of its eigenvectors $|u\rangle$ come with a *real number* $\theta \in [0, 1)$ satisfying $U|u\rangle = e^{2\pi i\theta}|u\rangle$, and θ is what we want to learn. That every eigenvalue of a unitary is a pure phase is exactly what allows an eigenvalue to be “kicked back” as a phase (Lemma 7.2.1). A real-amplitude theory could not encode s/r this way.

2.6 Functions of an operator

Definition 2.6.1 (Operator function). Let $A = \sum_k \lambda_k |a_k\rangle\langle a_k|$ be normal and $f : \sigma(A) \rightarrow \mathbb{C}$. Define $f(A) := \sum_k f(\lambda_k) |a_k\rangle\langle a_k|$.

The definition does not depend on the choice of eigenbasis, since grouping equal eigenvalues gives $f(A) = \sum_\lambda f(\lambda) P_\lambda$ and the spectral projectors are determined by A alone. It is consistent with the obvious meanings: for $f(z) = z^m$ it returns A^m , and for an everywhere-convergent power series it agrees with the series in A .

Example 2.6.2 (The exponential). $e^A := \sum_{n \geq 0} A^n / n!$. If $A^2 = \mathbb{I}$ then, splitting into even and odd parts,

$$e^{i\theta A} = \cos(\theta) \mathbb{I} + i \sin(\theta) A. \quad (2.5)$$

This one identity generates every rotation matrix in Chapter 4.

Proposition 2.6.3 (Unitaries and Hermitian generators). *U is unitary if and only if $U = e^{iA}$ for some Hermitian A . Moreover if $A|v\rangle = \lambda|v\rangle$ then $e^{iA}|v\rangle = e^{i\lambda}|v\rangle$: exponentiation preserves eigenvectors and exponentiates eigenvalues.*

Proof. If A is Hermitian, $e^{iA} = \sum_k e^{i\lambda_k} |a_k\rangle\langle a_k|$ with real λ_k , whose adjoint is its inverse. Conversely write $U = \sum_k e^{i\theta_k} |a_k\rangle\langle a_k|$ using [Theorem 2.5.1](#) and [Corollary 2.5.2\(b\)](#), and set $A := \sum_k \theta_k |a_k\rangle\langle a_k|$. $\square$

Corollary 2.6.4 (Powers of a unitary). *If $U|u\rangle = e^{2\pi i\theta}|u\rangle$ then $U^m|u\rangle = e^{2\pi im\theta}|u\rangle$ for every $m \in \mathbb{N}$.*

[Corollary 2.6.4](#) with $m = 2^k$ is the mechanism by which phase estimation extracts successive bits of θ : doubling shifts the binary expansion of θ one place left. Under convention (C3), counting qubit s carries the phase $2^{t-s}\theta \equiv 0.\theta_{t-s+1}\theta_{t-s+2}\cdots \pmod{1}$, so the *top* wire $s = 1$ probes deepest into the expansion and increasing s moves back towards the leading bits.

2.7 Tensor products

Two separate quantum systems, considered as one, do *not* live in the direct sum of their state spaces but in their tensor product. This is the mathematical origin of both the exponential size of quantum state spaces and of entanglement, so it deserves care.

Definition 2.7.1 (Tensor product). Let $\mathcal{H}_A$ and $\mathcal{H}_B$ have orthonormal bases $\{|a_j\rangle\}_{j=1}^m$ and $\{|b_k\rangle\}_{k=1}^n$. The *tensor product* $\mathcal{H}_A \otimes \mathcal{H}_B$ is the mn -dimensional Hilbert space with orthonormal basis the formal symbols $|a_j\rangle \otimes |b_k\rangle$, together with the bilinear extension of $\otimes$:

$$\begin{aligned} (\lambda|u\rangle) \otimes |v\rangle &= |u\rangle \otimes (\lambda|v\rangle) = \lambda(|u\rangle \otimes |v\rangle), \\ (|u\rangle + |u'\rangle) \otimes |v\rangle &= |u\rangle \otimes |v\rangle + |u'\rangle \otimes |v\rangle, \\ |u\rangle \otimes (|v\rangle + |v'\rangle) &= |u\rangle \otimes |v\rangle + |u\rangle \otimes |v'\rangle, \end{aligned} \tag{2.6}$$

and inner product determined by

$$\langle u \otimes v | u' \otimes v' \rangle = \langle u | u' \rangle \langle v | v' \rangle. \tag{2.7}$$

Why $\otimes$ and not $\oplus$

Two independent classical bits have $2 \times 2 = 4$ joint configurations, not $2 + 2$. The joint state space must contain all four basis configurations *and their superpositions*, so dimensions multiply: $\dim \mathcal{H}_n = 2^n$ for n qubits. This is the exponential that the algorithm exploits — and, as [Section 3.8](#) insists, exploiting it requires more than merely having it.

Notation 2.7.2. We freely write $|u\rangle \otimes |v\rangle = |u\rangle |v\rangle = |u, v\rangle = |uv\rangle$, and $|\psi\rangle^{\otimes n}$ for the n -fold product. Associativity is implicit.

Definition 2.7.3 (Tensor product of operators). For $A \in \mathcal{L}(\mathcal{H}_A)$, $B \in \mathcal{L}(\mathcal{H}_B)$, define $A \otimes B$ on product vectors by $(A \otimes B)(|u\rangle \otimes |v\rangle) := (A|u\rangle) \otimes (B|v\rangle)$, extended linearly.

Proposition 2.7.4 (Calculus of $\otimes$). *For operators of matching types,*

$$(A \otimes B)(C \otimes D) = (AC) \otimes (BD), \quad (A \otimes B)^\dagger = A^\dagger \otimes B^\dagger, \tag{2.8}$$

and $A \otimes B$ is unitary (resp. Hermitian, a projector) whenever A and B both are.

Proof. Both identities are checked on product vectors, where they follow from the definition and [Equation \(2.7\)](#), and then extended by linearity. The consequences follow: e.g. $(A \otimes B)^\dagger(A \otimes B) = (A^\dagger A) \otimes (B^\dagger B) = \mathbb{I} \otimes \mathbb{I} = \mathbb{I}$. $\square$

The first identity in [Equation \(2.8\)](#) is the algebraic content of the slogan *what happens on the left stays on the left*: operators acting on different factors never interfere, and in particular $(A \otimes \mathbb{I})(\mathbb{I} \otimes B) = (\mathbb{I} \otimes B)(A \otimes \mathbb{I}) = A \otimes B$. In circuit language, gates on disjoint wires commute and may be slid past one another freely.

Notation 2.7.5 (Kronecker product). In coordinates, with the ordered basis of $\mathcal{H}_A \otimes \mathcal{H}_B$ in lexicographic order, the matrix of $A \otimes B$ is the block matrix $[A_{jk}B]_{j,k}$.

Warning 2.7.6. $A \otimes B \neq B \otimes A$ in general — the two act on different factors. For instance $\mathbb{1} \otimes H \neq H \otimes \mathbb{1}$. *Order in a tensor product is position, not multiplication order.*

2.8 Trace, partial trace, and density operators

This section may be skipped on first reading; nothing in [Chapters 6](#) and [8](#) depends on it. It is included because it supplies the cleanest language for “the state of one register when we ignore the other”, which is exactly the situation of the work register in [Chapter 10](#).

Definition 2.8.1 (Trace). $\text{tr} A := \sum_k \langle e_k | A | e_k \rangle$ for any orthonormal basis $\{|e_k\rangle\}$ (the value does not depend on the basis). The properties used in practice are linearity, cyclicity $\text{tr}(AB) = \text{tr}(BA)$, invariance $\text{tr}(UAU^\dagger) = \text{tr} A$, and $\text{tr}(A|\psi\rangle\langle\psi|) = \langle\psi|A|\psi\rangle$ for unit $|\psi\rangle$.

Definition 2.8.2 (Density operator). A *density operator* on $\mathcal{H}$ is a positive semidefinite ρ with $\text{tr} \rho = 1$. If a system is in state $|\psi_i\rangle$ with probability p_i , its density operator is $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$. It is *pure* when $\rho = |\psi\rangle\langle\psi|$, equivalently $\text{tr}(\rho^2) = 1$; otherwise *mixed*.

Definition 2.8.3 (Partial trace). The *partial trace* over B is the linear map $\text{tr}_B : \mathcal{L}(\mathcal{H}_A \otimes \mathcal{H}_B) \rightarrow \mathcal{L}(\mathcal{H}_A)$ determined by $\text{tr}_B(|a\rangle\langle a'| \otimes |b\rangle\langle b'|) := \langle b'|b\rangle |a\rangle\langle a'|$. The *reduced state* of subsystem A is $\rho^A := \text{tr}_B(\rho)$.

The partial trace is the *unique* map reproducing all statistics of measurements performed on A alone: for every observable M on $\mathcal{H}_A$, $\text{tr}(M\rho^A) = \text{tr}((M \otimes \mathbb{1})\rho)$. That property, not the formula, is its definition in spirit.

Warning 2.8.4. tr_B is *not* cyclic: $\text{tr}_B(XY) \neq \text{tr}_B(YX)$ in general. This is a common slip.

Chapter 3

The postulates of quantum mechanics

Quantum mechanics is not derived here; it is *posited*. Four postulates fix what a state is, how it changes on its own, what happens when we look at it, and how systems combine. Everything a quantum computer does is an instance of these four statements, and the purpose of this chapter is to make that claim literal: by the end of [Chapter 4](#) every circuit diagram in this document will have a completely unambiguous mathematical meaning, assembled out of the postulates below and nothing else.

What is and is not needed

Defined here: complex Hilbert space and Dirac notation ([Chapter 2](#)); qubit, superposition, the Born rule; global versus relative phase; the tensor product and why $\dim = 2^n$; entanglement; unitary evolution; measurement of part of a composite system; no-cloning.

Not needed: the Schrödinger equation and Hamiltonians (mentioned once, in [Remark 3.2.1](#), purely to connect the discrete and continuous pictures); infinite-dimensional functional analysis; POVMs and general measurement operators; any *physical interpretation* of measurement.

Working definition. For present purposes, quantum computation is linear algebra over $\mathbb{C}$ together with one rule for reading out: the outcome x occurs with probability $|\alpha_x|^2$.

3.1 Postulate 1: states

Postulate 1 (State space). *Associated with any isolated physical system is a Hilbert space $\mathcal{H}$, its state space. The system is completely described by a state vector, a unit vector $|\psi\rangle \in \mathcal{H}$. Two unit vectors differing by a factor $e^{i\gamma}$ with $\gamma \in \mathbb{R}$ describe the same physical state.*

The last sentence says that a state is really an equivalence class — a *ray* — and $|\psi\rangle$ is a chosen representative. The factor $e^{i\gamma}$ is a *global phase*; [Proposition 3.7.2](#) proves it is genuinely unobservable, and [Section 4.2.3](#) identifies the one situation in which forgetting about it is fatal.

Definition 3.1.1 (Superposition, amplitude). If $\{|e_k\rangle\}$ is an orthonormal basis and $|\psi\rangle = \sum_k \alpha_k |e_k\rangle$ with $\sum_k |\alpha_k|^2 = 1$, we say $|\psi\rangle$ is a *superposition* of the $|e_k\rangle$ and call $\alpha_k \in \mathbb{C}$ the *amplitude* of $|e_k\rangle$ in $|\psi\rangle$.

An amplitude is not a probability

Amplitudes are complex; they may be negative or imaginary, and *they may cancel*. It is $|\alpha_k|^2$ — never α_k — that is a probability. The possibility of cancellation is called *interference* and is the entire source of quantum advantage. Two probabilities cannot cancel; two amplitudes can. That asymmetry is the whole subject.

3.2 Postulate 2: evolution

Postulate 2 (Unitary evolution). *The evolution of a closed quantum system between two times is described by a unitary operator U on its state space: $|\psi'\rangle = U|\psi\rangle$.*

That the operator must be unitary is *forced* by [Proposition 2.4.6](#): evolution must send unit vectors to unit vectors, for every state, and together with linearity that is exactly what unitarity says.

Remark 3.2.1 (Where unitaries come from). In continuous time a closed system obeys the Schrödinger equation $i\hbar d|\psi\rangle/dt = \mathbf{H}|\psi\rangle$, where $\mathbf{H}$ is a Hermitian operator called the Hamiltonian; its solution is $|\psi(t)\rangle = e^{-i\mathbf{H}t/\hbar}|\psi(0)\rangle$. By [Proposition 2.6.3](#) these are exactly the unitaries, so the discrete “apply a

gate” picture and the continuous “let it evolve” picture carry the same information. Nothing below needs Hamiltonians; we mention this only so that the symbol $\mathbf{H}$ is never confused with the Hadamard gate H , which is what H always denotes in this document.

Corollary 3.2.2 (Reversibility). *Every unitary is invertible, with $U^{-1} = U^\dagger$. Hence every step of a quantum computation other than measurement is reversible, and no information is destroyed by it.*

This is a strong constraint, and it has real teeth. Classical logic gates such as AND and OR are irreversible: from the output 0 of an AND gate one cannot recover the inputs. [Section 4.3](#) explains how classical computation is nevertheless embedded into the quantum model, and why doing it carelessly destroys the algorithm. The physical reading — an irreversible step erases information, and erasure costs $k_B T \ln 2$ per bit (Landauer’s principle) — is the thermodynamic shadow of the same fact.

3.3 Where unitarity comes from: time evolution derived

[Postulate 2](#) posits that evolution is unitary. It is worth seeing that this is not an arbitrary choice but is forced, and worth seeing how the discrete gate picture used throughout this document connects to the continuous dynamics of physics. Nothing in the rest of the document depends on this section; it is included so that the formalism is complete rather than free-floating.

3.3.1 The evolution operator and why it must be unitary

Let $|\psi(t)\rangle$ denote the state at time t and posit only that the state at time t is determined *linearly* by the state at an earlier time t_0 :

$$|\psi(t)\rangle = \hat{U}(t, t_0) |\psi(t_0)\rangle. \quad (3.1)$$

Proposition 3.3.1 (Unitarity is forced by conservation of probability). *If $\hat{U}(t, t_0)$ preserves the norm of every state vector, then $\hat{U}(t, t_0)$ is unitary. Moreover the family $\{\hat{U}(t, t_0)\}$ satisfies*

$$\hat{U}(t_2, t_0) = \hat{U}(t_2, t_1) \hat{U}(t_1, t_0), \quad \hat{U}(t_0, t_0) = \mathbb{1}, \quad \hat{U}(t, t_0)^{-1} = \hat{U}(t_0, t) = \hat{U}(t, t_0)^\dagger. \quad (3.2)$$

Proof. Normalisation of $|\psi(t)\rangle$ for all t gives $\langle \psi(t_0) | \hat{U}^\dagger \hat{U} | \psi(t_0) \rangle = \langle \psi(t_0) | \psi(t_0) \rangle$ for every state, so $\langle \psi | \hat{U}^\dagger \hat{U} - \mathbb{1} | \psi \rangle = 0$ for all $|\psi\rangle$; by the polarisation argument of [Definition 2.4.4](#) this forces $\hat{U}^\dagger \hat{U} = \mathbb{1}$, hence unitarity by [Proposition 2.4.6](#) and the finite-dimensional remark following it. The composition law is the statement that evolving from t_0 to t_1 and then to t_2 is evolving from t_0 to t_2 ; setting $t_2 = t_0$ gives the inverse. $\square$

The content of

Linearity plus “probabilities still sum to one” equals unitarity. There is no third assumption. This is why [Definition 4.1.1](#) may take “gate” to be synonymous with “unitary” without apology, and why [Corollary 3.2.2](#) — reversibility — is not a design decision one could have made differently.

3.3.2 The infinitesimal generator, and the Schrödinger equation

Proposition 3.3.2 (Hermitian generator). *Suppose $t \mapsto \hat{U}(t, t_0)$ is differentiable. Then there is a Hermitian operator $\mathbf{H}(t)$ — the Hamiltonian — with*

$$i\hbar \frac{\partial}{\partial t} \hat{U}(t, t_0) = \mathbf{H}(t) \hat{U}(t, t_0), \quad \text{equivalently} \quad i\hbar \frac{d}{dt} |\psi(t)\rangle = \mathbf{H}(t) |\psi(t)\rangle. \quad (3.3)$$

When $\mathbf{H}$ is time-independent the solution is $\hat{U}(t, t_0) = e^{-i\mathbf{H}(t-t_0)/\hbar}$.

Proof. Write $\hat{U}(t + \varepsilon, t) = \mathbb{1} - \frac{i\varepsilon}{\hbar} \mathbf{H}(t) + O(\varepsilon^2)$, which defines $\mathbf{H}(t)$ (the factor $-i/\hbar$ is a convention fixing units). Unitarity to first order in ε requires

$$\mathbb{1} = \hat{U}^\dagger \hat{U} = \left(\mathbb{1} + \frac{i\varepsilon}{\hbar} \mathbf{H}^\dagger \right) \left(\mathbb{1} - \frac{i\varepsilon}{\hbar} \mathbf{H} \right) + O(\varepsilon^2) = \mathbb{1} + \frac{i\varepsilon}{\hbar} (\mathbf{H}^\dagger - \mathbf{H}) + O(\varepsilon^2),$$

so $\mathbf{H}^\dagger = \mathbf{H}$: *unitarity of the finite evolution is exactly Hermiticity of its generator.* Now $\hat{U}(t + \varepsilon, t_0) - \hat{U}(t, t_0) = (\hat{U}(t + \varepsilon, t) - \mathbb{1}) \hat{U}(t, t_0) = -\frac{i\varepsilon}{\hbar} \mathbf{H}(t) \hat{U}(t, t_0) + O(\varepsilon^2)$; divide by ε and let $\varepsilon \rightarrow 0$. For constant $\mathbf{H}$, $e^{-i\mathbf{H}(t-t_0)/\hbar}$ solves Equation (3.3) with the right initial condition, and the solution is unique by linearity. $\square$

The bridge between physics and circuits

Proposition 2.6.3 said every unitary is e^{iA} for Hermitian A ; Proposition 3.3.2 says the A occurring in nature is (minus) the Hamiltonian times elapsed time over $\hbar$. So “apply the gate U ” and “switch on the Hamiltonian $\mathbf{H}$ for time t ” carry exactly the same information, and a circuit is a schedule of Hamiltonians. Everything after this chapter works in the gate picture and never needs $\mathbf{H}$ again — which is why we are free to write H for the Hadamard gate without ambiguity.

3.3.3 Observables, commutators, and the uncertainty relation

For completeness we record the two remaining pieces of standard formalism. Neither is used later, but their absence is often what makes a reader feel the treatment is not really quantum mechanics.

Definition 3.3.3 (Observable, expectation, commutator). An *observable* is a Hermitian operator A ; by Corollary 2.5.2(a) its eigenvalues are real, and by Theorem 2.5.1 it induces the measurement $\{P_\lambda\}$ of Postulate 3 with outcome λ . Its *expectation* in state $|\psi\rangle$ is $\langle A \rangle := \langle \psi | A | \psi \rangle = \sum_\lambda \lambda p(\lambda)$, and its *uncertainty* is $\Delta A := \sqrt{\langle A^2 \rangle - \langle A \rangle^2}$. The *commutator* is $[A, B] := AB - BA$.

Theorem 3.3.4 (Robertson uncertainty relation). For observables A, B and any state $|\psi\rangle$,

$$\Delta A \Delta B \geq \frac{1}{2} | \langle \psi | [A, B] | \psi \rangle |. \quad (3.4)$$

Proof. Put $|f\rangle = (A - \langle A \rangle) |\psi\rangle$ and $|g\rangle = (B - \langle B \rangle) |\psi\rangle$, so $\|f\| = \Delta A$ and $\|g\| = \Delta B$. Cauchy–Schwarz (Lemma 2.3.3) gives $\Delta A \Delta B \geq | \langle f | g \rangle | \geq | \operatorname{Im} \langle f | g \rangle | = \frac{1}{2} | \langle f | g \rangle - \langle g | f \rangle |$. Expanding, the $\langle A \rangle \langle B \rangle$ terms cancel and $\langle f | g \rangle - \langle g | f \rangle = \langle \psi | [A, B] | \psi \rangle$. $\square$

Proposition 3.3.5 (Heisenberg picture). Define $A_H(t) := \hat{U}(t, t_0)^\dagger A \hat{U}(t, t_0)$. Then expectations agree, $\langle \psi(t) | A | \psi(t) \rangle = \langle \psi(t_0) | A_H(t) | \psi(t_0) \rangle$, and for time-independent $\mathbf{H}$, $\frac{d}{dt} A_H = \frac{i}{\hbar} [\mathbf{H}, A_H]$.

Proof. The first claim is immediate from Equation (3.1). For the second, differentiate $A_H = e^{i\mathbf{H}t/\hbar} A e^{-i\mathbf{H}t/\hbar}$ using Proposition 3.3.2. $\square$

Remark 3.3.6 (Two pictures, one theory). One may push the time dependence onto the states (Schrödinger picture, used throughout this document: gates change the state) or onto the observables (Heisenberg picture: the state is fixed and the measured quantities evolve). They give identical predictions. The Heisenberg picture is the natural home of the stabiliser formalism behind Theorem 4.4.2, where one tracks how Pauli operators $P \mapsto CPC^\dagger$ evolve rather than tracking 2^n amplitudes.

3.4 Postulate 3: measurement

Postulate 3 (Projective measurement). A measurement on a system with state space $\mathcal{H}$ is specified by a family $\{P_m\}_{m \in O}$ of orthogonal projectors on $\mathcal{H}$, indexed by the set O of possible outcomes, satisfying

$$P_m P_{m'} = \delta_{mm'} P_m, \quad \sum_{m \in O} P_m = \mathbb{1}. \quad (3.5)$$

Performed on the state $|\psi\rangle$, the measurement yields outcome m with probability

$$p(m) = \langle \psi | P_m | \psi \rangle = \|P_m |\psi\rangle\|^2 \quad (\text{Born rule})$$

and, conditioned on that outcome, leaves the system in the state $|\psi_m\rangle = P_m |\psi\rangle / \|P_m |\psi\rangle\|$.

Proposition 3.4.1 (The probabilities are probabilities). *With the notation of Postulate 3, $p(m) \geq 0$ and $\sum_m p(m) = 1$.*

Proof. $p(m) = \langle \psi | P_m^\dagger P_m | \psi \rangle = \|P_m |\psi\rangle\|^2 \geq 0$ using $P_m^\dagger = P_m = P_m^2$. Summing and using $\sum_m P_m = \mathbb{I}$ gives $\sum_m p(m) = \langle \psi | \psi \rangle = 1$. $\square$

Definition 3.4.2 (Measuring in a basis). Given an orthonormal basis $\{|m\rangle\}$ of $\mathcal{H}$, *measuring in the basis $\{|m\rangle\}$* means performing the measurement with $P_m = |m\rangle\langle m|$. By Equation (Born rule), outcome m occurs with probability $|\langle m | \psi \rangle|^2$ and the state becomes $|m\rangle$.

This is the only kind of measurement we shall need, and almost always with $\{|m\rangle\}$ the computational basis.

Remark 3.4.3 (Observables, and what we do not need). Physicists usually package a measurement as a Hermitian operator $M = \sum_m m P_m$ (an *observable*), so that outcomes are eigenvalues and $\langle M \rangle = \langle \psi | M | \psi \rangle$ is the mean outcome. This adds nothing for our purposes: Theorem 2.5.1 shows the two descriptions are the same data. Likewise the general formalism (measurement operators $\{M_m\}$ with $\sum_m M_m^\dagger M_m = \mathbb{I}$, and POVMs) is strictly more general but not needed: every algorithm in this document ends with a computational-basis measurement and nothing else.

Proposition 3.4.4 (Repeatability). *Measuring twice in a row with the same $\{P_m\}$ gives the same outcome the second time, with probability 1.*

Proof. After outcome m the state is $P_m |\psi\rangle / \|P_m |\psi\rangle\|$; applying $P_{m'}$ to it gives $\delta_{mm'}$ times itself by Equation (3.5). $\square$

3.5 Postulate 4: composite systems

Postulate 4 (Composite systems). *The state space of a system composed of subsystems $1, \dots, n$ with state spaces $\mathcal{H}_1, \dots, \mathcal{H}_n$ is $\mathcal{H}_1 \otimes \dots \otimes \mathcal{H}_n$. If subsystem i is prepared in state $|\psi_i\rangle$, the joint state is $|\psi_1\rangle \otimes \dots \otimes |\psi_n\rangle$.*

Definition 3.5.1 (Product and entangled states). A state $|\psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ is a *product state* if $|\psi\rangle = |\alpha\rangle \otimes |\beta\rangle$ for some $|\alpha\rangle \in \mathcal{H}_A$, $|\beta\rangle \in \mathcal{H}_B$; otherwise it is *entangled*.

The definition is basis-independent: it asserts the non-existence of a factorisation, not the failure of one particular attempt.

Proposition 3.5.2. $|\Phi^+\rangle := \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ is entangled.

Proof. Suppose $|\Phi^+\rangle = (\alpha_0|0\rangle + \alpha_1|1\rangle) \otimes (\beta_0|0\rangle + \beta_1|1\rangle)$. Comparing coefficients of $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ gives $\alpha_0\beta_0 = \frac{1}{\sqrt{2}}$, $\alpha_0\beta_1 = 0$, $\alpha_1\beta_0 = 0$, $\alpha_1\beta_1 = \frac{1}{\sqrt{2}}$. The first forces $\alpha_0 \neq 0$ and $\beta_0 \neq 0$; then the second gives $\beta_1 = 0$ and the third gives $\alpha_1 = 0$, contradicting the fourth. $\square$

Remark 3.5.3 (The two-qubit determinant test). For $|\psi\rangle = \sum_{j,k \in \{0,1\}} c_{jk} |jk\rangle$ on two qubits, $|\psi\rangle$ is a product state if and only if $\det \begin{pmatrix} c_{00} & c_{01} \\ c_{10} & c_{11} \end{pmatrix} = 0$ — product means the coefficient matrix has rank ≤ 1 . This shortcut is specific to two qubits; in general one uses the Schmidt decomposition.

3.6 Measuring one part of a composite system

Quantum algorithms almost always measure one register and keep another. The following proposition is the rule for doing so. It is a direct consequence of [Postulate 3](#), and it is the exact statement that the final step of phase estimation uses.

Proposition 3.6.1 (Partial measurement). *Let $|\psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ be a unit vector and let $\{|a\rangle\}_{a \in O}$ be an orthonormal basis of $\mathcal{H}_A$. Write, uniquely,*

$$|\psi\rangle = \sum_{a \in O} |a\rangle \otimes |\chi_a\rangle, \quad |\chi_a\rangle \in \mathcal{H}_B \text{ (not normalised)}. \quad (3.6)$$

Measuring subsystem A in the basis $\{|a\rangle\}$ — formally, the measurement with projectors $P_a = |a\rangle\langle a| \otimes \mathbb{1}_B$ — yields outcome a with probability $p(a) = \|\chi_a\|^2$, and leaves the joint system in the state $|a\rangle \otimes |\chi_a\rangle / \|\chi_a\|$.

Proof. The P_a are projectors by [Proposition 2.7.4](#) and satisfy [Equation \(3.5\)](#) because $\{|a\rangle\langle a|\}$ does and $\sum_a |a\rangle\langle a| \otimes \mathbb{1} = \mathbb{1} \otimes \mathbb{1}$. Existence and uniqueness of [Equation \(3.6\)](#): expand $|\psi\rangle$ in the product basis $\{|a\rangle \otimes |b_k\rangle\}$ and group terms by a . Then $P_a |\psi\rangle = |a\rangle \otimes |\chi_a\rangle$, so $p(a) = \|P_a |\psi\rangle\|^2 = \|\chi_a\|^2$ by [Equation \(2.7\)](#), and [Postulate 3](#) gives the post-measurement state. $\square$

Read as an algorithm

To find out what measuring register A does: *factor the state so that every distinct basis state of A appears exactly once*, as $\sum_a |a\rangle \otimes |\chi_a\rangle$. The squared norms of the leftover vectors $|\chi_a\rangle$ are the outcome probabilities, and the surviving state of B is $|\chi_a\rangle$ renormalised.

Every analysis of a quantum algorithm in this document is an application of this recipe. When [Theorem 8.3.2](#) computes $\Pr[\ell]$, it is doing exactly this.

Example 3.6.2 (Correlation, not action at a distance). For $|\Phi^+\rangle = \frac{1}{\sqrt{2}}|0\rangle|0\rangle + \frac{1}{\sqrt{2}}|1\rangle|1\rangle$ we read off $|\chi_0\rangle = \frac{1}{\sqrt{2}}|0\rangle$ and $|\chi_1\rangle = \frac{1}{\sqrt{2}}|1\rangle$, so measuring the first qubit gives 0 or 1 with probability $\frac{1}{2}$ each, after which the second qubit is in the matching state with certainty. The outcomes are perfectly correlated, yet the marginal distribution on either side is uniform regardless of what the other side does: no information is transmitted.

3.7 Global phase, relative phase, and the qubit

Definition 3.7.1 (Qubit). A *qubit* is a quantum system whose state space is $\mathbb{C}^2$ with the distinguished orthonormal basis $\{|0\rangle, |1\rangle\}$. Its states are the unit vectors $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$, taken up to global phase. Two further states recur constantly:

$$|\pm\rangle := \frac{|0\rangle \pm |1\rangle}{\sqrt{2}}, \quad \text{so} \quad |0\rangle = \frac{|+\rangle + |-\rangle}{\sqrt{2}}, \quad |1\rangle = \frac{|+\rangle - |-\rangle}{\sqrt{2}}. \quad (3.7)$$

Proposition 3.7.2 (Global phase is unobservable). *For any $\gamma \in \mathbb{R}$, the states $|\psi\rangle$ and $e^{i\gamma}|\psi\rangle$ give the same probabilities for every measurement, and remain indistinguishable under every unitary evolution.*

Proof. For any projector P , $\langle \psi | e^{-i\gamma} P e^{i\gamma} | \psi \rangle = \langle \psi | P | \psi \rangle$, so [Equation \(Born rule\)](#) is unchanged. And $U(e^{i\gamma}|\psi\rangle) = e^{i\gamma}(U|\psi\rangle)$ by linearity, so the relation is preserved forever. Equivalently, both vectors have the same density operator. $\square$

Example 3.7.3 (Relative phase is observable). $|+\rangle$ and $|-\rangle$ give identical statistics in the computational basis (both $\frac{1}{2}, \frac{1}{2}$): a measurement *now* cannot tell them apart. Yet they are orthogonal, hence perfectly distinguishable — and concretely, with $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$,

$$H|+\rangle = |0\rangle, \quad H|-\rangle = |1\rangle.$$

The relative sign was invisible to one measurement and *decisive* after one gate.

The same structure, at scale

Example 3.7.3 is the whole algorithm in miniature. A relative phase between two amplitudes of one superposition is physical but not directly visible; a later operation that *recombines the branches* converts it into a probability. The inverse QFT plays the role of that second Hadamard at scale (Chapter 8). Shor's algorithm consists of manufacturing the right relative phases and then applying that conversion.

Proposition 3.7.4 (Bloch sphere). *Every qubit state has a representative of the form*

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle, \quad \theta \in [0, \pi], \phi \in [0, 2\pi), \quad (3.8)$$

and (θ, ϕ) is unique except at $\theta \in \{0, \pi\}$, where ϕ is irrelevant. The map sending $|\psi\rangle$ to the Bloch vector

$$\vec{r} := (\sin\theta \cos\phi, \sin\theta \sin\phi, \cos\theta)$$

is a bijection from qubit states onto the unit sphere of $\mathbb{R}^3$, the Bloch sphere.

Proof. Write $\alpha = r_\alpha e^{i\varphi_\alpha}$, $\beta = r_\beta e^{i\varphi_\beta}$ with $r_\alpha, r_\beta \geq 0$ and $r_\alpha^2 + r_\beta^2 = 1$; choose $\theta \in [0, \pi]$ with $r_\alpha = \cos(\theta/2)$, $r_\beta = \sin(\theta/2)$, possible since $\cos(\theta/2)$ decreases from 1 to 0 on that interval. Multiplying by the global phase $e^{-i\varphi_\alpha}$ gives Equation (3.8) with $\phi = \varphi_\beta - \varphi_\alpha$. Uniqueness and bijectivity are the standard spherical-coordinate statements. $\square$

Thus a qubit carries exactly two real parameters: four real numbers in (α, β) , minus one for normalisation, minus one for global phase. More generally a pure state of $\mathbb{C}^d$ costs $2d - 2$ real parameters.

Warning 3.7.5. There is no comparably useful picture for two or more qubits. The Bloch sphere is an aid to single-qubit intuition and nothing more.

3.8 Registers, and how integers are stored in them

Definition 3.8.1 (Register). An n -qubit register is a system of n qubits, with state space $(\mathbb{C}^2)^{\otimes n} \cong \mathbb{C}^{2^n}$ and computational basis $\{|x_1 x_2 \dots x_n\rangle : x \in \{0, 1\}^n\}$ of size 2^n .

Convention 3.8.2 (Integer encoding, big-endian). We identify $\{0, 1\}^n$ with $\{0, 1, \dots, 2^n - 1\}$ by

$$x = x_1 x_2 \dots x_n \longleftrightarrow x = \sum_{k=1}^n x_k 2^{n-k} = x_1 2^{n-1} + x_2 2^{n-2} + \dots + x_n 2^0, \quad (3.9)$$

and write $|x\rangle$ for either reading. So x_1 is the *most* significant bit and, by convention (C3), it sits on the *top* wire. For example $|3\rangle = |011\rangle$ and $|6\rangle = |110\rangle$ on three qubits.

The single most error-prone convention in the subject

The QFT circuit outputs its register in *bit-reversed* order relative to its input (Section 6.3), so a document that is careless here will produce a circuit that does not match its own formula. Published figures with reversed output labels exist. Qiskit uses the opposite convention from this document. Whenever you move between sources, check Equation (3.9) first, and verify the composed circuit end to end on a known basis state.

The output bottleneck

A register of n qubits is described by 2^n complex amplitudes, but *one measurement returns n classical bits and destroys the state*. The amplitudes are not directly readable, and by Theorem 3.9.1 the

state cannot be copied and re-measured.
Concretely: one layer of Hadamards produces

$$H^{\otimes n} |0\rangle^{\otimes n} = \bigotimes_{s=1}^n \frac{|0\rangle + |1\rangle}{\sqrt{2}} = \frac{1}{2^{n/2}} \sum_{x=0}^{2^n-1} |x\rangle,$$

in which every amplitude has the same modulus $2^{-n/2}$. Measuring it returns a uniformly random x — exactly what a classical coin gives. *An unmeasured uniform superposition carries zero information.* The exponentially large space is necessary but nowhere near sufficient; the work is done by *interference*, which redistributes those amplitudes before measurement.

At $n = 300$ the state space has dimension $2^{300} \approx 10^{90}$, more than the number of atoms in the observable universe — and a measurement still returns 300 classical bits. The design problem is arranging for those 300 bits to be the informative ones.

3.9 Two things quantum states cannot do

Theorem 3.9.1 (No-cloning). *There is no unitary U on $\mathcal{H} \otimes \mathcal{H}$ and state $|s\rangle \in \mathcal{H}$ such that $U(|\psi\rangle \otimes |s\rangle) = |\psi\rangle \otimes |\psi\rangle$ for every unit vector $|\psi\rangle \in \mathcal{H}$, when $\dim \mathcal{H} \geq 2$.*

Proof. Suppose such U exists and let $|\psi\rangle, |\phi\rangle$ be unit vectors. Unitaries preserve inner products ([Proposition 2.4.6](#)), so using [Equation \(2.7\)](#),

$$\langle \psi | \phi \rangle = \langle \psi | \phi \rangle \langle s | s \rangle = (\langle \psi | \otimes \langle s |) (|\phi\rangle \otimes |s\rangle) = (\langle \psi | \otimes \langle \psi |) (|\phi\rangle \otimes |\phi\rangle) = \langle \psi | \phi \rangle^2.$$

Hence $z = z^2$ for $z = \langle \psi | \phi \rangle$, so $z \in \{0, 1\}$: the two states are orthogonal ($z = 0$) or, by [Lemma 2.3.3](#), equal ($z = 1$). Since $\dim \mathcal{H} \geq 2$ admits pairs with $0 < |\langle \psi | \phi \rangle| < 1$ — for instance $|0\rangle$ and $|+\rangle$ — no such U exists. $\square$

Remark 3.9.2 (Copying basis states is fine). The map $|x\rangle \otimes |0\rangle \mapsto |x\rangle \otimes |x\rangle$ for $x \in \{0, 1\}$ is unitary — it is the CNOT gate. Extended linearly it sends $(\alpha|0\rangle + \beta|1\rangle) \otimes |0\rangle \mapsto \alpha|00\rangle + \beta|11\rangle$, which is *not* $|\psi\rangle \otimes |\psi\rangle = \alpha^2|00\rangle + \alpha\beta|01\rangle + \alpha\beta|10\rangle + \beta^2|11\rangle$. So: classical information can be copied, arbitrary quantum states cannot, and attempting to copy a superposition produces *entanglement* instead. This observation is not a curiosity — it is exactly the mechanism by which the oracle in [Chapter 10](#) entangles the two registers.

Proposition 3.9.3 (Non-orthogonal states cannot be told apart). *If $|\psi_1\rangle, |\psi_2\rangle$ are unit vectors with $0 < |\langle \psi_1 | \psi_2 \rangle| < 1$, no measurement identifies which of the two a given system is in with certainty. Orthogonal states, by contrast, are perfectly distinguishable: measure in a basis containing them.*

Together, [Theorem 3.9.1](#) and this proposition explain why the naive strategy “prepare the state, copy it many times, measure the copies” — which would trivialise phase estimation — is unavailable.

3.10 Interference, precisely

Interference is talked about loosely more often than it is stated. Here is the statement.

Proposition 3.10.1 (Interference). *Let $|\psi\rangle = \sum_x \alpha_x |x\rangle$ and let V be unitary with $V_{yx} = \langle y | V | x \rangle$. Then*

$$V|\psi\rangle = \sum_y \underbrace{\left(\sum_x V_{yx} \alpha_x \right)}_{\beta_y} |y\rangle, \quad \Pr[y] = |\beta_y|^2 = \left| \sum_x V_{yx} \alpha_x \right|^2. \quad (3.10)$$

For two contributions $A = \rho e^{i\theta}$ and $B = \sigma e^{i\phi}$,

$$|A + B|^2 = |A|^2 + |B|^2 + \underbrace{2\rho\sigma \cos(\theta - \phi)}_{\text{interference term}}. \quad (3.11)$$

The cross term is positive when the contributions are nearly aligned, negative when nearly opposite, and zero when orthogonal in phase. *Only the relative phase enters.*

The four rules of interference

(R1) Amplitudes are summed *before* $|\cdot|^2$ is taken; interference is the cross term in that square.

(R2) Only contributions landing on the *same* final basis state $|y\rangle$ are added.

(R3) If a label sits in an orthogonal unmeasured register, the cross term carries $\langle \chi_x | \chi_{x'} \rangle = 0$ and *vanishes*: which-path information kills interference.

(R4) A relative phase becomes observable only when a later operation recombines the branches.

Naming discipline. Do not write “the phases interfere”. Write: *the contributions* $V_{yx}\alpha_x$ and $V_{yx'}\alpha_{x'}$ *to* $|y\rangle$ *add with relative phase* $\theta - \phi$. If the two contributions cannot be named, there is no interference to discuss.

Rule (R3) is not a technicality; it is the reason [Section 4.3](#) insists that every arithmetic block be garbage-free, and the reason the elliptic-curve point addition of [Chapter 17](#) must recompute its own slope from its output. It is worth stating separately.

Proposition 3.10.2 (Which-path information destroys interference). *Suppose the joint state is $|\Psi\rangle = \sum_x \alpha_x |x\rangle |\chi_x\rangle$ with $\langle \chi_x | \chi_{x'} \rangle = 0$ for $x \neq x'$. Then for any unitary V acting on the first register alone, the outcome distribution of measuring the first register after V is*

$$\Pr[y] = \sum_x |V_{yx}\alpha_x|^2,$$

with no cross terms: exactly the classical mixture. The second register records which branch the first is in, and that record suffices to eliminate interference — whether or not anybody reads it.

Proof. $(V \otimes \mathbb{I})|\Psi\rangle = \sum_y |y\rangle \otimes (\sum_x V_{yx}\alpha_x |\chi_x\rangle)$. By [Proposition 3.6.1](#),

$$\Pr[y] = \left\| \sum_x V_{yx}\alpha_x |\chi_x\rangle \right\|^2 = \sum_{x,x'} V_{yx}\alpha_x \overline{V_{yx'}\alpha_{x'}} \langle \chi_{x'} | \chi_x \rangle = \sum_x |V_{yx}\alpha_x|^2$$

by orthogonality. □

Entanglement: where it sits, and what it is not

Entanglement appears in Shor’s algorithm at exactly one place — when the oracle correlates the counting and work registers ([Chapter 10](#), stage 3) — and what it *does* there is keep the r eigenphase branches from mixing. It is necessary bookkeeping, not the engine.

Independently, it cannot be sufficient: the Gottesman–Knill theorem ([Theorem 4.4.2](#)) exhibits highly entangled circuits that are classically simulable in polynomial time. Whatever makes a quantum computation hard to simulate must live outside the Clifford group.

The circuit model, gates, and what a gate costs

This chapter does three things. It fixes the *semantics* of circuit diagrams — what a picture denotes as an operator — so that every later diagram is unambiguous. It builds the embedding of classical computation into the reversible quantum model, together with the uncomputation discipline that [Proposition 3.10.2](#) makes mandatory. And it establishes the cost model: in a fault-tolerant architecture, Clifford gates are nearly free and the T gate is not, so T -count — not raw gate count — is the currency in which every resource estimate in [Chapter 18](#) is quoted.

4.1 Semantics of a circuit diagram

Definition 4.1.1 (Quantum circuit). A *quantum circuit* on q qubits is a finite sequence $G_1, G_2, \dots, G_L$ of gates, each G_i being a unitary acting on a specified subset of the wires (and as the identity on the others). It *denotes* the unitary

$$\llbracket G_1 G_2 \cdots G_L \rrbracket = G_L G_{L-1} \cdots G_2 G_1 \in U(2^q). \quad (4.1)$$

The four reading rules

- (D1) **Wires are qubits, time flows left to right.** A horizontal line is one qubit persisting through time, not a wire carrying a signal.
- (D2) **Composition is right to left.** A diagram showing G_1 then G_2 denotes $G_2 G_1$, because that is how operator application composes. This reversal is the single most common source of transcription error between a diagram and a matrix.
- (D3) **A gate on a subset of wires means $G \otimes \mathbb{I}$ on the rest,** with the tensor factors ordered by wire position ([Convention 3.8.2](#): top wire most significant).
- (D4) **Gates on disjoint wires commute** and may be slid past one another freely; this is $(A \otimes \mathbb{I})(\mathbb{I} \otimes B) = (\mathbb{I} \otimes B)(A \otimes \mathbb{I})$ from [Equation \(2.8\)](#).

What is forbidden

A circuit diagram may *not* contain: fan-out (copying a wire — forbidden by [Theorem 3.9.1](#)); fan-in (merging two wires into one); loops or feedback (the diagram is a DAG in time); or a gate that is not unitary. Measurement is allowed but is not a gate; it is drawn with a meter symbol and denotes [Postulate 3](#), not an element of [Equation \(4.1\)](#).

Definition 4.1.2 (Ancillas and garbage). A wire is an *ancilla* if it does not carry an input, i.e. it is initialised to a known state (always $|0\rangle$ here). An output wire is *garbage* if it is neither an input, nor a desired output, nor an ancilla restored to $|0\rangle$. Two equivalent accountings hold for a circuit on q qubits:

$$q = \text{inputs} + \text{ancillas} = \text{outputs} + \text{garbage},$$

and solving the second for the garbage term is the standard way to check a design.

Proposition 4.1.3 (Deferred measurement). *Any circuit containing a mid-circuit measurement whose outcome classically controls a later gate is equivalent — as a distribution over final classical outcomes — to a circuit in which the measurement is moved to the end and the classical control is replaced by a quantum control. Conversely (the principle of implicit measurement), a controlled gate whose control is measured immediately afterwards and never reused may be replaced by a classically controlled gate applied after measuring that control.*

Proof sketch. For the forward direction, replace “measure qubit c , then apply V if the outcome was 1” by “apply c - V controlled on c , then measure c ”. The controlled gate leaves the joint state $\alpha|0\rangle|\psi\rangle + \beta|1\rangle V|\psi\rangle$; measuring c afterwards yields 0 with probability $|\alpha|^2$ and residual state $|\psi\rangle$, and 1 with probability $|\beta|^2$ and residual $V|\psi\rangle$ — exactly the original distribution and post-states, by [Proposition 3.6.1](#). The converse runs the argument backwards; it is legitimate because no later gate is controlled by c , so c is diagonal-in-the-computational-basis from that point on. $\square$

[Proposition 4.1.3](#) is what makes the *semiclassical* (Griffiths–Niu) quantum Fourier transform of [Section 8.6](#) exact rather than approximate, and it is the mechanism behind Beauregard’s $2n + 3$ -qubit circuit.

4.2 The gate zoo

4.2.1 One-qubit gates

Definition 4.2.1 (Pauli operators).

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (4.2)$$

Each is both Hermitian and unitary, and each squares to I . In action, $X|x\rangle = |x \oplus 1\rangle$ and $Z|x\rangle = (-1)^x|x\rangle$ for $x \in \{0, 1\}$: X is the quantum NOT (bit flip) and Z is a phase flip. Because $X^2 = Y^2 = Z^2 = I$, [Equation \(2.5\)](#) applies to each.

Gate	Matrix	Action on $ b\rangle$	Where it is used here
I	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	$ b\rangle$	padding; the “do nothing” branch of c - U
X	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	$ b \oplus 1\rangle$	preparing $ 1\rangle$ in the work register; two’s-complement negation; folding a constant bit $a_i = 1$ into a circuit
Y	$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$	$i(-1)^b b \oplus 1\rangle$	not used directly; needed only to define the Pauli group
Z	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	$(-1)^b b\rangle$	$= R_1$; the $k = 1$ case of the QFT rotation
H	$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	$\frac{1}{\sqrt{2}}(0\rangle + (-1)^b 1\rangle)$	the $H^{\otimes t}$ layer; first and last gate of every QFT factor; the X -basis measurement in the temporary-AND
$P(\theta)$	$\begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}$	$e^{i\theta b} b\rangle$	the general phase gate; all QFT rotations are instances
S	$\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$	$i^b b\rangle$	$= P(\pi/2) = R_2$; Clifford generator
T	$\begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$	$e^{i\pi b/4} b\rangle$	$= P(\pi/4) = R_3$; the unit of fault-tolerant cost
R_k	$\begin{pmatrix} 1 & 0 \\ 0 & e^{2\pi i/2^k} \end{pmatrix}$	$e^{2\pi i b/2^k} b\rangle$	$= P(2\pi/2^k)$; the QFT rotation ladder, $k = 2, \dots, n$

The rotation gates, needed for [Theorem 4.2.3](#), are $R_x(\theta) = e^{-i\theta X/2}$, $R_y(\theta) = e^{-i\theta Y/2}$, $R_z(\theta) = e^{-i\theta Z/2} = \text{diag}(e^{-i\theta/2}, e^{i\theta/2})$, so that $P(\theta) = e^{i\theta/2}R_z(\theta)$; every $U \in U(2)$ factors as $U = e^{i\alpha}R_z(\beta)R_y(\gamma)R_z(\delta)$.

R_k beyond $k = 3$

$R_1 = Z$, $R_2 = S$, $R_3 = T$. Beyond $k = 3$ the R_k are *not* in Clifford+ T and must be synthesised approximately, at cost $\approx 3\log_2(1/\epsilon)$ T gates apiece. That single fact is why the approximate QFT

of Section 6.4 exists, and why the Draper adder of Section 12.5 loses to ripple-carry arithmetic by a factor of about 30.

4.2.2 Two- and three-qubit gates

Definition 4.2.2 (Controlled gate). On $1 + m$ qubits,

$$\text{c-}U := |0\rangle\langle 0| \otimes \mathbb{I}_{2^m} + |1\rangle\langle 1| \otimes U, \quad (4.3)$$

so $\text{c-}U |c\rangle |\psi\rangle = |c\rangle U^c |\psi\rangle$ for $c \in \{0, 1\}$. By linearity this determines the action on superpositions of the control — which is the entire content of phase kickback (Lemma 7.2.1).

Gate	Matrix	Action	Where it is used here
CNOT = c-X	4×4 perm.	$ a\rangle b\rangle \mapsto a\rangle b \oplus a\rangle$	copying a basis value; every sum bit of the adder; the MAJ/UMA blocks
CZ = c-Z	$\text{diag}(1, 1, 1, -1)$	$(-1)^{ab} a\rangle b\rangle$	the free correction in temporary-AND uncomputation
c- $P(\theta)$	$\text{diag}(1, 1, 1, e^{i\theta})$	$e^{i\theta ab} a\rangle b\rangle$	every controlled rotation of QFT and QFT [†]
SWAP	4×4 perm.	$ a\rangle b\rangle \mapsto b\rangle a\rangle$	the QFT's closing bit reversal; the in-place modular multiplier; binary shifts
TOF = c-c-X	8×8 perm.	$ a\rangle b\rangle c\rangle \mapsto a\rangle b\rangle c \oplus ab\rangle$	every carry in every adder; the unit in which arithmetic cost is reported
FREDKIN	8×8 perm.	controlled swap	controlling the swap layer of the in-place multiplier
$C^k(X)$	2^{k+1} perm.	X iff all k controls are 1	doubly-controlled modular additions inside a controlled U_a

4.2.3 Controlling a gate makes a global phase local

The one place global phase matters

By Proposition 3.7.2 the gates U and $e^{i\alpha}U$ are physically indistinguishable. But $\text{c-}U$ and $\text{c-}(e^{i\alpha}U)$ are *not*: the second applies $e^{i\alpha}$ only on the $|1\rangle$ branch of the control, where it is a *relative* phase. Controlling a gate promotes its global phase to an observable one. This is exactly the mechanism of phase kickback, and it is why Theorem 4.2.3 needs an explicit $P(\alpha)$ on the control wire.

Theorem 4.2.3 (Barenco et al.: the ABC decomposition). *For every $U \in U(2)$ there exist $A, B, C \in SU(2)$ and $\alpha \in \mathbb{R}$ with*

$$ABC = \mathbb{I}, \quad U = e^{i\alpha} A X B X C. \quad (4.4)$$

Then $\text{c-}U$ is realised with two CNOTs and three single-qubit gates, plus a phase $P(\alpha)$ on the control.

Proof. If the control is $|0\rangle$ the target sees $ABC = \mathbb{I}$. If it is $|1\rangle$ the target sees $AXBXC = e^{-i\alpha}U$, and the $P(\alpha)$ on the control supplies the missing $e^{i\alpha}$ — which by Section 4.2.3 is a relative phase, hence physical, hence necessary. For the construction, write $U = e^{i\alpha} R_z(\beta) R_y(\gamma) R_z(\delta)$ and take

$$A = R_z(\beta) R_y(\frac{\gamma}{2}), \quad B = R_y(-\frac{\gamma}{2}) R_z(-\frac{\delta+\beta}{2}), \quad C = R_z(\frac{\delta-\beta}{2}),$$

then verify $ABC = \mathbb{I}$ and $AXBXC = R_z(\beta) R_y(\gamma) R_z(\delta)$ using $X R_y(\theta) X = R_y(-\theta)$ and $X R_z(\theta) X = R_z(-\theta)$. $\square$

Proposition 4.2.4 (Controlled phase gates, the case the QFT needs). $c\text{-}P(\theta) = \text{diag}(1, 1, 1, e^{i\theta})$ is implemented by $P(\theta/2)$ on the control, $\text{CNOT}_{c \rightarrow t}$, $P(-\theta/2)$ on the target, $\text{CNOT}_{c \rightarrow t}$, $P(\theta/2)$ on the target: two CNOTs and three phase gates.

Proof. The accumulated phase on $|c, t\rangle$ is $\exp(\frac{i\theta}{2}[c - (t \oplus c) + t])$. Evaluating the bracket: $(0, 0) \mapsto 0 - 0 + 0 = 0$; $(1, 0) \mapsto 1 - 1 + 0 = 0$; $(0, 1) \mapsto 0 - 1 + 1 = 0$; $(1, 1) \mapsto 1 - 0 + 1 = 2$, giving $e^{i\theta}$. $\square$

Remark 4.2.5 (Control and target are interchangeable). As a matrix, $c\text{-}R_k = \text{diag}(1, 1, 1, e^{2\pi i/2^k})$ is symmetric under exchanging the two qubits. This is a general feature of controlled-diagonal gates, and it means QFT diagrams may be drawn with either qubit carrying the control dot.

4.3 Reversible classical computation

4.3.1 Two-bit reversible gates are not universal

Theorem 4.3.1 (Two bits do not suffice). *Every reversible circuit built from gates acting on at most two bits computes an affine map $\mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$, $x \mapsto Mx + c$. Consequently none computes $(x, y, z) \mapsto (x, y, z \oplus xy)$, nor $\wedge$ — even with ancillas and garbage allowed.*

Proof. *Step 1: every 1- and 2-bit reversible gate is affine.* For one bit the only bijections of $\mathbb{F}_2$ are $x \mapsto x$ and $x \mapsto x + 1$. For two bits, count: the reversible 2-bit gates form S_4 , of order 24, and $|AGL_2(\mathbb{F}_2)| = (2^2 - 1)(2^2 - 2) \cdot 4 = 24$ as well. Every affine bijection permutes the four points, so $AGL_2(\mathbb{F}_2) \leq S_4$; equal orders force equality.

Step 2: affineness is preserved under embedding into more wires and under composition, since $M_2(M_1x + c_1) + c_2 = (M_2M_1)x + (M_2c_1 + c_2)$.

Step 3. Every output bit is therefore a polynomial of degree ≤ 1 over $\mathbb{F}_2$; but xy has degree 2. $\square$

The order coincidence, and its consequence

$|S_4| = |AGL_2(\mathbb{F}_2)| = 24$ is a coincidence of small numbers. For three bits $|S_8| = 40320$ while $|AGL_3(\mathbb{F}_2)| = 168 \cdot 8 = 1344$, so the overwhelming majority of 3-bit reversible gates are *not* affine — and the Toffoli gate is one of them. That is exactly why the step from two bits to three buys universality: *non-linearity is what computation needs*.

Proposition 4.3.2 (Toffoli is universal for reversible classical computation). $\{\text{TOF}\}$ suffices: $\text{TOF}(x, y, 1) = (x, y, \text{nand}(x, y))$ and $\text{TOF}(1, x, 0) = (1, x, x)$ give NAND and copy. Replacing each gate of a size- s classical circuit by its reversible counterpart gives $O(s)$ gates and $O(s)$ ancillas.

4.3.2 Computing a classical function reversibly

A classical $f : \{0, 1\}^n \rightarrow \{0, 1\}^m$ is generally not injective, so $|x\rangle \mapsto |f(x)\rangle$ is not unitary. The standard fix uses an extra register:

$$U_f : |x\rangle |z\rangle \mapsto |x\rangle |z \oplus f(x)\rangle, \quad U_f^2 = \mathbb{I}. \quad (4.5)$$

This is unitary (a permutation of basis states) and is built from Toffoli and CNOT gates. Applied to a superposition,

$$U_f \left(\frac{1}{\sqrt{2^n}} \sum_x |x\rangle \right) |0\rangle = \frac{1}{\sqrt{2^n}} \sum_x |x\rangle |f(x)\rangle \quad \text{— one call, all inputs.}$$

Garbage must be erased, and this is not optional

A naive circuit leaves intermediate scratch bits behind: $|x\rangle |f(x)\rangle |g(x)\rangle$. By rule (R3) — formally, **Proposition 3.10.2** — that garbage tags the branch and destroys the interference the algorithm depends on. This is not a matter of tidiness or qubit economy. A circuit that leaves garbage computes a different, and wrong, distribution.

Theorem 4.3.3 (Bennett: compute–copy–uncompute). *Let C be a reversible circuit computing $|x\rangle|0\rangle|0\rangle \mapsto |x\rangle|f(x)\rangle|g(x)\rangle$ with scratch g . Then the composite circuit*

$$|x\rangle|y\rangle|0\rangle|0\rangle \xrightarrow{C} |x\rangle|y\rangle|f(x)\rangle|g(x)\rangle \xrightarrow{\text{copy}} |x\rangle|y \oplus f(x)\rangle|f(x)\rangle|g(x)\rangle \xrightarrow{C^{-1}} |x\rangle|y \oplus f(x)\rangle|0\rangle|0\rangle$$

realises U_f exactly, with all scratch restored, at twice the gate count of C plus m CNOTs.

Proof. The copy step is m CNOTs from the f -register into the y -register; it does not touch the x - or g -registers. Hence C^{-1} receives exactly the state C produced on those registers and inverts it. The y -register is untouched by C^{-1} because C never acted on it. $\square$

The discipline, stated once

Every arithmetic block in Chapter 12 and every field operation in Chapter 17 is engineered this way, and every one of them ends by recomputing its own flags from its output. When you read the modular adder of Theorem 12.3.1 or the point addition of Algorithm 17.4.1, the step that looks mysterious is always the same step: the one that reconstructs a scratch value from data that survives, so that it can be cleared.

4.4 Universality and the cost model

4.4.1 Clifford+ T and Gottesman–Knill

Definition 4.4.1 (Pauli and Clifford groups). The n -qubit Pauli group is $\mathcal{P}_n = \{i^k P_1 \otimes \cdots \otimes P_n\}$ with $P_j \in \{I, X, Y, Z\}$. The Clifford group is its normaliser, $\mathcal{C}_n = \{C \in U(2^n) : CPC^\dagger \in \mathcal{P}_n \forall P \in \mathcal{P}_n\}/U(1)$, generated by $\{H, S, \text{CNOT}\}$.

$T \notin \mathcal{C}_1$: indeed $TXT^\dagger = \frac{1}{\sqrt{2}}(X + Y)$, which is not a scalar multiple of any Pauli. But $T^2 = S \in \mathcal{C}_1$.

Theorem 4.4.2 (Gottesman–Knill). *A circuit built only from computational-basis preparation, Clifford gates, and computational-basis measurement (including adaptive Clifford corrections) can be simulated exactly in time polynomial in the number of qubits and gates.*

Proof idea. Track the stabiliser group rather than 2^n amplitudes: for a stabiliser state it is generated by n commuting Paulis, storable in $O(n^2)$ bits. A Clifford gate maps generators to generators, $P \mapsto CPC^\dagger$, an $O(n)$ update each, so $O(n^2)$ per gate. $\square$

Why this settles the “entanglement is the speed-up” question

Bell and GHZ states are stabiliser states, produced by Clifford circuits, and classically simulable. So entanglement alone *cannot* be the source of quantum advantage. Whatever makes a computation hard to simulate must live outside the Clifford group; for Clifford+ T that is the T gate.

In the surface code this is not a coincidence but the same fact seen twice. Paulis are bookkeeping, H and S are single-patch operations, CNOT is lattice surgery — all cheap. T is not transversal and must be injected from a magic state $|A\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{i\pi/4}|1\rangle)$, produced by distillation or cultivation, which dominates the qubit budget. *The gate that places a circuit outside the classically simulable class is the same gate that dominates its fault-tolerant cost.*

Theorem 4.4.3 (Solovay–Kitaev). *Any finite universal gate set closed under inverses approximates any single-qubit unitary to precision ϵ using $O(\log^c(1/\epsilon))$ gates, with $c \approx 4$ for generic sets. For z -rotations in Clifford+ T , the optimal ancilla-free synthesis of Ross–Selinger achieves*

$$R_z(\theta) \text{ to precision } \epsilon \approx 3 \log_2(1/\epsilon) \text{ } T \text{ gates.} \quad (4.6)$$

4.4.2 The Toffoli gate in Clifford+T

Proposition 4.4.4 (*T-count of a Toffoli is 7*). *The Toffoli gate decomposes into 7 T-type gates, 6 CNOTs and 2 Hadamards, with no ancilla, at T-depth 3 (Amy–Maslov–Mosca–Roetteler) or T-depth 1 with 4 ancillas (Selinger).*

Why exactly seven. $\text{TOF} = H_c C^2(Z) H_c$, and with $a, b, c \in \{0, 1\}$ the basis labels,

$$C^2(Z) = \exp\left(\frac{i\pi}{4} [a + b + c - (a \oplus b) - (a \oplus c) - (b \oplus c) + (a \oplus b \oplus c)]\right),$$

since the bracket vanishes except on $|111\rangle$, where it equals 4. Each of the seven terms is a $\pi/4$ phase on a parity of the labels — that is, a T gate conjugated by CNOTs. Hence exactly seven. $\square$

Proposition 4.4.5 (Multiply-controlled NOT). *For $k \geq 3$, $C^k(X)$ decomposes as: (a) with $k - 2$ clean ancillas, $2k - 3$ Toffoli gates; (b) with one borrowed ancilla in an arbitrary unknown state, $O(k)$ Toffolis; (c) with no ancilla at all, $\Theta(k^2)$ elementary gates.*

Proof of (a). With ancillas $b_1, \dots, b_{k-2}$, compute the conjunction in a binary chain

$$\text{TOF}(c_1, c_2; b_1), \quad \text{TOF}(b_1, c_3; b_2), \quad \dots, \quad \text{TOF}(b_{k-3}, c_{k-1}; b_{k-2}),$$

so that $b_{k-2} = c_1 \cdots c_{k-1}$; apply $\text{TOF}(b_{k-2}, c_k; \text{target})$; run the chain backwards to restore the ancillas. Total $(k - 2) + 1 + (k - 2) = 2k - 3$. This is Theorem 4.3.3 with the copy step replaced by one controlled operation. $\square$

4.4.3 The temporary logical-AND: Toffolis in pairs cost 4 T

Theorem 4.4.6 (Measurement-based uncomputation of an AND). *Let $|\Psi\rangle = \sum_{x,y \in \{0,1\}} \alpha_{xy} |x\rangle |y\rangle |xy\rangle$ be any state in which the third qubit holds the AND of the first two. Measure the third qubit in the X basis $\{|+\rangle, |-\rangle\}$. Then:*

- (1) *each outcome occurs with probability exactly $\frac{1}{2}$;*
- (2) *on outcome $+$, the remaining state is $\sum_{x,y} \alpha_{xy} |x\rangle |y\rangle$ — the AND is erased;*
- (3) *on outcome $-$, the state is $\sum_{x,y} (-1)^{xy} \alpha_{xy} |x\rangle |y\rangle$, which a single CZ corrects.*

The whole uncomputation uses zero T gates, since CZ is Clifford.

Proof. Projecting the third qubit onto $|\pm\rangle$ gives, up to normalisation, $\frac{1}{\sqrt{2}} \sum_{x,y} (\pm 1)^{xy} \alpha_{xy} |x\rangle |y\rangle$, using $\langle + | t \rangle = 1/\sqrt{2}$ and $\langle - | t \rangle = (-1)^t/\sqrt{2}$. The squared norm is $\frac{1}{2}$ either way, giving (1). For $+$ all signs are $+1$, giving (2). For $-$ the factor $(-1)^{xy}$ is exactly the action of $CZ = \text{diag}(1, 1, 1, -1)$, and $CZ^2 = \mathbb{1}$, giving (3). $\square$

The consequence, which reshapes every count

Computing an AND costs 4 T; uncomputing it costs *none*. A Toffoli occurring as a compute/uncompute pair therefore costs 4 T in total rather than $2 \times 7 = 14$.

Almost every Toffoli in Shor's algorithm is half of such a pair — each carry in the CDKM adder is computed by a MAJ block and uncomputed by the matching UMA block. So the n -bit adder costs $4n + O(1)$ T gates, not $14n$. A free measurement beats a unitary.

4.5 What to count: two cost regimes

	NISQ	Fault-tolerant
dominant error source	two-qubit gates, decoherence	magic-state production
count this	CNOT-count, CNOT-depth	T -count, T -depth, Toffoli count
free	— (nothing)	Clifford gates
expensive	every gate	T , TOF, arbitrary rotations
volume proxy	KQ_{CNOT}	KQ_T , spacetime volume
preferred adder	minimise two-qubit gates (Draper wins: zero ancillas)	minimise Toffolis, avoid rotations (ripple-carry wins by $\approx 30\times$)

Here $KQ = \text{depth} \times q$ is a spacetime-volume fidelity proxy. A circuit is plausibly runnable without error correction when $1/KQ > \epsilon$, the machine's per-operation failure rate.

The governing exchange rate

Object	Cost
Clifford gate (H , S , CNOT, CZ, Paulis)	≈ 0
TOF standalone	$7T$, T -depth 3
TOF as a compute/uncompute pair	$4T$ for the pair
FREDKIN	$7T$
$C^k(X)$, clean ancillas	$(2k - 3) \times \text{TOF}$
$R_z(\theta)$ to precision ϵ	$3 \log_2(1/\epsilon) T$

At $\epsilon = 10^{-6}$ one arbitrary rotation is $\approx 60 T$ gates; at $\epsilon = 10^{-15}$, ≈ 150 . A Toffoli costs 7, or 4 in a pair. **An arbitrary rotation is one to two orders of magnitude more expensive than a Toffoli.** This single fact decides both remaining design choices in [Chapter 12](#): the approximate QFT replaces the exact one, and ripple-carry arithmetic replaces the Fourier (Draper) adder.

Reported costs often omit uncomputation

Garbage must be uncomputed before qubits can be reused *or before any interference step*, so quoted counts for garbage-producing designs are optimistic, sometimes by a factor of two. Every circuit quoted in this document includes its uncomputation and has zero garbage.

Separately: T -count is itself only a *proxy*. The honest currency is spacetime volume (logical qubits $\times$ logical time steps). T -count tracks it well only while magic-state production is the bottleneck — with magic state cultivation the factories shrink and idle logical qubits become a comparable share. See [Section 18.2](#).

Part II

The Fourier Machinery

Chapter 5

Where the Fourier basis comes from

The quantum Fourier transform is usually introduced by writing down its matrix and verifying that it is unitary. That is correct and completely unilluminating: it makes the QFT look like a lucky guess. This chapter takes the opposite route and *derives* the Fourier basis, showing that it is not a design choice but a consequence of the spectral theorem applied to a single, very simple operator — the cyclic shift.

The payoff is stated in advance, because it organises everything that follows.

What this chapter establishes

1. A set of matrices is simultaneously diagonalisable iff they mutually commute ([Theorem 5.1.2](#)); and if one of them has *simple* spectrum, its eigenvectors already diagonalise the whole set ([Lemma 5.1.3](#)).
2. The *circulant* matrices — equivalently, the shift-invariant linear maps — are exactly the polynomials in the cyclic shift S , and they mutually commute ([Theorem 5.2.4](#)).
3. S has simple spectrum: its eigenvalues are the r -th roots of unity, forced by nothing more than “go once around the cycle” ([Theorem 5.3.1](#)).
4. Therefore the eigenvectors of S — the *characters* of $\mathbb{Z}/r\mathbb{Z}$ — simultaneously diagonalise every shift-invariant operator, and the matrix of that change of basis is the discrete Fourier transform ([Corollary 5.3.4](#)). The DFT is discovered, not postulated.
5. Multiplication by a modulo N , restricted to the orbit of 1, is the cyclic shift ([Proposition 5.5.3](#)). Hence the hidden period r has been rewritten as the denominator of an eigenvalue’s phase — and [Chapter 8](#) does nothing but measure that phase.

5.1 Simultaneous diagonalisation

Definition 5.1.1. A set $\mathcal{M}$ of $n \times n$ matrices is *simultaneously diagonalisable* if there is a single invertible V such that $V^{-1}AV$ is diagonal for every $A \in \mathcal{M}$.

Theorem 5.1.2 (Commuting is necessary and (for diagonalisable matrices) sufficient). *A set of diagonalisable matrices is simultaneously diagonalisable if and only if they mutually commute.*

Proof. Necessity. Diagonal matrices commute. If $A = V\Lambda_a V^{-1}$ and $B = V\Lambda_b V^{-1}$ then $AB = V\Lambda_a \Lambda_b V^{-1} = V\Lambda_b \Lambda_a V^{-1} = BA$.

Sufficiency. We prove the case needed below ([Lemma 5.1.3](#)) and sketch the rest. Suppose $AB = BA$ and A is diagonalisable. Let $|v\rangle$ satisfy $A|v\rangle = \lambda|v\rangle$. Then

$$A(B|v\rangle) = BA|v\rangle = \lambda B|v\rangle,$$

so B maps the λ -eigenspace of A into itself. Hence B restricts to each eigenspace of A ; diagonalising each restriction (possible since B is diagonalisable) and collecting the resulting bases gives a basis diagonalising both. Iterating handles any finite commuting family. $\square$

Lemma 5.1.3 (Simple spectrum determines the basis). *If A has n distinct eigenvalues, then A and B are simultaneously diagonalisable if and only if they commute; and in that case the diagonalising basis is (up to scaling) exactly the eigenbasis of A — there is no freedom left.*

Proof. By the displayed computation above, $B|v_i\rangle$ lies in the λ_i -eigenspace of A . Since the eigenvalues are distinct, that eigenspace is one-dimensional, so $B|v_i\rangle = \gamma_i|v_i\rangle$ for some scalar γ_i . Thus each $|v_i\rangle$ is

automatically an eigenvector of B , possibly with a different eigenvalue. Uniqueness up to scaling is the statement that a one-dimensional eigenspace has a unique direction. $\square$

The constructive recipe

To simultaneously diagonalise a commuting family $\mathcal{M}$: *find one member with simple spectrum and compute its eigenvectors*. Those eigenvectors diagonalise everything in $\mathcal{M}$ at once.

In [Section 5.2](#) the family is the shift-invariant operators and the distinguished member is the shift S itself. Solving that one eigenvalue problem produces the DFT. This is the sense in which the Fourier transform is *discovered rather than postulated*.

5.2 The cyclic shift and the circulants

Throughout, $\mathbb{Z}/r\mathbb{Z} = \{0, 1, \dots, r-1\}$ with arithmetic modulo r , and $\mathbb{C}^r$ has the standard basis $\{|k\rangle\}_{k \in \mathbb{Z}/r\mathbb{Z}}$. It is useful to picture a vector as a set of numbers arranged at equidistant points around a circle: a function on the discrete circle. Modular arithmetic is then angle addition, and $\mathbb{Z}/r\mathbb{Z}$ is isomorphic to the group μ_r of r -th roots of unity $\rho_m := e^{2\pi i m/r}$ under complex multiplication — an isomorphism we shall use constantly, and which is the entire reason the algorithm works.

Definition 5.2.1 (Cyclic shift). $S \in \mathbb{C}^{r \times r}$ is defined by $S|k\rangle = |k+1 \bmod r\rangle$, i.e. $S = \sum_k |k+1\rangle\langle k|$, with entries $S_{k\ell} = \delta_{k-\ell, 1}$. More generally $S^t|k\rangle = |k+t\rangle$.

Proposition 5.2.2 (Elementary properties). S is unitary, $S^r = \mathbb{1}$, $S^\dagger = S^{-1} = S^{r-1}$ with $S^\dagger|k\rangle = |k-1\rangle$, and $t \mapsto S^t$ is a group homomorphism $(\mathbb{Z}/r\mathbb{Z}, +) \rightarrow U(r)$.

Proof. S permutes an orthonormal basis, hence is unitary by [Proposition 2.4.6\(iii\)](#); $S^r|k\rangle = |k+r\rangle = |k\rangle$. The rest is immediate. $\square$

Definition 5.2.3 (Circulant). For $a = (a_0, \dots, a_{r-1}) \in \mathbb{C}^r$ the *circulant* C_a has entries $(C_a)_{k\ell} = a_{k-\ell \bmod r}$; equivalently

$$C_a = \sum_{m \in \mathbb{Z}/r\mathbb{Z}} a_m S^m, \quad (5.1)$$

a polynomial in the shift. Concretely, each column of C_a is a cyclic shift of the previous one:

$$C_a = \begin{pmatrix} a_0 & a_{r-1} & \cdots & a_1 \\ a_1 & a_0 & \cdots & a_2 \\ \vdots & & \ddots & \vdots \\ a_{r-1} & a_{r-2} & \cdots & a_0 \end{pmatrix}.$$

Theorem 5.2.4 (Circulants are exactly the shift-invariant maps). For $M \in \mathbb{C}^{r \times r}$ the following are equivalent: (i) M is a circulant; (ii) $M = \sum_m a_m S^m$ for some a ; (iii) $MS = SM$ (M is shift-invariant: shifting the input shifts the output). In particular the circulants form a commutative algebra, closed under products and inverses.

Proof. (i) $\Leftrightarrow$ (ii): the (k, ℓ) entry of $\sum_m a_m S^m$ is $\sum_m a_m \delta_{k-\ell, m} = a_{k-\ell}$. (ii) $\Rightarrow$ (iii): powers of S commute with S , and so does any linear combination. (iii) $\Rightarrow$ (i): from $MS = SM$ we get $M = S^{-1}MS$, whose (k, ℓ) entry is $M_{k+1, \ell+1}$; so $M_{k\ell}$ depends only on $k - \ell$, which is (i). Commutativity of the algebra follows from (ii) and $S^m S^{m'} = S^{m+m'}$. $\square$

Remark 5.2.5 (Circular convolution). Under [Equation \(5.1\)](#), applying C_a to x computes the *circular convolution* $(a \otimes x)_k = \sum_\ell a_{k-\ell} x_\ell$. So [Theorem 5.2.4](#) says: circular convolution operators = shift-invariant operators = polynomials in S . Diagonalising them all at once is exactly the convolution theorem, and [Corollary 5.3.4](#) is its derivation.

5.3 Diagonalising the shift: roots of unity are forced

We now solve the one eigenvalue problem that [Lemma 5.1.3](#) says we need. Notice how little input it requires: no Fourier analysis is assumed, only “the cycle closes”.

Theorem 5.3.1 (Spectrum and eigenvectors of the shift). *The eigenvalues of S on $\mathbb{C}^r$ are exactly the r -th roots of unity $\omega_r^s = e^{2\pi i s/r}$, $s = 0, 1, \dots, r-1$, each simple. The corresponding normalised eigenvectors are*

$$|u_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} \omega_r^{-sk} |k\rangle, \quad S|u_s\rangle = \omega_r^s |u_s\rangle. \quad (5.2)$$

They form an orthonormal basis of $\mathbb{C}^r$.

Proof. Derivation, not verification. Seek $Sv = \lambda v$ with $v = \sum_k c_k |k\rangle$. Since $Sv = \sum_k c_k |k+1\rangle = \sum_k c_{k-1} |k\rangle$, the eigenvalue equation reads

$$c_{k-1} = \lambda c_k \quad \text{for all } k \bmod r \quad \implies \quad c_k = \lambda^{-k} c_0. \quad (5.3)$$

Consistency around the cycle, $c_0 = c_r = \lambda^{-r} c_0$, forces

$$\lambda^r = 1 \quad \implies \quad \lambda = \omega_r^s, \quad s = 0, 1, \dots, r-1.$$

These are r distinct values, so the spectrum is simple and each eigenspace is one-dimensional. Normalising ($|c_k| = 1/\sqrt{r}$, forced by $|\lambda| = 1$) gives [Equation \(5.2\)](#).

Check. $S|u_s\rangle = \frac{1}{\sqrt{r}} \sum_k \omega_r^{-sk} |k+1\rangle = \frac{1}{\sqrt{r}} \sum_{k'} \omega_r^{-s(k'-1)} |k'\rangle = \omega_r^s |u_s\rangle$.

Orthonormality. $\langle u_s | u_{s'} \rangle = \frac{1}{r} \sum_k \omega_r^{(s-s')k} = \delta_{ss'}$ by [Lemma A.0.1](#). □

Remark 5.3.2 ($|\lambda| = 1$ is automatic). One need not impose it: S permutes an orthonormal basis, hence is unitary and norm-preserving, so $|\lambda| \|v\| = \|Sv\| = \|v\|$. Every eigenvalue of a unitary is a pure phase — which is precisely what allows an eigenvalue to be kicked back as a phase in [Chapter 7](#).

What has just happened

The hidden period r has been rewritten as *the denominator of an eigenvalue's phase*. Finding r is now the problem of measuring a number of the form s/r . [Chapter 6](#) through [Chapter 10](#) do nothing else.

And note what has *not* appeared: no quantum mechanics. [Theorem 5.3.1](#) is a fact about $r \times r$ complex matrices.

5.3.1 The eigenvectors are the characters

Definition 5.3.3 (Character). A *character* of the abelian group $(\mathbb{Z}/r\mathbb{Z}, +)$ is a group homomorphism $\chi : \mathbb{Z}/r\mathbb{Z} \rightarrow \mathbb{C}^\times$. The characters are exactly $\chi_s(k) = \omega_r^{sk}$ for $s \in \mathbb{Z}/r\mathbb{Z}$.

The eigenvector $|u_s\rangle$ has coordinates $\chi_{-s}(k)/\sqrt{r}$. There is a one-line reason characters *must* be the eigenvectors of a translation:

Why characters diagonalise translation

A character turns *addition of its argument* into *multiplication of its value*: $\chi(k+t) = \chi(k)\chi(t)$. So translating a character can only multiply it by the scalar $\chi(t)$ — i.e. a character is automatically an eigenvector of translation, with eigenvalue $\chi(t)$. Everything else in this chapter is bookkeeping around that sentence.

Corollary 5.3.4 (The DFT, discovered). *Let F_r be the change-of-basis matrix to the eigenbasis, $F_r := \sum_s |s\rangle\langle u_s|$, so that $F_r |u_s\rangle = |s\rangle$ and, entrywise,*

$$(F_r)_{sk} = \overline{(u_s)_k} = \frac{\omega_r^{sk}}{\sqrt{r}}$$

— the rows of F_r are the bras $\langle u_s|$, hence the conjugated coefficients. Then F_r is unitary and

$$S = F_r^\dagger D F_r, \quad D = \text{diag}(1, \omega_r, \omega_r^2, \dots, \omega_r^{r-1}). \quad (5.4)$$

Consequently every circulant is diagonalised by the same basis,

$$C_a = F_r^\dagger \text{diag}(\hat{a}(0), \dots, \hat{a}(r-1)) F_r, \quad \hat{a}(s) = \sum_{m=0}^{r-1} a_m \omega_r^{ms}, \quad (5.5)$$

and the eigenvalue formula $\hat{a}$ is the discrete Fourier transform of the vector a .

Proof. Unitarity of F_r is the orthonormality in Theorem 5.3.1. For Equation (5.4), apply both sides to $|s\rangle$: $F_r S F_r^\dagger |s\rangle = F_r S |u_s\rangle = \omega_r^s F_r |u_s\rangle = \omega_r^s |s\rangle = D |s\rangle$, so $F_r S F_r^\dagger = D$, i.e. $S = F_r^\dagger D F_r$. For Equation (5.5): by Theorem 5.2.4 every circulant is $\sum_m a_m S^m$, and by Lemma 5.1.3 it is diagonal in the same basis; its eigenvalue on $|u_s\rangle$ is $\sum_m a_m (\omega_r^s)^m = \hat{a}(s)$, since $S^m |u_s\rangle = \omega_r^{sm} |u_s\rangle$. $\square$

A sign convention that is easy to get backwards

The eigenvector $|u_s\rangle$ carries ω_r^{-sk} (Equation (5.2)) while the matrix F_r carries ω_r^{+sk} , because F_r is built from the bras $\langle u_s|$, and passing from a ket to a bra conjugates (Proposition 2.2.3). Both signs are needed, in these two places, and swapping them silently replaces S by $S^\dagger$. Verified numerically in Appendix D.

The DFT is a change of basis, not a signal transform

Corollary 5.3.4 inverts the usual pedagogy. One does not define the DFT and then notice it diagonalises convolution. One asks: *what basis simultaneously diagonalises every shift-invariant operator?* — and the answer, forced by Lemma 5.1.3 and Theorem 5.3.1, is the character basis. The DFT is the matrix of that change of basis, and the classical formula $\hat{a}(s) = \sum_m a_m \omega_r^{ms}$ falls out as the *eigenvalue* of C_a .

This is why Chapter 6 can say the QFT is not a trick: any procedure that wants to see a period must land in this basis, because the spectral theorem leaves it no alternative.

Corollary 5.3.5 (Convolution theorem). $\widehat{a \circledast x} = \hat{a} \cdot \hat{x}$ pointwise. Indeed C_a acts as multiplication by $\hat{a}(s)$ on the s -th Fourier coordinate, by Equation (5.5).

Remark 5.3.6 (Other conventions). Using the eigenvectors of S rather than $S^\dagger$ (or of S^p with $\gcd(p, r) = 1$) yields the same theorem with ω_r replaced by ω_r^{-1} (resp. ω_r^p) — the familiar sign-convention freedom of the DFT. Any of them works; the standing convention of this document is the one fixed in the front matter and restated as Definition 6.1.1, and it is used without further comment.

5.4 What a Fourier coefficient computes

For $f : \mathbb{Z}/Q\mathbb{Z} \rightarrow \mathbb{C}$, the *unitary* Fourier coefficient

$$\tilde{f}(y) := \frac{1}{\sqrt{Q}} \sum_x f(x) \omega_Q^{-xy}$$

takes each sample $f(x)$, rotates it *backwards* by the angle $2\pi xy/Q$, and sums. (This differs from the eigenvalue formula $\hat{a}(s) = \sum_m a_m \omega_r^{ms}$ of Equation (5.5) by the normalisation $Q^{-1/2}$ and by the sign of the

exponent: $\hat{a}$ is what a circulant's eigenvalue *is*, while $\tilde{f}$ is the amplitude the inverse transform *produces*. Both are called “the DFT”; they are related by $\tilde{f} = \bar{\hat{a}}/\sqrt{Q}$. The convention used for circuits is fixed once, in [Definition 6.1.1](#).) Multiplying by a conjugate rotating mode and summing is a *test for phase matching*:



The three regimes, in advance

- If the unwinding rate y/Q *matches* the signal's rate, every arrow points the same way and they add coherently: $|\hat{f}(y)|$ is large.
- If it mismatches by a non-zero multiple of $1/Q$, the arrows are the Q -th roots of unity taken k at a time — a *balanced closed polygon* — and the sum is exactly 0. That is [Lemma A.0.1](#).
- If it *nearly* matches, the arrows form a short arc: the sum is large but not maximal.

[Theorem 8.3.2](#) shows the inverse QFT is exactly this picture with Q arrows and common increment $2\pi\delta_y$, and the closed-form intensity is the Fejér kernel — literally the diffraction pattern of a Q -slit grating.

5.5 The bridge: modular multiplication *is* the shift

We can now connect the algebra to the arithmetic. This is the single observation that makes Shor's algorithm a Fourier algorithm.

Definition 5.5.1 (The modular multiplication operator). Fix N and a with $\gcd(a, N) = 1$, and let $n = \lceil \log_2 N \rceil$. Define U_a on the n -qubit register by

$$U_a |y\rangle := \begin{cases} |ay \bmod N\rangle, & 0 \leq y < N, \\ |y\rangle, & N \leq y < 2^n. \end{cases} \quad (5.6)$$

Proposition 5.5.2 (U_a is unitary, and its powers are cheap). U_a is a permutation of the computational basis, hence unitary. Moreover $U_a U_b = U_{ab \bmod N}$, so $U_a^p = U_{a^p \bmod N}$; in particular

$$U_a^{2^j} = U_{a^{2^j} \bmod N}, \quad (5.7)$$

and the constant $a^{2^j} \bmod N$ is computed classically by j repeated squarings.

Proof. On $\{0, \dots, N-1\}$ the map $y \mapsto ay \bmod N$ is injective: $ay \equiv ay'$ gives $a(y - y') \equiv 0$, and multiplying by a^{-1} (which exists by [Proposition 1.1.4](#)) gives $y = y'$. An injective self-map of a finite set is a bijection; on the padding range it is the identity. The composition law holds on basis states and both sides are linear. $\square$

The real source of the exponential speed-up

[Equation \(5.7\)](#) says $U_a^{2^j}$ costs *one* modular multiplication, not 2^j of them. This — not the Fourier transform — is where the exponential advantage comes from. A ladder of $t = 2n + O(1)$ controlled powers reaches exponent $2^t \approx N^2$ at a cost of t multiplications. Without [Equation \(5.7\)](#) there would be no algorithm.

Proposition 5.5.3 (U_a restricted to an orbit is the cyclic shift). *Let $r = \text{ord}_N(a)$ and let $\mathcal{O} := \{a^k \bmod N : k = 0, 1, \dots, r-1\} \subseteq (\mathbb{Z}/N\mathbb{Z})^\times$ be the orbit of 1, a set of exactly r distinct elements. Let $\mathcal{H}_{\mathcal{O}} := \text{span}\{|a^k \bmod N\rangle : 0 \leq k < r\}$, of dimension r . Then U_a preserves $\mathcal{H}_{\mathcal{O}}$, and the unitary identification*

$$V : \mathcal{H}_{\mathcal{O}} \rightarrow \mathbb{C}^r, \quad V |a^k \bmod N\rangle := |k\rangle \quad (k \in \mathbb{Z}/r\mathbb{Z})$$

conjugates U_a into the cyclic shift:

$$V U_a|_{\mathcal{H}_{\mathcal{O}}} V^\dagger = S. \quad (5.8)$$

Proof. The elements $a^0, \dots, a^{r-1}$ are pairwise distinct: if $a^k \equiv a^{k'}$ with $0 \leq k < k' < r$ then $a^{k'-k} \equiv 1$ with $0 < k' - k < r$, contradicting minimality of r . So V maps an orthonormal basis to an orthonormal basis and is unitary. Next, $U_a |a^k\rangle = |a^{k+1}\rangle$, and $a^r \equiv 1$ gives $a^{k+r} \equiv a^k$, so the index is well defined modulo r and $\mathcal{H}_{\mathcal{O}}$ is preserved. Finally $V U_a V^\dagger |k\rangle = V U_a |a^k\rangle = V |a^{k+1}\rangle = |k+1 \bmod r\rangle$. $\square$

Theorem 5.5.4 (Eigenvectors and eigenvalues of U_a). *For $s \in \{0, 1, \dots, r-1\}$ define*

$$|u_s\rangle := \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi i s k / r} |a^k \bmod N\rangle. \quad (5.9)$$

Then $\{|u_s\rangle\}_{s=0}^{r-1}$ is an orthonormal basis of $\mathcal{H}_{\mathcal{O}}$ and

$$U_a |u_s\rangle = e^{2\pi i s / r} |u_s\rangle. \quad (5.10)$$

Proof. Transport [Theorem 5.3.1](#) through the unitary V of [Proposition 5.5.3](#): indeed $V |u_s\rangle$ is exactly the $|u_s\rangle$ of [Equation \(5.2\)](#). Directly: $U_a |u_s\rangle = \frac{1}{\sqrt{r}} \sum_k e^{-2\pi i s k / r} |a^{k+1}\rangle = \frac{1}{\sqrt{r}} \sum_{k'} e^{-2\pi i s (k'-1) / r} |a^{k'}\rangle = e^{2\pi i s / r} |u_s\rangle$. $\square$

Corollary 5.5.5 (The Fourier basis is forced, not chosen). *$U_a|_{\mathcal{H}_{\mathcal{O}}}$ is unitary with r distinct eigenvalues, so by [Theorem 2.5.1](#) and [Lemma 5.1.3](#) its eigenbasis is unique up to phases. Any procedure that wants to see the period must land in that basis. Moreover, by [Lemma 5.1.3](#) the same basis simultaneously diagonalises every operator commuting with U_a .*

Summary of

Statement	Consequence
U_a on the orbit of 1 is the shift S on $\mathbb{C}^r$ (Proposition 5.5.3)	multiplication modulo N is translation in $\mathbb{Z}/r\mathbb{Z}$, wearing a disguise supplied by $k \mapsto a^k$
S has eigenvalues ω_r^s , forced by $\lambda^r = 1$ (Theorem 5.3.1)	the eigenvalues of U_a on $\mathcal{H}_{\mathcal{O}}$ are exactly the r -th roots of unity
eigenvectors are the characters (Corollary 5.3.4)	they are the columns of the Fourier matrix; the eigenbasis is unique
$\mathbb{Z}/r\mathbb{Z} \cong \mu_r$	the order r is <i>readable off the eigenvalues</i> , as the denominator of s/r

Shor's algorithm is nothing but: build the hidden shift, estimate one of its eigenphases, read r off the denominator.

Chapter 6

The quantum Fourier transform

Chapter 5 established that the Fourier basis is the unique eigenbasis of the cyclic shift, hence the unique basis in which a hidden period becomes visible. This chapter makes the change of basis into a *circuit*: $\Theta(n^2)$ gates exactly, or $O(n \log n)$ approximately, on n qubits — against $\Theta(n2^n)$ for the classical FFT on the same 2^n amplitudes.

What the QFT is not

It is *not* a fast DFT. The Q Fourier coefficients are created coherently as amplitudes and are never made classically accessible: one measurement returns a single index y with probability $|\hat{f}(y)|^2$ and destroys the rest. There is no known way to extract the spectrum, and by Theorem 3.9.1 one cannot copy the state and re-measure. The QFT is a change of basis performed *in situ*, useful only when a subsequent measurement is arranged to be informative.

6.1 Definition and first properties

Definition 6.1.1 (QFT). For $Q \geq 2$ and indices in $\mathbb{Z}/Q\mathbb{Z}$, with $\omega_Q = e^{2\pi i/Q}$,

$$\text{QFT}_Q |x\rangle = \frac{1}{\sqrt{Q}} \sum_{y=0}^{Q-1} \omega_Q^{xy} |y\rangle, \quad \text{QFT}_Q^\dagger |x\rangle = \frac{1}{\sqrt{Q}} \sum_{y=0}^{Q-1} \omega_Q^{-xy} |y\rangle. \quad (6.1)$$

Throughout, $Q = 2^t$ so that the index register is t qubits.

Proposition 6.1.2 (Unitarity). QFT_Q is unitary.

Proof. $\langle \text{QFT}_Q x' | \text{QFT}_Q x \rangle = \frac{1}{Q} \sum_{y,y'} \omega_Q^{-x'y'} \omega_Q^{xy} \langle y' | y \rangle = \frac{1}{Q} \sum_y \omega_Q^{(x-x')y} = \delta_{x'x}$ by Lemma A.0.1. So QFT maps an orthonormal basis to an orthonormal basis; apply Proposition 2.4.6(iii). $\square$

Proposition 6.1.3 (Special values). $\text{QFT}_Q |0\rangle = \frac{1}{\sqrt{Q}} \sum_y |y\rangle = H^{\otimes t} |0\rangle^{\otimes t}$, and $\text{QFT}_2 = H$. More generally $\text{QFT}_Q |x\rangle$ is a state all of whose amplitudes have modulus $Q^{-1/2}$: the QFT of a basis state carries its information entirely in relative phases.

6.1.1 The QFT diagonalises addition

This is the structural statement that makes the QFT the right tool, and it is Chapter 5 restated for $Q = 2^t$.

Theorem 6.1.4 (Diagonalisation of translation). Let $T_a |b\rangle = |b+a \bmod Q\rangle$ be translation by a . Then

$$\text{QFT}_Q T_a \text{QFT}_Q^\dagger = \text{diag}(\omega_Q^{ak}), \quad \text{equivalently} \quad T_a = \text{QFT}_Q^\dagger \text{diag}(\omega_Q^{ak}) \text{QFT}_Q. \quad (6.2)$$

Proof. $T_a = S^a$ where S is the cyclic shift on $\mathbb{Z}/Q\mathbb{Z}$, and QFT_Q is the matrix F_Q of Corollary 5.3.4. Applying both sides to $|k\rangle$:

$$\text{QFT}_Q T_a \text{QFT}_Q^\dagger |k\rangle = \text{QFT}_Q S^a |u_k\rangle = \omega_Q^{ak} \text{QFT}_Q |u_k\rangle = \omega_Q^{ak} |k\rangle,$$

using $S^a |u_k\rangle = \omega_Q^{ak} |u_k\rangle$ from Theorem 5.3.1. Conjugating gives the second form. $\square$

Remark 6.1.5 (Direct check of the sign). Explicitly, $\text{QFT}^\dagger |k\rangle = Q^{-1/2} \sum_x \omega_Q^{-xk} |x\rangle$; applying T_a and substituting $x' = x + a$ gives $Q^{-1/2} \sum_{x'} \omega_Q^{-(x'-a)k} |x'\rangle = \omega_Q^{ak} \text{QFT}^\dagger |k\rangle$, and a final QFT returns $\omega_Q^{ak} |k\rangle$. The exponent is $+ak$, not $-ak$; this sign is checked numerically in [Appendix D](#) and it is the sign the Draper adder of [Section 12.5](#) uses.

Read from right to left, and it is an algorithm

To *add*, Fourier transform, apply phases, transform back. That is the Draper adder of [Section 12.5](#). It is the most elegant adder known and — for reasons that are entirely about fault-tolerant cost, not about correctness — the wrong one to build.

6.2 The product form

The circuit rests on one identity, which says that the QFT of a basis state *factorises*.

Theorem 6.2.1 (Product form). *For $Q = 2^n$ and a basis state $|j\rangle = |j_1 j_2 \cdots j_n\rangle$ in big-endian order ([Convention 3.8.2](#)),*

$$\text{QFT}_{2^n} |j_1 \cdots j_n\rangle = \frac{1}{2^{n/2}} \bigotimes_{\ell=1}^n \left(|0\rangle + e^{2\pi i (0.j_{n-\ell+1} \cdots j_n)} |1\rangle \right), \quad (6.3)$$

where $0.j_m \cdots j_n := \sum_{s \geq 1} j_{m+s-1} 2^{-s}$ is a binary fraction.

Proof. Write $k = k_1 \cdots k_n$ with $k = \sum_{\ell=1}^n k_\ell 2^{n-\ell}$. Then

$$\text{QFT} |j\rangle = \frac{1}{2^{n/2}} \sum_{k=0}^{2^n-1} e^{2\pi i j k / 2^n} |k\rangle = \frac{1}{2^{n/2}} \sum_{k_1, \dots, k_n \in \{0,1\}} \prod_{\ell=1}^n e^{2\pi i j k_\ell 2^{-\ell}} |k_1 \cdots k_n\rangle,$$

using $k/2^n = \sum_{\ell} k_\ell 2^{-\ell}$ and the multiplicativity of the exponential. The sum over the n independent bits of a product factorises, giving $\bigotimes_{\ell} (|0\rangle + e^{2\pi i j 2^{-\ell}} |1\rangle) / 2^{n/2}$. Finally $j 2^{-\ell} \equiv 0.j_{n-\ell+1} \cdots j_n \pmod{1}$, because the terms of $j 2^{-\ell}$ coming from bits $j_1, \dots, j_{n-\ell}$ are integers and drop out of $e^{2\pi i (\cdot)}$. $\square$

Two readings of one identity

[Equation \(6.3\)](#) says the QFT of a *basis* state is always a *product* state — never entangled. That is exactly what permits a small circuit: n wires can be prepared independently. (Applied to a superposition, the QFT generally *does* produce entanglement.)

Read in the other direction, the same identity is the phase-estimation ladder of [Theorem 7.4.1](#): a tensor product of $(|0\rangle + e^{2\pi i 2^{l-s} \varphi} |1\rangle)$ over s equals the ramp $\sum_x e^{2\pi i x \varphi} |x\rangle$. Recognising that the product form and the kickback ladder are the same identity read in opposite directions halves what has to be remembered — and explains why $\text{QFT}^\dagger$ is exactly the right thing to apply to the ladder.

6.3 The circuit

Lemma 6.3.1 (Building one output factor). *Fix $s \in \{1, \dots, n\}$ and suppose qubits $s, s+1, \dots, n$ are in the basis state $|j_s j_{s+1} \cdots j_n\rangle$. Apply to qubit s first a Hadamard, then, for each $k = 2, 3, \dots, n - s + 1$, a controlled- R_k with control qubit $s+k-1$ and target qubit s . Then qubit s ends in $\frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i (0.j_s j_{s+1} \cdots j_n)} |1\rangle)$, and qubits $s+1, \dots, n$ are unchanged.*

Proof. $H |j_s\rangle = \frac{1}{\sqrt{2}} (|0\rangle + (-1)^{j_s} |1\rangle) = \frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i 0.j_s} |1\rangle)$, since $0.j_s = j_s/2$. A controlled- R_k with control $s+k-1$ — a basis state, so there is no back-action — multiplies the $|1\rangle$ component by $e^{2\pi i j_{s+k-1}/2^k}$.

Accumulating over $k = 2, \dots, n - s + 1$,

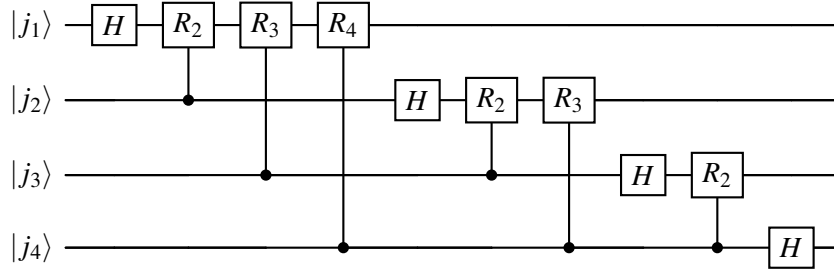
$$\exp\left(2\pi i \sum_{k=2}^{n-s+1} \frac{j_{s+k-1}}{2^k}\right) = \exp\left(2\pi i \sum_{p=1}^{n-s} \frac{j_{s+p}}{2^{p+1}}\right) = e^{2\pi i (0.0j_{s+1}j_{s+2}\dots j_n)}.$$

Combined with $e^{2\pi i 0.j_s}$ and $0.j_s + 0.0j_{s+1}\dots j_n = 0.j_sj_{s+1}\dots j_n$, the claim follows. $\square$

Order matters

Lemma 6.3.1 requires qubits $s + 1, \dots, n$ to still be *basis states* when they act as controls. So the circuit processes qubit 1 first, then qubit 2, and so on: once qubit s has been Hadamarded it is no longer a basis state and may not be used as a control.

Algorithm 6.3.2 (QFT on n qubits). 1. For $s = 1, 2, \dots, n$: (a) apply H to qubit s ; (b) for $k = 2, 3, \dots, n - s + 1$, apply $c\text{-}R_k$ with control qubit $s + k - 1$ and target qubit s .
2. Reverse the qubit order with $\lfloor n/2 \rfloor$ swaps (qubit $s \leftrightarrow$ qubit $n + 1 - s$).



The QFT on four qubits (bit-reversal swaps not drawn). Each H writes the leading bit of a binary fraction; each controlled R_k appends the k -th bit without disturbing the wire supplying it.

Theorem 6.3.3 (Correctness and cost). *Algorithm 6.3.2* implements QFT_{2^n} using n Hadamards, $\binom{n}{2}$ controlled rotations and $\lfloor n/2 \rfloor$ swaps:

$$\frac{n(n+1)}{2} + \left\lfloor \frac{n}{2} \right\rfloor = \Theta(n^2) \text{ gates,}$$

at depth $\Theta(n)$ if commuting rotations are parallelised.

Proof. Apply **Lemma 6.3.1** for $s = 1, \dots, n$ in order; the hypothesis of the lemma is maintained because the gates acting on qubit s never alter qubits $s + 1, \dots, n$ (controls of a controlled gate are unchanged). The result is **Equation (6.3)** but with output factor ℓ appearing on wire $s = n - \ell + 1$, which the closing swaps repair. Counting: wire s receives one H and $n - s$ rotations, so $\sum_s (n - s) = \binom{n}{2}$. $\square$

Remark 6.3.4 (Why the swaps). Output factor ℓ of **Equation (6.3)** depends on the *tail* $j_{n-\ell+1} \dots j_n$, so the natural circuit emits the qubits in reversed order. The swaps repair that — or they may be dropped and the relabelling done classically, which is free. This is the mechanical origin of **Section 3.8**.

Remark 6.3.5 (The inverse). $\text{QFT}^\dagger$ is the same circuit run backwards with $R_k \mapsto R_k^\dagger$, i.e. every rotation angle negated. There is nothing else to derive: $(\prod_i G_i)^\dagger = \prod_i G_{L-i}^\dagger$, and $H^\dagger = H$, $\text{SWAP}^\dagger = \text{SWAP}$.

Two checks to run before trusting an implementation

1. Build the $2^n \times 2^n$ matrix of the circuit for $n = 3, 4$ and compare entrywise with $(F_{2^n})_{kj} = \omega^{kj} / \sqrt{2^n}$.
2. Check $\text{QFT}^\dagger \circ \text{QFT} = \mathbb{I}$ numerically. A failure here is almost always the swap layer or the bit

order.

Both were run for this document; see [Appendix D](#).

6.4 The approximate QFT

The rotation R_k is exponentially close to the identity for large k : $\|R_k - \mathbb{1}\| = |e^{2\pi i/2^k} - 1| \leq 2\pi/2^k$. Coppersmith observed that these gates may simply be *dropped*.

Definition 6.4.1. AQFT_m is [Algorithm 6.3.2](#) with every $c\text{-}R_k$ for $k > m$ omitted.

Theorem 6.4.2 (Coppersmith). $\|\text{QFT}_{2^n} - \text{AQFT}_m\| \leq \frac{2\pi n}{2^m}$, and AQFT_m uses $O(nm)$ gates. Taking $m = \lceil \log_2(2\pi n/\epsilon) \rceil$ gives error at most ϵ with $O(n \log(n/\epsilon))$ gates.

Proof. For products of unitaries, $\|V - \tilde{V}\| \leq \sum_\ell \|V_\ell - \tilde{V}_\ell\|$: indeed, writing $V - \tilde{V}$ as a telescoping sum $\sum_\ell V_L \cdots V_{\ell+1} (V_\ell - \tilde{V}_\ell) \tilde{V}_{\ell-1} \cdots \tilde{V}_1$ and using that unitaries have operator norm 1. There are fewer than n omitted rotations per wire and n wires; each contributes at most $2\pi 2^{-k} \leq 2\pi 2^{-m}$, and the geometric tail $\sum_{k>m} 2\pi 2^{-k} = 2\pi 2^{-m}$ sums to less than twice the first term. $\square$

Why every serious resource estimate uses the approximate QFT

In the Clifford+ T cost model of [Section 4.5](#), an arbitrary-angle rotation costs $\approx 3 \log_2(1/\epsilon)$ T gates — one to two orders of magnitude more than a Toffoli. Dropping rotations with $k \gtrsim \log_2 n + 2$ turns the QFT from $\Theta(n^2)$ *expensive* rotations into $O(n \log n)$ gates with negligible distortion.

Combined with [Theorem 12.7.1](#), the consequence is stark: at $n = 2048$ the approximate QFT is $\approx 10^5$ gates against $\approx 2.6 \times 10^9$ Toffoli gates of modular exponentiation — a ratio of 3×10^4 . **The QFT is 0.003% of Shor's algorithm.**

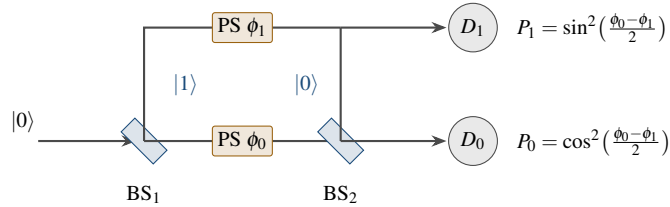
Chapter 7

Phase kickback and the interferometer picture

Chapter 5 turned the hidden period into an eigenvalue $e^{2\pi is/r}$. That eigenvalue is, by Proposition 3.7.2, invisible: a global phase on the eigenvector. This chapter supplies the one gate that makes it visible — the controlled unitary — and the picture that explains *why* it works: every quantum algorithm in this document is a multi-particle interferometer.

7.1 The Mach–Zehnder interferometer as a one-qubit circuit

A photon meets a beam-splitter, propagates by two paths through phase shifters, and meets a second beam-splitter which directs it to one of two detectors.



Labelling the lower path $|0\rangle$ and the upper $|1\rangle$, and taking the beam-splitter action to be $|0\rangle \mapsto \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, $|1\rangle \mapsto \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ — that is, a Hadamard — the particle undergoes

$$\begin{aligned} |0\rangle &\xrightarrow{\text{BS}_1} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \xrightarrow{\text{PS}} \frac{1}{\sqrt{2}}(e^{i\phi_0}|0\rangle + e^{i\phi_1}|1\rangle) \\ &\xrightarrow{\text{BS}_2} e^{i\frac{\phi_0+\phi_1}{2}} \left(\cos \frac{\phi_0-\phi_1}{2} |0\rangle + i \sin \frac{\phi_0-\phi_1}{2} |1\rangle \right), \end{aligned} \quad (7.1)$$

so the detection probabilities are $P_0 = \frac{1}{2}(1 + \cos \phi)$ and $P_1 = \frac{1}{2}(1 - \cos \phi)$ with $\phi = \phi_0 - \phi_1$. The overall factor $e^{i(\phi_0+\phi_1)/2}$ is a global phase and is irrelevant — *the interference pattern depends only on the difference*.

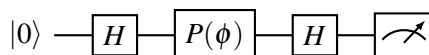
Three ingredients, and the one that is usually overlooked

1. The first beam-splitter **prepares a superposition** of possible paths.
2. The phase shifters **modify the relative phase** between the paths.
3. The second beam-splitter **recombines** the paths, *erasing all information about which path was taken*.

The third is the one that is usually skipped and is essential. If anything in the apparatus recorded which path the particle took, rule (R3) of Section 3.10 — formally Proposition 3.10.2 — would eliminate the cross term and Equation (7.1) would collapse to a 50:50 classical mixture with no dependence on ϕ at all.

The whole of Chapter 12 is an exercise in obeying point 3.

As a circuit, the interferometer is simply



7.2 Simulating the phase shifter with a controlled gate

Now replace the phase shifter by a controlled- U acting on an auxiliary register prepared in an eigenstate. This is the single move on which every algorithm in this document rests.

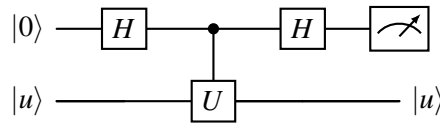
Lemma 7.2.1 (Phase kickback). *Let $U|u\rangle = e^{2\pi i\varphi}|u\rangle$ with $\varphi \in [0, 1)$. Then for every one-qubit control state,*

$$c-U(\alpha|0\rangle + \beta|1\rangle) \otimes |u\rangle = (\alpha|0\rangle + \beta e^{2\pi i\varphi}|1\rangle) \otimes |u\rangle. \quad (7.2)$$

Proof. By linearity and $c-U|c\rangle|\psi\rangle = |c\rangle U^c|\psi\rangle$ (Definition 4.2.2),

$$c-U(\alpha|0\rangle + \beta|1\rangle)|u\rangle = \alpha|0\rangle U^0|u\rangle + \beta|1\rangle U^1|u\rangle = \alpha|0\rangle|u\rangle + \beta e^{2\pi i\varphi}|1\rangle|u\rangle,$$

and the scalar $e^{2\pi i\varphi}$ moves from the second tensor factor to the first because scalars commute with everything; factoring $|u\rangle$ out gives the claim. $\square$



Reading it off: $\Pr[0] = \cos^2(\pi\varphi)$, $\Pr[1] = \sin^2(\pi\varphi)$. This is an *exact* simulation of the Mach–Zehnder interferometer, with the auxiliary register playing the role of the phase shifter — and it is the kernel of every quantum algorithm here.

What the gate does, and what it does not

Nothing is moved and nothing is copied. The eigenvalue was already a property of the pair $(U, |u\rangle)$; the gate *re-attaches* it to the control, where it is a *relative* phase and therefore observable, unlike a global phase on $|u\rangle$. The target is returned unchanged, so it can drive an entire ladder of such gates — it is a catalyst.

Four correct statements and one incorrect inference

Correct.

- The eigenvalue becomes a relative phase *precisely because* it multiplies the $|1\rangle$ branch and not the $|0\rangle$ branch.
- The target factors out *only because* $|u\rangle$ is an eigenstate; then $U|u\rangle$ is $|u\rangle$ up to a scalar.
- If the target is $|\psi\rangle = \sum_s c_s |u_s\rangle$, then $c-U(|+\rangle \otimes |\psi\rangle) = \frac{1}{\sqrt{2}} \sum_s c_s (|0\rangle + e^{2\pi i\varphi_s} |1\rangle) |u_s\rangle$, which is *entangled* unless all φ_s coincide.
- Entanglement appears exactly when the target mixes eigenstates with distinct eigenphases — which is what occurs in Shor’s algorithm.

Incorrect: “ $c-U$ forces or projects the target into an eigenstate.” It does not. $c-U$ is a fixed linear gate; it acts on whatever target it is given. Nothing is measured, projected, or selected. The factorisation in Lemma 7.2.1 works only because $|u\rangle$ was *already* an eigenstate.

Proposition 7.2.2 (The asymmetry of “control” and “target” is an illusion). *For diagonal U the labels are interchangeable. Concretely $c-Z = CZ = \text{diag}(1, 1, 1, -1)$ is symmetric under exchanging the two qubits, and so is every $c-R_k$. More generally, in the X basis a CNOT acts as a CNOT with control and target reversed: $(H \otimes H) \text{CNOT} (H \otimes H) = \text{CNOT}_{\text{rev}}$, the CNOT with control and target exchanged.*

Proof. For the last claim, check on basis states: $\text{CNOT}|x\rangle|y\rangle = |x\rangle|x \oplus y\rangle$ becomes, after conjugation by $H^{\otimes 2}$, the map $|a\rangle|b\rangle \mapsto |a \oplus b\rangle|b\rangle$. One computes the matrix element $\langle a'b'| (H \otimes H) \text{CNOT} (H \otimes H) |ab\rangle$ using $H|x\rangle = 2^{-1/2} \sum_z (-1)^{xz} |z\rangle$, and evaluates the resulting character sums with Lemma A.0.1. $\square$

Why this matters conceptually

The words “control” and “target” suggest a causal direction that is not there. A controlled gate is a symmetric interaction; *which* qubit appears to be acted upon depends on the basis one chooses to describe it in. Kickback looks surprising only if one takes the causal reading literally.

7.3 When kickback fails

Proposition 7.3.1 (Entanglement destroys the phase). *Let $|\psi\rangle = \sum_s c_s |u_s\rangle$ with $U|u_s\rangle = e^{2\pi i \varphi_s} |u_s\rangle$ and suppose the φ_s are not all equal. Then after $c\text{-}U(|+\rangle \otimes |\psi\rangle)$ the control qubit alone is in the mixed state with reduced density operator*

$$\rho_{\text{ctrl}} = \frac{1}{2} \begin{pmatrix} 1 & \bar{\kappa} \\ \kappa & 1 \end{pmatrix}, \quad \kappa = \sum_s |c_s|^2 e^{2\pi i \varphi_s},$$

and $|\kappa| < 1$ strictly. Measuring the control in the X basis gives $\Pr[0] = \frac{1}{2}(1 + \text{Re } \kappa)$: the fringe visibility is $|\kappa|$, not 1.

Proof. Trace out the target using [Section 2.8](#) and the orthonormality of the $|u_s\rangle$. Strictness of $|\kappa| < 1$ is the strict triangle inequality for a convex combination of distinct points on the unit circle. $\square$

This failure is the algorithm

Proposition 7.3.1 reads like a defect, but it is exactly the situation of Shor’s algorithm: the work register is $|1\rangle = \frac{1}{\sqrt{r}} \sum_s |u_s\rangle$, a superposition of *all* r eigenvectors with distinct eigenphases ([Theorem 10.1.1](#)). The resulting state *is* entangled, and the counting register carries not one clean phase but a uniform mixture of the r ramps $e^{2\pi i x s/r}$.

That is not a problem, because we never wanted one particular s — any s will do, and [Theorem 8.5.1](#) shows the statistics are exactly those of drawing s uniformly at random. *Kickback “failing” in the presence of several eigenphases is precisely the mechanism that lets us dispense with knowing r in advance.*

7.4 The kickback ladder

Theorem 7.4.1 (The ladder). *Let $Q = 2^t$ and $U|u\rangle = e^{2\pi i \varphi} |u\rangle$. Initialise t counting qubits in $|0\rangle^{\otimes t}$ and apply $H^{\otimes t}$; then for $s = 1, \dots, t$ apply $c\text{-}U^{2^{t-s}}$ with the s -th counting qubit as control (most significant first). The resulting state is*

$$\bigotimes_{s=1}^t \frac{|0\rangle + e^{2\pi i 2^{t-s} \varphi} |1\rangle}{\sqrt{2}} \otimes |u\rangle = \frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} e^{2\pi i x \varphi} |x\rangle \otimes |u\rangle. \quad (7.3)$$

Proof. After $H^{\otimes t}$ the state is $Q^{-1/2} \sum_x |x\rangle \otimes |u\rangle$. By [Corollary 2.6.4](#), $U^{2^{t-s}}|u\rangle = e^{2\pi i 2^{t-s} \varphi} |u\rangle$, so [Lemma 7.2.1](#) puts $e^{2\pi i 2^{t-s} \varphi}$ on the $|1\rangle$ branch of qubit s and leaves the target untouched, giving the left-hand side. For the equality, expand the tensor product over $x = x_1 \cdots x_t$ with $x = \sum_s x_s 2^{t-s}$:

$$\bigotimes_{s=1}^t \frac{|0\rangle + e^{2\pi i 2^{t-s} \varphi} |1\rangle}{\sqrt{2}} = \frac{1}{\sqrt{Q}} \sum_{x_1, \dots, x_t \in \{0,1\}} \prod_s e^{2\pi i x_s 2^{t-s} \varphi} |x_1 \cdots x_t\rangle = \frac{1}{\sqrt{Q}} \sum_x e^{2\pi i x \varphi} |x\rangle,$$

using $\prod_s e^{2\pi i x_s 2^{t-s} \varphi} = e^{2\pi i \varphi \sum_s x_s 2^{t-s}} = e^{2\pi i x \varphi}$. $\square$

What each rung inspects, and why it is cheap

Qubit s carries the phase $2^{t-s}\varphi \pmod{1}$, so across the register the phases are $2^{t-1}\varphi, 2^{t-2}\varphi, \dots, \varphi$ from the top wire down. Doubling shifts the binary expansion of φ one place left, and $2^{t-s}\varphi \equiv 0.\varphi_{t-s+1}\varphi_{t-s+2}\dots$: the *top* wire $s = 1$ therefore probes deepest into the expansion, and the bottom wire $s = t$ sees only the leading bit $0.\varphi_1$. This is [Corollary 2.6.4](#) doing all the work.

And $U^{2^{t-s}}$ is obtained from $t - s$ classical repeated squarings ([Equation \(5.7\)](#)), not from 2^{t-s} applications of U . Without that, the ladder would cost $2^t - 1$ operations and there would be no algorithm.

Measure now and you learn nothing

Every amplitude of $Q^{-1/2} \sum_x e^{2\pi i x \varphi} |x\rangle$ has modulus $Q^{-1/2}$. Measuring at this point gives probability $1/Q$ for every x and tells you *nothing* about φ .

All the information is in relative phases. Converting phase into magnitude is the job of the inverse QFT — which is exactly the second beam-splitter of [Section 7.1](#), at scale.

7.5 From two paths to Q paths

The interferometer of [Section 7.1](#) had two paths and produced a two-slit pattern $\frac{1}{2}(1 + \cos \phi)$. The ladder produces Q paths — one per basis state $|x\rangle$ — with a common phase increment $2\pi\varphi$ between consecutive paths. Recombining them with $\text{QFT}^\dagger$ produces the Q -slit pattern.

	Mach–Zehnder	Phase estimation
splitter	$\text{BS}_1 = H$	$H^{\otimes t}$, a Q -way splitter
paths	2	$Q = 2^t$
phase written	ϕ by a phase shifter	$2\pi x \varphi$ by the kickback ladder
recombiner	$\text{BS}_2 = H$	$\text{QFT}_Q^\dagger$
pattern	$\frac{1}{2}(1 + \cos \phi)$, two-slit	Fejér kernel, Q -slit (Theorem 8.3.2)
readout	which detector fired	the index ℓ

The unifying statement

A quantum computer is a multi-particle interferometer whose phase shifts result from the operations of quantum logic gates. Every algorithm treated in this document is an instance of one pattern: *prepare a superposition of computational paths; arrange for the paths to acquire phases that encode the answer; recombine so that the wrong answers cancel and the right one survives.*

Single-particle interference has a classical wave analogue. *Multi*-particle interference, where the number of paths grows as 2^t with the number of qubits, does not.

Chapter 8

Quantum phase estimation

Everything is now in place. [Chapter 5](#) turned a hidden period into an eigenphase; [Chapter 7](#) wrote that eigenphase as a ramp across a register; [Chapter 6](#) built the transform that converts phase into magnitude. This chapter assembles them and computes, exactly, what the resulting measurement returns.

8.1 The problem and the circuit

Problem 8.1.1 (Phase estimation). **Input.** A unitary U on m qubits, given as a black box together with the ability to apply the controlled operations $c-U^{2^\ell}$ for $\ell = 0, 1, 2, \dots$; and one copy of an eigenvector $|u\rangle$ of U , so that $U|u\rangle = e^{2\pi i\varphi}|u\rangle$ with $\varphi \in [0, 1)$. **Output.** An estimate $\tilde{\varphi}$ of φ .

Remark 8.1.2 (Why $\varphi \in [0, 1)$ and why $e^{2\pi i\varphi}$). Every eigenvalue of a unitary has modulus 1 ([Corollary 2.5.2\(b\)](#)), so it can be written uniquely as $e^{2\pi i\varphi}$ with $\varphi \in [0, 1)$. The 2π is absorbed so that φ is a number in $[0, 1)$ whose binary expansion $\varphi = 0.a_1a_2a_3\dots$ is what the algorithm returns. The phase is defined only modulo 1, matching the fact that $\mathbb{Z}/2^t\mathbb{Z}$ -indexed outcomes are defined only modulo 2^t .

What is and is not assumed

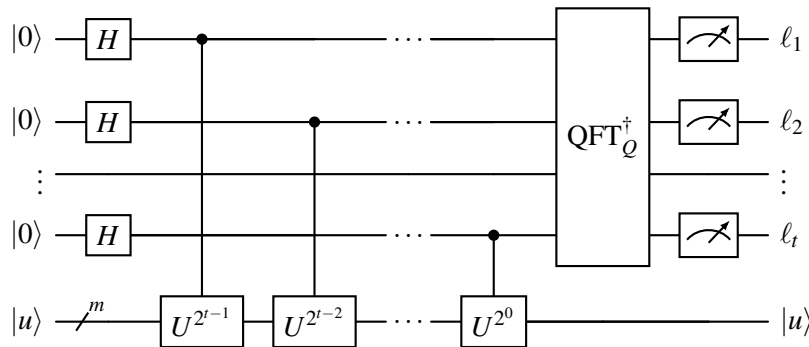
QPE does *not* require knowing U as a matrix, or $|u\rangle$, or φ . It *does* require:

- (i) a circuit for $c-U^{2^\ell}$ whose size grows at most *polynomially* in ℓ — **not** 2^ℓ applications of $c-U$, which would destroy the speed-up; and
- (ii) a preparable input for the second register.

Condition (i) is a genuine restriction. It holds when U is a group action that composes cheaply — e.g. modular multiplication, by [Equation \(5.7\)](#) — or when $U = e^{-iHt}$ for a simulable Hamiltonian. Condition (ii) looks fatal for order finding, since writing down a single $|u_s\rangle$ needs r ; [Theorem 8.5.1](#) dissolves it, and that is what makes Shor’s algorithm possible.

Algorithm 8.1.3 (Quantum phase estimation). Let $t \geq 1$ be the number of counting qubits and $Q = 2^t$.

1. Prepare $|0\rangle^{\otimes t} \otimes |u\rangle$ and apply $H^{\otimes t}$ to the counting register.
2. For $s = 1, \dots, t$ apply $c-U^{2^{t-s}}$ with control the s -th counting qubit (the kickback ladder, [Theorem 7.4.1](#)).
3. Apply $\text{QFT}_Q^\dagger$ to the counting register.
4. Measure the counting register in the computational basis, obtaining $\ell \in \{0, \dots, Q-1\}$; output $\tilde{\varphi} := \ell/Q$.



Left block: the Q -way beam splitter. Middle: the kickback ladder writing the ramp $e^{2\pi i x \varphi}$. Right: the recombining element $\text{QFT}^\dagger$ and readout. The eigenvector register is returned unchanged and can be reused.

Theorem 8.1.4 (State evolution). Write $|\Psi_j\rangle$ for the state after step j of [Algorithm 8.1.3](#). Then

$$|\Psi_1\rangle = \frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} |x\rangle \otimes |u\rangle, \quad (8.1)$$

$$|\Psi_2\rangle = \frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} e^{2\pi i x \varphi} |x\rangle \otimes |u\rangle = \bigotimes_{s=1}^t \frac{|0\rangle + e^{2\pi i 2^{t-s} \varphi} |1\rangle}{\sqrt{2}} \otimes |u\rangle, \quad (8.2)$$

$$|\Psi_3\rangle = \sum_{\ell=0}^{Q-1} \alpha_\ell |\ell\rangle \otimes |u\rangle, \quad \alpha_\ell = \frac{1}{Q} \sum_{x=0}^{Q-1} e^{2\pi i x(\varphi - \ell/Q)}. \quad (8.3)$$

In particular the eigenvector register is never entangled with the counting register and emerges in the state $|u\rangle$.

Proof. Equation (8.1) is [Proposition 6.1.3](#); Equation (8.2) is [Theorem 7.4.1](#). For Equation (8.3), apply $\text{QFT}^\dagger |x\rangle = Q^{-1/2} \sum_{\ell} e^{-2\pi i x \ell / Q} |\ell\rangle$ to each $|x\rangle$ and collect coefficients of $|\ell\rangle$. Factorisation of $|u\rangle$ persists at every step because every gate acts either on the counting register alone or is a $c\text{-}U^P$ with $|u\rangle$ an eigenvector. $\square$

8.2 The exact case

Theorem 8.2.1 (QPE is exact on t -bit phases). Suppose φ has a terminating binary expansion of length at most t , i.e. $\varphi = a/Q$ for an integer $a \in \{0, \dots, Q-1\}$. Then $|\Psi_3\rangle = |a\rangle \otimes |u\rangle$ and [Algorithm 8.1.3](#) outputs $\tilde{\varphi} = \varphi$ with probability 1.

Proof. By [Equation \(8.3\)](#) with $\varphi = a/Q$, $\alpha_\ell = \frac{1}{Q} \sum_x \omega_Q^{x(a-\ell)} = \delta_{a\ell}$ by [Lemma A.0.1](#), since $|a - \ell| < Q$ forces $a \equiv \ell \pmod{Q}$ to mean $a = \ell$.

Equivalently and more conceptually: by [Equation \(6.3\)](#) the state $|\Psi_2\rangle$ is the Fourier basis state $\text{QFT}_Q |a\rangle$ tensored with $|u\rangle$; applying $\text{QFT}_Q^\dagger$ returns $|a\rangle$ by unitarity. $\square$

The whole algorithm in one sentence

The kickback ladder writes the number $a = Q\varphi$ into the *phases* of the counting register; the inverse QFT rewrites it into the *labels*; the measurement reads the labels. The two encodings are related by the isomorphism $\mathbb{Z}/Q\mathbb{Z} \cong \mu_Q$ of [Chapter 5](#), and $\text{QFT}^\dagger$ is that isomorphism made unitary.

8.3 The general case: the Fejér kernel

In general $Q\varphi \notin \mathbb{Z}$ — indeed φ need not be rational — and the output becomes a probability distribution concentrated near $Q\varphi$. We compute it *exactly*.

Notation 8.3.1 (Detuning). For $\ell \in \{0, \dots, Q-1\}$ set $\delta_\ell := \varphi - \ell/Q \pmod{1}$, chosen in $[-\frac{1}{2}, \frac{1}{2})$. Because α_ℓ in [Equation \(8.3\)](#) depends on $\varphi - \ell/Q$ only through $e^{2\pi i x(\cdot)}$ with $x \in \mathbb{Z}$, it is a function of δ_ℓ alone.

Theorem 8.3.2 (Exact output distribution). For every ℓ ,

$$\alpha_\ell = \begin{cases} 1, & \delta_\ell = 0, \\ \frac{1}{Q} \cdot \frac{1 - e^{2\pi i Q \delta_\ell}}{1 - e^{2\pi i \delta_\ell}}, & \delta_\ell \neq 0, \end{cases} \quad (8.4)$$

and consequently

$$\Pr[\ell] = |\alpha_\ell|^2 = \frac{1}{Q^2} \cdot \frac{\sin^2(\pi Q \delta_\ell)}{\sin^2(\pi \delta_\ell)} \quad (8.5)$$

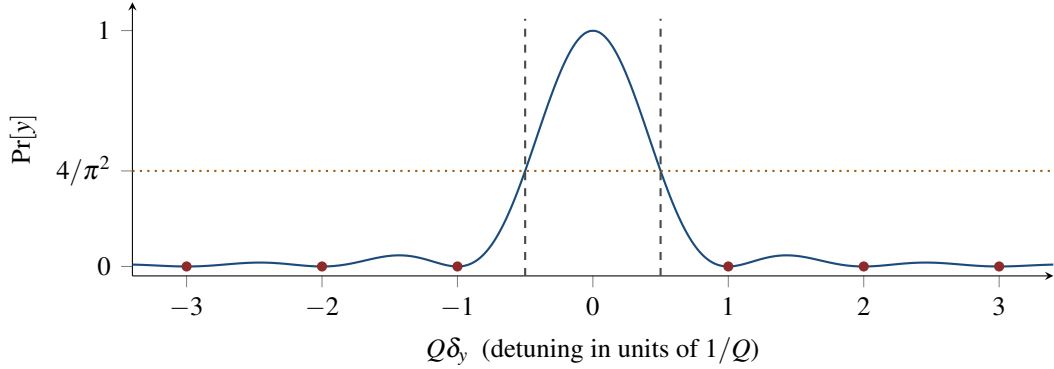
with the value at $\delta_\ell = 0$ understood as the limit 1. The right-hand side is the Fejér kernel: the intensity pattern of a Q -slit diffraction grating.

Proof. If $\delta_\ell = 0$ every summand in Equation (8.3) equals 1 and $\alpha_\ell = 1$. Otherwise $e^{2\pi i \delta_\ell} \neq 1$ and the geometric sum (Lemma A.0.2) gives $\sum_{x=0}^{Q-1} e^{2\pi i \delta_\ell x} = \frac{1 - e^{2\pi i \delta_\ell Q}}{1 - e^{2\pi i \delta_\ell}}$. For the modulus use $|1 - e^{i\theta}| = 2|\sin(\theta/2)|$ (Lemma A.0.3) twice: $|\alpha_\ell| = \frac{1}{Q} \cdot \frac{2|\sin(\pi Q \delta_\ell)|}{2|\sin(\pi \delta_\ell)|}$, and square. $\square$

Corollary 8.3.3 (Consistency check). $\sum_\ell \Pr[\ell] = 1$, and if $Q\phi \in \mathbb{Z}$ the distribution collapses to the point mass of Theorem 8.2.1.

Proof. Normalisation holds because $\text{QFT}^\dagger$ is unitary and $|\Psi_2\rangle$ is a unit vector. If $Q\phi = a \in \mathbb{Z}$ then $\delta_\ell = (a - \ell)/Q$, so $\sin(\pi Q \delta_\ell) = \sin(\pi(a - \ell)) = 0$ for every ℓ with $\delta_\ell \neq 0$, and $\delta_\ell = 0$ only for $\ell = a$. $\square$

8.3.1 Three regimes



The Fejér kernel in the large- Q limit, $\Pr \approx \text{sinc}^2(\pi Q \delta_y)$. Red dots: the exact zeros at integer detunings. Dashed lines bound the nearest grid point, $|\delta_y| \leq 1/(2Q)$, where the $4/\pi^2$ guarantee applies.

The contributions to α_ℓ are the Q unit vectors $1, e^{2\pi i \delta_\ell}, \dots, e^{2\pi i (Q-1) \delta_\ell}$.

1. **Exact match**, $\delta_\ell = 0$: every vector is 1, so $\alpha_\ell = 1$ — all the probability.
2. **Exact non-matching mode**, $\delta_\ell = k/Q$ with $k \not\equiv 0$: the vectors are the Q -th roots of unity taken k at a time, a *balanced closed polygon*, so $\alpha_\ell = 0$ exactly. This is Lemma A.0.1.
3. **Near match**, $|\delta_\ell|$ small: the vectors rotate by a small angle and form an *arc*, so the sum stays large. Theorem 8.3.4 makes this quantitative.

Theorem 8.3.4 (The $4/\pi^2$ guarantee). Let $b \in \{0, \dots, Q-1\}$ be an integer nearest to $Q\phi$ modulo Q , so that $|\delta_b| \leq \frac{1}{2Q}$. Then

$$\Pr[b] \geq \frac{4}{\pi^2} = 0.4052\dots \quad (8.6)$$

Consequently Algorithm 8.1.3 returns the best t -bit approximation of ϕ with probability at least $4/\pi^2$; repeating $O(\log(1/\eta))$ times and taking the most frequent outcome gives it with probability $\geq 1 - \eta$.

Proof. If $\delta_b = 0$ then $\Pr[b] = 1$, so assume $\delta := \delta_b \neq 0$ with $0 < |\delta| \leq \frac{1}{2Q}$. Write $\theta := \pi Q |\delta| \in (0, \pi/2]$. From Equation (8.5),

$$\sqrt{\Pr[b]} = \frac{1}{Q} \cdot \frac{|\sin(\pi Q \delta)|}{|\sin(\pi \delta)|} = \frac{1}{Q} \cdot \frac{\sin \theta}{|\sin(\pi \delta)|}.$$

Bound numerator and denominator separately. *Denominator:* $|\sin x| \leq |x|$ for all real x (Lemma A.0.4(a)), so $|\sin(\pi\delta)| \leq \pi|\delta|$. *Numerator:* since $\theta \in (0, \pi/2]$, Jordan's inequality $\sin \theta \geq \frac{2}{\pi}\theta$ (Lemma A.0.4(b)) gives $\sin \theta \geq \frac{2}{\pi}\pi Q|\delta| = 2Q|\delta|$. Therefore

$$\sqrt{\Pr[b]} \geq \frac{1}{Q} \cdot \frac{2Q|\delta|}{\pi|\delta|} = \frac{2}{\pi},$$

and squaring gives Equation (8.6).

For the amplification statement one first needs that b is the *unique mode* — which does not follow from $p \geq 4/\pi^2$ alone, since $4/\pi^2 < \frac{1}{2}$. It follows instead from the shape of Equation (8.5): the numerator $\sin^2(\pi Q\delta_\ell) = \sin^2(\pi Q\varphi)$ is the same for every ℓ , because $Q\delta_\ell = Q\varphi - \ell$ differs from $Q\varphi$ by an integer. Hence

$$\Pr[\ell] = \frac{\sin^2(\pi Q\varphi)}{Q^2} \cdot \frac{1}{\sin^2(\pi\delta_\ell)}$$

is a strictly decreasing function of $|\delta_\ell|$ on $[-\frac{1}{2}, \frac{1}{2})$, so it is maximised exactly at the grid point nearest $Q\varphi$ (with a tie only in the boundary case $|\delta_b| = 1/(2Q)$). With b the unique mode and $\Pr[b] \geq 4/\pi^2$, the standard Chernoff argument applies: taking the mode of $O(\log(1/\eta))$ independent runs returns b except with probability $\leq \eta$. $\square$

Remark 8.3.5 (The bound is tight). Taking $|\delta| = \frac{1}{2Q}$ exactly gives $\Pr[b] = (Q \sin(\pi/2Q))^{-2}$, which decreases to $4/\pi^2$ as $t \rightarrow \infty$: the values are 0.41053 at $t = 3$, 0.40537 at $t = 6$ and 0.405285 at $t = 10$. So no constant larger than $4/\pi^2$ works uniformly in t .

Proposition 8.3.6 (The two straddling values carry 81%). *Let $b_0 = \lfloor Q\varphi \rfloor$ and $b_1 = b_0 + 1 \bmod Q$, and put $\theta := Q\varphi - b_0 \in [0, 1)$. Then*

$$\Pr[b_0] + \Pr[b_1] \geq \frac{8}{\pi^2} = 0.8105\dots \quad (8.7)$$

Proof. $\delta_{b_0} = \theta/Q$ and $\delta_{b_1} = (\theta - 1)/Q$. Since $\sin^2(\pi(\theta - 1)) = \sin^2(\pi\theta)$, Equation (8.5) together with $|\sin x| \leq |x|$ in each denominator gives

$$\Pr[b_0] + \Pr[b_1] \geq \frac{1}{\pi^2} \sin^2(\pi\theta) \underbrace{\left(\frac{1}{\theta^2} + \frac{1}{(1-\theta)^2} \right)}_{=: f(\theta)}.$$

A one-variable calculus exercise shows $\sin^2(\pi\theta)f(\theta)$ attains its minimum on $(0, 1)$ at $\theta = \frac{1}{2}$, where it equals $1 \cdot (4 + 4) = 8$. $\square$

8.4 Precision: how many qubits for how much accuracy

Theorem 8.3.4 guarantees the single nearest grid point. In applications one usually wants a stronger statement: $\tilde{\varphi}$ within 2^{-n} of φ , with failure probability at most ε . Since $\Pr[\cdot]$ has heavy $1/\ell^2$ tails, buying accuracy means buying extra qubits.

Theorem 8.4.1 (Precision–qubit trade-off). *Fix $n \geq 1$ and $\varepsilon \in (0, \frac{1}{2})$, and run Algorithm 8.1.3 with*

$$t = n + \left\lceil \log_2 \left(2 + \frac{1}{2\varepsilon} \right) \right\rceil \quad (8.8)$$

counting qubits. Then the outcome ℓ satisfies $|\ell/Q - \varphi| < 2^{-n}$ (as a distance modulo 1) with probability at least $1 - \varepsilon$.

Proof. Let $b := \lfloor Q\varphi \rfloor$, so $\delta := \varphi - b/Q \in [0, 2^{-t})$ and $0 \leq Q\delta < 1$. Index the outcomes by $\ell' := (b + \ell) \bmod Q$ where ℓ runs over $-Q/2 < \ell \leq Q/2$. By Equation (8.3),

$$\alpha_\ell = \frac{1}{Q} \sum_{x=0}^{Q-1} e^{2\pi i x(\delta - \ell/Q)} = \frac{1}{Q} \cdot \frac{1 - e^{2\pi i(Q\delta - \ell)}}{1 - e^{2\pi i(\delta - \ell/Q)}}$$

when the denominator is non-zero.

Bound the amplitude. The numerator has modulus at most 2. For the denominator, $|1 - e^{i\theta}| \geq 2|\theta|/\pi$ for $|\theta| \leq \pi$ (Lemma A.0.3), applied with $\theta = 2\pi(\delta - \ell/Q)$, which satisfies $|\theta| \leq \pi$ in the stated range of ℓ . Hence $|1 - e^{2\pi i(\delta - \ell/Q)}| \geq 4|\delta - \ell/Q|$ and

$$|\alpha_\ell| \leq \frac{2}{Q \cdot 4|\delta - \ell/Q|} = \frac{1}{2|Q\delta - \ell|}. \quad (8.9)$$

Sum the tails. Let $e \geq 2$ be an integer; the probability of an outcome with $|\ell| > e$ is, by Equation (8.9),

$$p \leq \frac{1}{4} \left[\sum_{\ell \leq -(e+1)} \frac{1}{(Q\delta - \ell)^2} + \sum_{\ell \geq e+1} \frac{1}{(Q\delta - \ell)^2} \right].$$

Since $0 \leq Q\delta < 1$: for $\ell \leq -(e+1)$ we have $|Q\delta - \ell| \geq |\ell|$, and for $\ell \geq e+1$ we have $|Q\delta - \ell| \geq \ell - 1$. Therefore

$$p \leq \frac{1}{4} \left[\sum_{j \geq e+1} \frac{1}{j^2} + \sum_{j \geq e} \frac{1}{j^2} \right] \leq \frac{1}{2} \sum_{j \geq e} \frac{1}{j^2} \leq \frac{1}{2} \int_{e-1}^{\infty} \frac{dx}{x^2} = \frac{1}{2(e-1)}.$$

Choose e . Put $e := 2^{t-n} - 1$. If the outcome satisfies $|\ell| \leq e$ then

$$\left| \frac{\ell'}{Q} - \varphi \right| \leq \frac{|\ell|}{Q} + \delta < \frac{2^{t-n} - 1}{2^t} + \frac{1}{2^t} = 2^{-n},$$

as required. It therefore suffices that $\frac{1}{2(e-1)} = \frac{1}{2(2^{t-n}-2)} \leq \varepsilon$, i.e. $2^{t-n} \geq 2 + \frac{1}{2\varepsilon}$, which is Equation (8.8). (Note $e \geq 2$ holds automatically, since $2 + \frac{1}{2\varepsilon} > 3$ for $\varepsilon < \frac{1}{2}$ forces $t - n \geq 2$.) $\square$

Reading the formula

Equation (8.8) splits the cost into two independent parts: n qubits **buy resolution** (2^{-n}), and $\lceil \log_2(2 + \frac{1}{2\varepsilon}) \rceil$ extra qubits **buy confidence** ($1 - \varepsilon$).

The confidence term is tiny: $\varepsilon = 10^{-2}$ costs 6 qubits, $\varepsilon = 10^{-6}$ costs 19. Confidence is exponentially cheap; resolution is not, because each extra bit of resolution doubles the number of applications of U .

8.5 Input that is not an eigenvector

This is the case that makes Shor's algorithm work, and it deserves a precise statement.

Theorem 8.5.1 (QPE on a superposition of eigenvectors). *Let $U = \sum_u e^{2\pi i \varphi_u} |u\rangle\langle u|$ with $\{|u\rangle\}$ an orthonormal eigenbasis, and run Algorithm 8.1.3 on the input $|0\rangle^{\otimes t} \otimes |\psi\rangle$ where $|\psi\rangle = \sum_u c_u |u\rangle$, $\sum_u |c_u|^2 = 1$. Then the state before measurement is*

$$|\Psi_3\rangle = \sum_u c_u \left(\sum_\ell \alpha_\ell^{(u)} |\ell\rangle \right) \otimes |u\rangle, \quad \alpha_\ell^{(u)} = \frac{1}{Q} \sum_{x=0}^{Q-1} e^{2\pi i x(\varphi_u - \ell/Q)}, \quad (8.10)$$

and measuring the counting register yields ℓ with probability

$$\Pr[\ell] = \sum_u |c_u|^2 |\alpha_\ell^{(u)}|^2. \quad (8.11)$$

That is: the algorithm behaves exactly as if an eigenvector $|u\rangle$ had first been drawn at random with probability $|c_u|^2$, and QPE had then been run on that eigenvector.

Proof. Algorithm 8.1.3 is a linear map, and Theorem 8.1.4 computed its action on each $|0\rangle^{\otimes t} |u\rangle$ separately; superposing gives Equation (8.10). For the probability, the $|u\rangle$ are orthonormal, so by Proposition 3.6.1

$$\Pr[\ell] = \left\| \sum_u c_u \alpha_\ell^{(u)} |u\rangle \right\|^2 = \sum_u |c_u|^2 |\alpha_\ell^{(u)}|^2,$$

the cross terms vanishing by $\langle u|u'\rangle = \delta_{uu'}$. □

The commuting-measurement argument

Here is the same statement in a form worth internalising. Imagine measuring the second register in the eigenbasis $\{|u\rangle\}$ *before* running the circuit: one obtains $|u\rangle$ with probability $|c_u|^2$ and the algorithm proceeds as in Theorem 8.1.4.

But this hypothetical measurement *commutes with every gate* of Algorithm 8.1.3 — each is either supported on the counting register or is a $c-U^P$, and U^P is diagonal in $\{|u\rangle\}$ — so it may equally be performed *after* the circuit, where it cannot influence the already-recorded outcome of the counting register. Therefore the two experiments have identical statistics on the counting register.

This is the cleanest way to see Equation (8.11), and it is exactly the argument that justifies feeding $|1\rangle$ into the order-finding circuit in Chapter 10.

What can go wrong

Theorem 8.5.1 is a statement about the *counting register only*. Two caveats.

- (i) The output is generally *entangled*: for a given ℓ with several contributing u , the second register is left in the unnormalised state $\sum_u c_u \alpha_\ell^{(u)} |u\rangle$, not in a single eigenvector. This matters if one intends to reuse it.
- (ii) If several φ_u are closer together than the resolution $1/Q$, the peaks overlap and the outcome cannot be attributed to a single u . Increasing t separates them, at the cost of more applications of U .

8.6 Iterative phase estimation and the semiclassical QFT

Algorithm 8.1.3 uses t counting qubits simultaneously. A variant due in essence to Kitaev, and made exact by the Griffiths–Niu *semiclassical QFT*, uses a *single* counting qubit at the price of t rounds of measurement and classical feedback. It is the method of choice on hardware with few qubits, and it is what makes Beauregard’s $2n + 3$ -qubit circuit possible.

The observation is that every control of the inverse QFT is a qubit that is measured immediately afterwards and never used again. By Proposition 4.1.3, such a controlled gate may instead be replaced by a *classically* controlled gate applied after measuring that control. Doing this to every rotation in $\text{QFT}^\dagger$ turns the two-qubit gates into classically conditioned one-qubit rotations.

Algorithm 8.6.1 (Iterative QPE). Assume $\varphi = 0.a_1 a_2 \cdots a_t$ exactly. Maintain a classical register Φ , initially 0. For $\ell = t - 1, t - 2, \dots, 0$:

1. prepare one qubit in $|0\rangle$ and apply H ;
2. apply $c-U^{2^\ell}$ with this qubit as control and the eigenvector register as target;
3. apply the correction $P(-2\pi\Phi)$ to the qubit;
4. apply H and measure, obtaining the bit $a_{\ell+1}$;
5. update $\Phi \leftarrow 0.0a_{\ell+1}a_{\ell+2} \cdots a_t$ (a classical shift-and-append).

Output $\varphi = 0.a_1 a_2 \cdots a_t$.

Proposition 8.6.2 (Correctness). If $\varphi = 0.a_1 \cdots a_t$, Algorithm 8.6.1 outputs $a_1, \dots, a_t$ with probability 1, using one counting qubit, $2^t - 1$ applications of U , and t measurements.

Proof. By Lemma 7.2.1 the phase kicked back at round ℓ is $e^{2\pi i 2^\ell \varphi}$, and

$$2^\ell \varphi = \sum_{s=1}^t a_s 2^{\ell-s} \equiv \sum_{s=\ell+1}^t a_s 2^{\ell-s} = 0.a_{\ell+1}a_{\ell+2}\cdots a_t \pmod{1},$$

because the terms with $s \leq \ell$ are integers. At round $\ell = t - 1$ this is the single bit $0.a_t$, so the qubit is in $\frac{1}{\sqrt{2}}(|0\rangle + (-1)^{a_t}|1\rangle)$ and H maps it to $|a_t\rangle$: the bit is read out exactly. Inductively, at round ℓ the bits $a_{\ell+2}, \dots, a_t$ are already known, and by construction $\Phi = 0.0a_{\ell+2}\cdots a_t$ equals $0.a_{\ell+1}\cdots a_t - 0.a_{\ell+1}$; subtracting it leaves the qubit in $\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 0.a_{\ell+1}}|1\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + (-1)^{a_{\ell+1}}|1\rangle)$, which H maps to $|a_{\ell+1}\rangle$. The count $\sum_{\ell=0}^{t-1} 2^\ell = 2^t - 1$ is unchanged from Algorithm 8.1.3. $\square$

Remark 8.6.3. For general φ the exactness fails and each round must be repeated $O(\log(t/\eta))$ times with a majority vote; this is Kitaev's original scheme, recovering the same precision with $O(t \log t)$ measurements. The qubit–time trade-off (one qubit, t rounds versus t qubits, one round) is the same trade-off as between a serial and a parallel FFT.

8.7 Resource summary

Resource	Cost of Algorithm 8.1.3
counting qubits	$t = n + \lceil \log_2(2 + \frac{1}{2\varepsilon}) \rceil$ (Theorem 8.4.1)
eigenvector qubits	m (the size of the register U acts on)
applications of U	$\sum_{\ell=0}^{t-1} 2^\ell = 2^t - 1$
calls to $c-U^{2^\ell}$	t (each built by ℓ classical squarings)
gates for $\text{QFT}_Q^\dagger$	$\Theta(t^2)$ exactly; $O(t \log t)$ approximately (Theorem 6.4.2)
measurements	t (all at the end; or t rounds, Algorithm 8.6.1)
success probability	$\geq 4/\pi^2$ for the nearest grid point (Theorem 8.3.4); $\geq 1 - \varepsilon$ for accuracy 2^{-n} (Theorem 8.4.1)

Summary of

1. The ladder of controlled powers writes φ into a phase ramp (Theorem 7.4.1).
2. If $Q\varphi \in \mathbb{Z}$, that state is a Fourier basis state and $\text{QFT}^\dagger$ reads it out exactly (Theorem 8.2.1).
3. Otherwise the outcome distribution is the Q -slit diffraction pattern Equation (8.5), sharply peaked at $Q\varphi$, with $\Pr[\text{nearest}] \geq 4/\pi^2$ (Theorem 8.3.4) and $\geq 8/\pi^2$ for the two straddling values.
4. Precision 2^{-n} with confidence $1 - \varepsilon$ costs $t = n + \lceil \log_2(2 + \frac{1}{2\varepsilon}) \rceil$ qubits (Theorem 8.4.1).
5. Feeding in a *superposition* of eigenvectors samples one of them (Theorem 8.5.1). **This is the door through which Shor's algorithm enters.**

The dominant cost is almost always the $2^t - 1$ applications of U , not the QFT.

Part III

Factoring

Chapter 9

The classical reduction: factoring to order finding

Shor's algorithm is a *classical* algorithm with one quantum subroutine. This chapter builds the classical part: the reduction of factoring to order finding, and the proof that a randomly chosen base works with probability at least $\frac{1}{2}$. No quantum mechanics appears.

Two features of the reduction are worth flagging in advance, because they are what make the algorithm practical rather than merely asymptotic. First, every candidate the quantum subroutine produces is *verified classically*, so the algorithm never returns a wrong factor — it is Las Vegas, not Monte Carlo. Second, there are *two independent* failure probabilities, both bounded away from 0, and they must not be conflated: the choice of base a may be unlucky, and a given quantum run may return a useless outcome.

9.1 Non-trivial square roots of 1 split N

Lemma 9.1.1 (Square roots of 1). *Let $N \geq 3$ and let x satisfy*

$$x^2 \equiv 1 \pmod{N}, \quad x \not\equiv 1, \quad x \not\equiv -1 \pmod{N}.$$

Then $\gcd(x-1, N)$ and $\gcd(x+1, N)$ are both non-trivial factors of N .

Proof. From $x^2 - 1 = (x-1)(x+1) \equiv 0$ we get $N \mid (x-1)(x+1)$. Put $d := \gcd(x-1, N)$.

- If $d = N$ then $N \mid x-1$, i.e. $x \equiv 1$ — excluded.
- If $d = 1$ then N and $x-1$ are coprime, so from $N \mid (x-1)(x+1)$ and Gauss's lemma (Lemma 1.2.2), $N \mid x+1$, i.e. $x \equiv -1$ — also excluded.

Hence $1 < d < N$ and $d \mid N$. The same argument applied to $-x$ (which also satisfies the hypotheses) gives the claim for $\gcd(x+1, N)$. $\square$

How order finding manufactures such an x

Pick a coprime to N and let $r = \text{ord}_N(a)$, so $a^r \equiv 1$. If

(i) r is **even**, and

(ii) $a^{r/2} \not\equiv -1 \pmod{N}$,

put $x := a^{r/2} \pmod{N}$. Then $x^2 = a^r \equiv 1$; $x \not\equiv 1$, else $r/2$ would already be a period, contradicting the minimality of r ; and $x \not\equiv -1$ by (ii). So Lemma 9.1.1 applies, and $\gcd(a^{r/2} \pm 1, N)$ splits N . This matches the factorisation $a^r - 1 = (a^{r/2} - 1)(a^{r/2} + 1)$.

9.2 A random base usually works

It remains to show that conditions (i)–(ii) hold often for a random a . This is the one place where the structure of $(\mathbb{Z}/N\mathbb{Z})^\times$ genuinely matters, and it is why the easy cases must be stripped first.

Theorem 9.2.1 (A random base works). *Let $N = p_1^{\alpha_1} \cdots p_m^{\alpha_m}$ be odd and composite with $m \geq 2$ distinct prime factors, and let a be drawn uniformly from $(\mathbb{Z}/N\mathbb{Z})^\times$, with $r = \text{ord}_N(a)$. Then*

$$\Pr \left[r \text{ even and } a^{r/2} \not\equiv -1 \pmod{N} \right] \geq 1 - \frac{1}{2^{m-1}} \geq \frac{1}{2}. \quad (9.1)$$

Proof. Write $N_j := p_j^{\alpha_j}$. By the Chinese remainder theorem (Theorem 1.3.1), $(\mathbb{Z}/N\mathbb{Z})^\times \cong (\mathbb{Z}/N_1\mathbb{Z})^\times \times \cdots \times (\mathbb{Z}/N_m\mathbb{Z})^\times$, and drawing a uniformly from $(\mathbb{Z}/N\mathbb{Z})^\times$ is the same as drawing its components a_j independently and uniformly. Let $r_j := \text{ord}_{N_j}(a_j)$, so $r = \text{lcm}(r_1, \dots, r_m)$ by Corollary 1.3.2, and write $v_2(\cdot)$ for the exponent of 2 in an integer, $d_j := v_2(r_j)$, $d := v_2(r) = \max_j d_j$.

Step 1: the bad event forces all d_j to be equal.

Case r odd. Then every r_j divides r , hence is odd, so $d_1 = \cdots = d_m = 0$.

Case r even and $a^{r/2} \equiv -1 \pmod{N}$. Reducing modulo N_j gives $a_j^{r/2} \equiv -1 \not\equiv 1 \pmod{N_j}$ (using N_j odd, so $-1 \not\equiv 1$), while $a_j^{r_j} \equiv 1$. Hence $r_j \mid r$ but $r_j \nmid r/2$. If we had $d_j < d$, then r_j would divide $r/2$ — its odd part divides the odd part of r , which equals that of $r/2$, and $v_2(r_j) = d_j \leq d - 1 = v_2(r/2)$ — a contradiction. So $d_j = d$ for every j .

Step 2: each d_j is spread out. For p an odd prime, $(\mathbb{Z}/p^\alpha\mathbb{Z})^\times$ is cyclic (Theorem 1.3.3), say of order $n_j := \phi(N_j) = p_j^{\alpha_j-1}(p_j - 1)$, which is even. Write $c_j := v_2(n_j) \geq 1$ and fix a generator g_j . Writing $a_j = g_j^{k_j}$ with k_j uniform in $\mathbb{Z}/n_j\mathbb{Z}$, we have $r_j = n_j / \gcd(n_j, k_j)$, so

$$d_j = v_2(r_j) = c_j - \min(v_2(k_j), c_j).$$

Since k_j is uniform modulo the even number n_j , $\Pr[v_2(k_j) = e] = 2^{-(e+1)}$ for $0 \leq e < c_j$. Therefore

$$\Pr[d_j = c_j] = \frac{1}{2}, \quad \Pr[d_j = c_j - e] = 2^{-(e+1)} \quad (0 \leq e < c_j), \quad \text{so} \quad \Pr[d_j = d] \leq \frac{1}{2} \text{ for every } d.$$

Step 3: combine. Fix any index j_0 . Using independence and Step 2 for the $m - 1$ indices $j \neq j_0$,

$$\Pr[d_1 = \cdots = d_m] = \sum_{d \geq 0} \prod_{j=1}^m \Pr[d_j = d] \leq \sum_{d \geq 0} \Pr[d_{j_0} = d] \cdot \left(\frac{1}{2}\right)^{m-1} = \frac{1}{2^{m-1}}.$$

By Step 1 the bad event is contained in $\{d_1 = \cdots = d_m\}$, so it has probability at most $2^{-(m-1)} \leq \frac{1}{2}$ for $m \geq 2$. $\square$

Remark 9.2.2 (Where the hypotheses were used). A sharper analysis gives $1 - 2^{-m}$; we only need a constant. Note precisely where the hypotheses entered: N odd makes each n_j even (so $c_j \geq 1$), and $m \geq 2$ gives at least one factor of $\frac{1}{2}$. Both are guaranteed by removing even N and prime powers beforehand — which is why those two preconditions, so often omitted in informal accounts, are load-bearing.

The two preconditions people omit

Before the quantum part is ever invoked:

1. **Strip factors of 2.** If N is even, return 2. (Recall from Theorem 1.3.3 that $(\mathbb{Z}/2^\alpha\mathbb{Z})^\times$ is *not* cyclic for $\alpha \geq 3$, so Step 2 above would fail.)
2. **Test whether $N = b^c$ is a perfect power** for $c \leq \log_2 N$, by computing $\lfloor N^{1/c} \rfloor$ and testing. If so, return b . Theorem 9.2.1 needs $m \geq 2$ distinct odd primes.

Both are classical and cost $O(n^2 \text{ poly } \log n)$ time.

9.3 The complete classical algorithm

Algorithm 9.3.1 (Factoring, given an order-finding oracle). **Input:** a composite N .

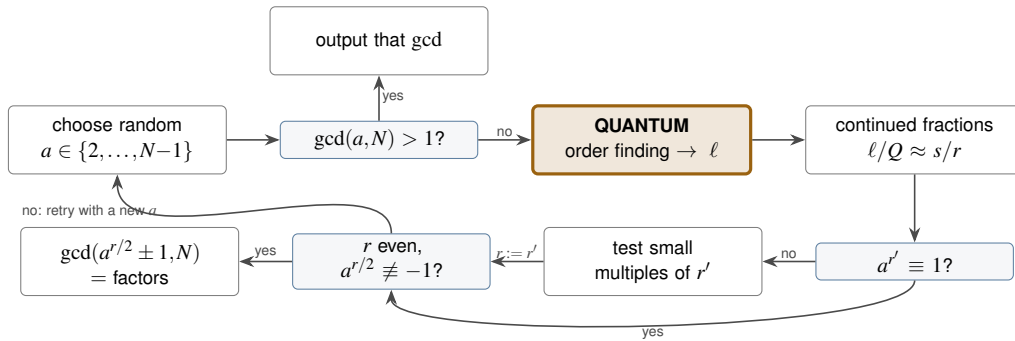
1. If N is even, return 2.
2. If $N = b^c$ for integers $b \geq 2$, $c \geq 2$, return b .
3. Choose $a \in \{2, \dots, N - 1\}$ uniformly at random.
4. Compute $d := \gcd(a, N)$ by Euclid's algorithm. If $d > 1$, return d . (A lucky guess — the quantum subroutine was never needed.)

5. Call ORDERFINDING(N, a) to get $r = \text{ord}_N(a)$.
6. If r is odd, or $a^{r/2} \equiv -1 \pmod{N}$, go to step 3.
7. Compute $d := \gcd(a^{r/2} - 1, N)$. If $1 < d < N$ return d ; otherwise return $\gcd(a^{r/2} + 1, N)$.

Theorem 9.3.2 (Correctness). *On odd composite non-prime-power input N , each pass through steps 3–7 of Algorithm 9.3.1 returns a non-trivial factor with probability at least $\frac{1}{2}$, so the expected number of passes is at most 2. All classical steps run in $O(n^2 \text{ polylog } n)$ time.*

Proof. If step 4 triggers, the output is correct by definition of $\gcd$. Otherwise $a \in (\mathbb{Z}/N\mathbb{Z})^\times$ is uniform, so by Theorem 9.2.1 conditions (i)–(ii) hold with probability $\geq \frac{1}{2}$, and then Lemma 9.1.1 applies to $x = a^{r/2} \pmod{N}$, so step 7 outputs a non-trivial factor. Steps 1–2 and 4 use Euclid’s algorithm and integer root extraction; step 7 uses modular exponentiation by repeated squaring. $\square$

9.4 The pipeline, with the quantum part isolated



Reading the diagram

Only the shaded box is quantum. Everything else — $\gcd$, continued fractions, verification, factor extraction — is cheap classically *and verifies its own output*, so the procedure never returns a wrong factor: it is **Las Vegas**, not Monte Carlo.

Two independent failure probabilities, both bounded away from 0, must not be conflated:

1. the choice of a may be unlucky (r odd or $a^{r/2} \equiv -1$) — probability $\leq \frac{1}{2}$ by Theorem 9.2.1;
2. a given quantum run may return a useless outcome ($\ell = 0$, or a convergent that is only a proper divisor of r) — quantified in Chapter 11.

Failures cost one repetition each; that constant overhead is why Shor’s algorithm is still polynomial time.

9.5 Two running examples

$N = 15, a = 2$ — the dyadic case

$1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 1 \pmod{15}$, so $r = \text{ord}_{15}(2) = 4$.

(i) $r = 4$ even $\checkmark$ (ii) $2^2 = 4 \not\equiv -1 \equiv 14 \checkmark$

$\gcd(3, 15) = 3$, $\gcd(5, 15) = 5$.

$r = 4$ is a power of two, so phase estimation is *exact* and the continued-fraction step is **vacuous**.

$N = 21, a = 2$ — the non-dyadic case

$1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 16 \rightarrow 11 \rightarrow 1 \pmod{21}$, so $r = \text{ord}_{21}(2) = 6$.

(i) $r = 6$ even $\checkmark$ (ii) $2^3 = 8 \not\equiv -1 \equiv 20 \checkmark$

$\gcd(7, 21) = 7$, $\gcd(9, 21) = 3$.

$6 \nmid Q$ for any power of two, so the peaks broaden into a diffraction pattern and the continued-fraction step is **non-trivial**.

Why $N = 15$ is not a representative test case

$(\mathbb{Z}/15\mathbb{Z})^\times \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$, so $\text{ord}_{15}(a) \in \{1, 2, 4\}$ *always*: the period is a power of two whichever base is chosen. The continued-fraction step therefore does literally no work for $N = 15$ — the hardest step of the algorithm disappears.

Nearly every published hardware demonstration uses $N = 15$. [Chapter 14](#) works $N = 21$ analytically instead, and [Section 14.8](#) discusses what the demonstrations do and do not show.

Chapter 10

Order finding is phase estimation

This is the chapter where the three strands meet. [Chapter 5](#) showed that U_a restricted to the orbit of 1 is the cyclic shift, with eigenvalues $e^{2\pi is/r}$. [Chapter 8](#) showed how to measure an eigenphase. The only remaining obstacle is circular: phase estimation wants an eigenvector as input, but writing down a single $|u_s\rangle$ requires knowing r — the very quantity being computed. [Theorem 10.1.1](#) removes the circularity in one line.

Problem 10.0.1 (Order finding). Given N and a with $\gcd(a, N) = 1$, output $r = \text{ord}_N(a)$.

Convention 10.0.2. Throughout, $n := \lceil \log_2 N \rceil$ is the bit length of N , so $N < 2^n$ and every element of $\mathbb{Z}/N\mathbb{Z}$ fits in an n -qubit register.

10.1 The free input

Theorem 10.1.1 (The input $|1\rangle$ is free). With $|u_s\rangle$ as in [Equation \(5.9\)](#),

$$\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |u_s\rangle = |1\rangle, \quad (10.1)$$

and more generally, for every $k_0 \in \mathbb{Z}/r\mathbb{Z}$,

$$\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} e^{2\pi i s k_0 / r} |u_s\rangle = |a^{k_0} \bmod N\rangle. \quad (10.2)$$

Proof. Substituting [Equation \(5.9\)](#) and exchanging the order of summation,

$$\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} e^{2\pi i s k_0 / r} |u_s\rangle = \frac{1}{r} \sum_{k=0}^{r-1} \left(\underbrace{\sum_{s=0}^{r-1} e^{2\pi i s (k_0 - k) / r}}_{= r \delta_{k, k_0} \text{ by Lemma A.0.1}} \right) |a^k\rangle = |a^{k_0}\rangle.$$

Taking $k_0 = 0$ and $a^0 = 1$ gives [Equation \(10.1\)](#). □

The trick that makes Shor's algorithm possible

Preparing any single $|u_s\rangle$ requires knowing r — and r is exactly what we are trying to compute. [Theorem 10.1.1](#) dissolves the circularity: *the uniform superposition of all r eigenvectors is the computational basis state $|0 \cdots 01\rangle$, which is free to prepare with a single X gate.*

By [Theorem 8.5.1](#), running QPE on $|1\rangle$ is statistically identical to running it on a uniformly random $|u_s\rangle$, $s \in \{0, \dots, r-1\}$. So a run returns an estimate of s/r for a uniformly random s — which is all we need, since *any* s coprime to r recovers r . The work register is never measured.

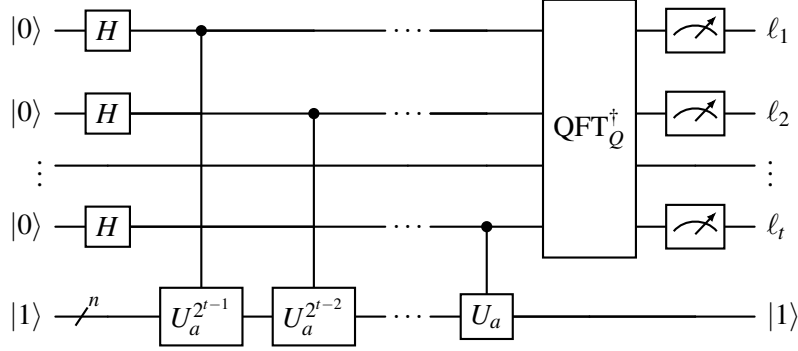
10.2 The order-finding circuit

Algorithm 10.2.1 (Quantum order finding). Input N with $n = \lceil \log_2 N \rceil$, and a with $\gcd(a, N) = 1$. Fix $\varepsilon \in (0, \frac{1}{2})$ and set

$$t = 2n + 1 + \left\lceil \log_2 \left(2 + \frac{1}{2\varepsilon} \right) \right\rceil, \quad Q = 2^t. \quad (10.3)$$

(This is the high-confidence choice; the minimal choice $t = \lceil 2 \log_2 N \rceil$ also works, at a lower per-run success probability. See [Section 11.1](#).)

1. Prepare $|0\rangle^{\otimes t} \otimes |1\rangle_n$ and apply $H^{\otimes t}$ to the first register.
2. Apply the ladder $c-U_a^{2^{t-s}}$, $s = 1, \dots, t$ — i.e. modular exponentiation.
3. Apply $\text{QFT}_Q^\dagger$ to the first register and measure it, obtaining ℓ .
4. Run the continued-fraction algorithm on ℓ/Q to obtain s'/r' in lowest terms with $r' < N$.
5. If $a^{r'} \equiv 1 \pmod{N}$, output r' . Otherwise (optionally testing small multiples $2r', 3r', \dots$) return to step 1.



Reading the circuit

The top (most significant) counting wire drives $U_a^{2^{t-1}}$, the bottom drives U_a . Each $U_a^{2^j} = U_{a^{2^j \bmod N}}$ is **one modular multiplication by a classical constant** (Proposition 5.5.2). The work register is a *catalyst*: it returns to $|1\rangle$ and is never measured.

Sizes: $n = \lceil \log_2 N \rceil$ work qubits and $t = 2n + O(1)$ counting qubits, the requirement being $Q \geq N^2$ (Corollary 11.1.1). With the minimal choice $t = \lceil 2 \log_2 N \rceil$ and $N = 21$: $n = 5$ and $441 \leq Q$ forces $Q = 512$, $t = 9$ — the sizes used in Chapter 14.

Proposition 10.2.2 (Repeated squaring). $U_a^{2^\ell} = U_{a_\ell}$ where $a_\ell := a^{2^\ell} \bmod N$, and the constants $a_0, a_1, \dots, a_{t-1}$ are computed classically by $a_0 = a$, $a_{\ell+1} = a_\ell^2 \bmod N$, in t modular squarings. Consequently the ladder implements the modular exponentiation map

$$|x\rangle |1\rangle \mapsto |x\rangle |a^x \bmod N\rangle, \quad x \in \{0, \dots, Q-1\}, \quad (10.4)$$

using t controlled modular multiplications.

Proof. $U_a^p = U_{a^p \bmod N}$ is Proposition 5.5.2; take $p = 2^\ell$. On the basis state $|x\rangle = |x_1 \dots x_t\rangle$ the ladder applies $\prod_s U_a^{x_s 2^{t-s}} = U_a^x$ to the second register, and $U_a^x |1\rangle = |a^x \bmod N\rangle$. $\square$

This circuit and the kickback ladder are the same circuit

The ladder of controlled powers $c-U^{2^\ell}$ that phase estimation calls for is the modular exponentiation that Shor's original 1994 formulation calls for. Each box is one in-place modular multiplication by a classically precomputed constant, at $\Theta(n^2)$ Toffoli gates (Chapter 12).

10.3 The state before the inverse QFT, in two bases

Theorem 10.3.1 (Two descriptions of one state). *After the controlled modular multiplications and before the inverse QFT, the joint state is*

$$\underbrace{\frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} |x\rangle \otimes |a^x \bmod N\rangle}_{\text{periodic-function form}} = \underbrace{\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} \left(\frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} e^{2\pi i x s / r} |x\rangle \right) \otimes |u_s\rangle}_{\text{eigenphase form}}. \quad (10.5)$$

Proof. Inverting the eigenvector formula with Equation (10.2) gives $|a^x \bmod N\rangle = \frac{1}{\sqrt{r}} \sum_s e^{2\pi i s x / r} |u_s\rangle$. Substituting into the left-hand side and exchanging the sums gives the right-hand side. $\square$

Why this matters

In the *orbit* basis the work register holds the periodic sequence $a^x \bmod N$, whose period the inverse QFT will detect. In the *character* basis it holds one eigenphase branch $|u_s\rangle$, each tagged with a ramp of frequency s/r .

Because it is literally the same vector, both descriptions give *identical* counting-register statistics. Whether the work register is measured (period finding) or ignored (phase estimation) changes the description, not the mechanism. Shor's 1994 presentation and the modern phase-estimation presentation are the same circuit.

10.4 The state through the circuit

Stage	Joint state (counting $\otimes$ work)	Basis	Ent.?	Where r lives	Measure now $\rightarrow$
1. initial	$ 0\rangle^{\otimes t} \otimes 1\rangle$	comp.	no	nowhere yet	0; work $ 1\rangle$
2. after $H^{\otimes t}$	$\frac{1}{\sqrt{Q}} \sum_x x\rangle \otimes 1\rangle$	comp.	no	nowhere	uniform x
3. after the $c-U_a$ ladder	$\frac{1}{\sqrt{Q}} \sum_x x\rangle a^x \bmod N\rangle$	orbit	yes	period of a^x	correlated pair
4. same state, eigenbasis	$\frac{1}{\sqrt{r}} \sum_s \left(\frac{1}{\sqrt{Q}} \sum_x e^{2\pi i s x / r} x\rangle \right) u_s\rangle$	character	yes	denominators of s/r	same distribution
5. ladder reading	each branch: ramp $e^{2\pi i s x / r}$, uniform $ \cdot $	character	yes	ramp frequency s/r	still uniform
6. after QFT †	$\frac{1}{\sqrt{r}} \sum_s \sum_y A_s(y) y\rangle u_s\rangle$	comp./char.	yes	peaks at $y \approx Qs/r$	$\ell \approx Qs/r$
7. measure counting	$ \ell\rangle \otimes u_s\rangle$ (one branch)	comp./char.	no	in $\ell/Q \approx s/r$	ℓ obtained

Reading the table

Entanglement appears at **stage 3**, when the oracle correlates the registers, and disappears at stage 7 only because measurement projects onto one branch.

r is *never* in a single amplitude. It is a *global* property — the period of the sequence, equivalently the common denominator of the eigenphases, equivalently the spacing of the peaks. That is exactly why one measurement gives a *rational approximation* and needs continued fractions to be promoted to an exact fraction.

10.5 Correctness and cost

Theorem 10.5.1 (Correctness and cost of Algorithm 10.2.1). *Each iteration of Algorithm 10.2.1 outputs $r = \text{ord}_N(a)$ with probability at least*

$$(1 - \varepsilon) \cdot \frac{\phi(r)}{r} = \Omega\left(\frac{1}{\log \log N}\right), \quad (10.6)$$

so $O(\log \log N)$ iterations suffice in expectation — or $O(1)$ using the two-run lcm variant of Proposition 1.5.2. Each iteration uses $t + n = O(n)$ qubits and $O(n^3)$ gates, dominated by the modular exponentiation. The output is always verified in step 5, so the algorithm is **Las Vegas**: it never returns a wrong answer.

Proof. By Theorem 10.1.1 the second register is $|1\rangle = \frac{1}{\sqrt{r}} \sum_s |u_s\rangle$, and by Theorem 8.5.1 the measurement outcome is distributed as if a uniformly random $s \in \{0, \dots, r-1\}$ had been chosen and QPE run on $|u_s\rangle$, whose eigenphase is $\varphi_s = s/r$ by Theorem 5.5.4. With the choice Equation (10.3), Theorem 8.4.1 (applied with “ n ” = $2n+1$) guarantees $|\ell/Q - s/r| < 2^{-(2n+1)}$ with probability $\geq 1 - \varepsilon$, which is exactly the hypothesis of Corollary 11.1.1, so step 4 recovers s/r in lowest terms. If in addition $\gcd(s, r) = 1$

— probability $\phi(r)/r$ by [Proposition 1.5.1](#), independent of the QPE outcome — then $r' = r$ and step 5 accepts.

Costs: $t = O(n)$ counting qubits and n target qubits; $O(n^3)$ gates for step 2 ([Theorem 12.7.1](#)), $O(t^2)$ or $O(t \log t)$ for step 3 ([Theorems 6.3.3](#) and [6.4.2](#)), and $\text{poly}(n)$ classical post-processing. Step 5 is an explicit verification, whence the Las Vegas property. $\square$

Corollary 10.5.2 (Shor’s theorem). *There is a randomised quantum algorithm that outputs a non-trivial factor of a composite n -bit integer N using $O(n)$ qubits and $\tilde{O}(n^3)$ gates, with success probability $\geq \frac{2}{3}$; the answer is verifiable classically.*

Proof. Combine [Theorem 9.3.2](#) (expected 2 bases a) with [Theorem 10.5.1](#) (expected $O(\log \log N)$ order-finding runs per base, or $O(1)$ with the lcm variant), and amplify by repetition. $\square$

Remark 10.5.3 (Why this step, and not the QFT, is the bottleneck). The QFT costs $O(t^2)$, or $O(t \log t)$ approximately; the modular exponentiation costs $O(n^3)$. *Shor’s algorithm is dominated by arithmetic, not by Fourier analysis.* The Fourier transform contributes the idea; the modular exponentiation contributes the cost. [Chapter 12](#) makes this precise, and [Chapter 18](#) quantifies it: at $n = 2048$ the ratio is about 3×10^4 to one.

10.6 The original formulation, and why it is the same circuit

Shor’s 1994 presentation did not mention phase estimation. It is worth seeing that the two descriptions produce the same circuit, because each explains something the other hides.

Algorithm 10.6.1 (Period finding, original form). 1. Prepare $\frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} |x\rangle \otimes |0\rangle_n$.
 2. Compute $|x\rangle |0\rangle \mapsto |x\rangle |a^x \bmod N\rangle$.
 3. (Optionally) measure the second register.
 4. Apply $\text{QFT}_Q^\dagger$ to the first register and measure it.

Proposition 10.6.2 (Equivalence). *Algorithm 10.6.1 and Algorithm 10.2.1 are the same circuit and have the same output distribution. Step 3 is optional: measuring the second register, or not, does not change the counting-register statistics.*

Proof. Steps 1–2 of [Algorithm 10.6.1](#) are steps 1–2 of [Algorithm 10.2.1](#): by [Proposition 10.2.2](#) the ladder of controlled powers is the modular exponentiation map applied to $|1\rangle$ (up to the trivial relabelling $|0\rangle \rightarrow |1\rangle$ of the work register’s initial state, which is one X gate). [Theorem 10.3.1](#) identifies the two descriptions of the resulting state.

For the optionality of step 3: if the second register is measured, obtaining some value a^{k_0} , the first register collapses (by [Proposition 3.6.1](#)) to the uniform superposition over $\{x : x \equiv k_0 \bmod r\}$, whose $\text{QFT}^\dagger$ concentrates on multiples of Q/r . If it is not measured, [Theorem 10.3.1](#) and [Theorem 8.5.1](#) give the same distribution. Formally, the measurement commutes with $\text{QFT}^\dagger \otimes \mathbb{I}$, which acts on the other factor. $\square$

What each formulation makes obvious

Period finding makes obvious *why the modular exponentiation is there*: it creates a periodic sequence in the work register.

Phase estimation makes obvious *why the inverse QFT is there and why $t = 2n + O(1)$* : it is measuring an eigenphase to $2n + 1$ bits of precision.

Neither is more correct. Holding both is what makes the algorithm feel inevitable rather than clever.

Chapter 11

Post-processing: from ℓ to r

Phase estimation returns $\tilde{\phi} = \ell/Q$, a number *close to* s/r — not s/r itself. This chapter supplies the classical step that recovers the exact rational, explains why the counting register must have $2n + O(1)$ qubits rather than $n + O(1)$, and quantifies precisely what a single run yields.

11.1 Legendre's theorem and the register-size rule

The Fejér peak of [Theorem 8.3.2](#) concentrates ℓ on the grid point nearest $Q\phi$, so $|\ell/Q - s/r| \leq \frac{1}{2Q}$. The question is how small that must be for s/r to be uniquely recoverable. [Theorem 1.4.3](#) answers it: the threshold is $1/(2q^2)$ where q is the denominator sought.

Corollary 11.1.1 (Why $t = 2n + O(1)$). *Suppose $r < N$ and phase estimation returns ℓ with*

$$\left| \frac{\ell}{Q} - \frac{s}{r} \right| < \frac{1}{2N^2} \quad (\text{as a distance modulo } 1).$$

Then running continued fractions on ℓ/Q and keeping the last convergent with denominator $< N$ returns s'/r' : the fraction s/r written in lowest terms. In particular $r' \mid r$, and $r' = r$ whenever $\gcd(s, r) = 1$.

In particular $Q \geq N^2$ suffices, i.e.

$$t \geq 2n, \quad \text{so} \quad t = 2n + O(1). \quad (11.1)$$

Proof. Write $s/r = s'/r'$ in lowest terms, so $r' \leq r < N$. Then

$$\left| \frac{\ell}{Q} - \frac{s'}{r'} \right| < \frac{1}{2N^2} \leq \frac{1}{2r'^2},$$

so [Theorem 1.4.3](#) applies with $x = \ell/Q$ and $p/q = s'/r'$: the fraction s'/r' is one of the convergents, say $s'/r' = p_k/q_k$.

It remains to check that the *last* convergent with denominator $< N$ is this one, i.e. that no later convergent also has denominator $< N$. Suppose $q_{k+1} < N$ as well. The standard approximation bound for convergents,

$$\left| x - \frac{p_k}{q_k} \right| \geq \frac{1}{q_k(q_{k+1} + q_k)},$$

would then give $|x - s'/r'| > 1/(2N^2)$, contradicting the hypothesis. Hence $q_{k+1} \geq N$ and p_k/q_k is the last admissible convergent, so the algorithm returns exactly (s', r') . Finally $r' = r/\gcd(s, r)$ divides r , with equality iff $\gcd(s, r) = 1$. $\square$

Two legitimate choices of t , and what each buys

The bound just proved is a statement about *precision*: it needs $|\ell/Q - s/r| < 1/(2N^2)$. There are two standard ways to guarantee that, and they are different design points — a source of genuine confusion in the literature.

$$t = \lceil 2 \log_2 N \rceil, \text{ i.e. } N^2 \leq Q < 4N^2$$

The *minimal* choice. The nearest grid point to Qs/r is then within $1/(2Q) \leq 1/(2N^2)$ of it, and [Theorem 8.3.4](#) gives that grid point with probability $\geq 4/\pi^2$ per branch. This is the choice used in the worked example of [Chapter 14](#) ($N = 21$: $Q = 512$, $t = 9$), because it is the one whose distribution is small enough to display.

$$t = 2n + 1 + \lceil \log_2(2 + \frac{1}{2\varepsilon}) \rceil$$

The *high-confidence* choice, [Equation \(10.3\)](#). Here [Theorem 8.4.1](#), applied with accuracy $2^{-(2n+1)}$, gives $|\ell/Q - s/r| < 2^{-(2n+1)} \leq 1/(2N^2)$ with probability $\geq 1 - \varepsilon$ — not merely $4/\pi^2$. The extra $1 + \lceil \log_2(2 + \frac{1}{2\varepsilon}) \rceil$ qubits buy confidence, exponentially cheaply.

Both give $t = 2n + O(1)$ and both make [Corollary 11.1.1](#) apply; they differ only in the per-run success probability, and hence in the expected number of repetitions. [Algorithm 10.2.1](#) states the second because it yields the cleaner correctness theorem.

Answering “why $Q \approx N^2$ and not $Q \approx N$?”

The denominator being sought can be as large as $r < N$, and Legendre’s threshold scales as the *square* of the denominator. So the required precision is better than $1/(2N^2)$, and $Q \geq N^2$ follows. This one inequality is the entire reason the counting register is twice the size of the work register — and hence the reason the circuit performs $2n$ rather than n modular multiplications.

It is also, incidentally, where a real saving is available: the Ekerå–Håstad variant reduces the number of phase-estimation rounds from $2n$ to $n/2$ at the cost of a lattice-enumeration post-processing step, dividing every Toffoli count by 4 without saving qubits ([Section 18.4](#)).

11.2 The recovery procedure

Algorithm 11.2.1 (Recovering the order). Given a measured ℓ :

1. Form ℓ/Q and expand it as a continued fraction. If $\ell = 0$, abort: this outcome carries no information.
2. List the convergents p_k/q_k ([Proposition 1.4.2](#)).
3. Keep those with $0 < q_k < N$.
4. Let $r' := q_k$ for the last such convergent, and test $a^{r'} \bmod N$.
5. If $a^{r'} \equiv 1$, accept r' .
6. Otherwise r' is only a *divisor* of r ; test small multiples $2r', 3r', \dots$ up to $\lfloor N/r' \rfloor$, accepting the first kr' with $a^{kr'} \equiv 1$.
7. Alternatively, combine two runs: $\text{lcm}(r'_1, r'_2)$ ([Proposition 1.5.2](#)).
8. **Verify before extracting factors.**

```
def convergents(num, den):
    cf, n, d = [], num, den
    while d:
        cf.append(n // d)
        n, d = d, n - (n // d) * d
    h1, h2, k1, k2 = 1, 0, 0, 1
    out = []
    for ai in cf:
        h, k = ai*h1 + h2, ai*k1 + k2
        out.append((h, k))
        h2, h1, k2, k1 = h1, h, k1, k
    return out      # (numerator, denominator) pairs

def recover_order(ell, Q, N, a):
    if ell == 0: return None      # carries no information
    cands = [k for (h, k) in convergents(ell, Q) if 0 < k < N]
```

```

if not cands: return None
rp = cands[-1] # last convergent with denominator < N
for m in range(1, N // rp + 1): # small multiples
    if pow(a, m*rp, N) == 1: return m*rp
return None

```

11.3 What is actually recovered

Proposition 11.3.1 (The recovered denominator is a divisor of r). *The machine samples s uniformly in $\{0, \dots, r-1\}$, and the convergent is returned in lowest terms. What is recovered is therefore*

$$r' = \frac{r}{\gcd(s, r)}, \quad (11.2)$$

a divisor of r . The value r itself is obtained outright exactly when $\gcd(s, r) = 1$, with probability $\phi(r)/r = \Omega(1/\log \log r)$. For $r = 4$ this is $\frac{1}{2}$; for $r = 6$ it is $\frac{1}{3}$.

Proof. Immediate from [Corollary 11.1.1](#) and [Proposition 1.5.1](#). □

This is not a failure mode

All other outcomes still yield a *divisor*, repaired classically by testing small multiples or taking the lcm of two runs. And because step 8 of [Algorithm 11.2.1](#) *verifies*, a wrong r is never used: the algorithm is Las Vegas. It may take a few runs, but it never returns a wrong answer. That is why a high per-run success probability is not required, and why the constant-factor overhead does not affect the asymptotic claim.

Why the denominator is never simply trusted

The measured ℓ carries s/r in *lowest terms*; the reduction by $\gcd(s, r)$ is *invisible* from ℓ alone. There is no way to tell, from the measurement, whether the convergent you obtained is r or a proper divisor of it. The classical verification $a^{r'} \stackrel{?}{=} 1$ is what makes the whole algorithm never return a wrong answer.

11.4 Two related questions, answered precisely

Why continued fractions rather than rounding?

Continued fractions *are* the optimal way to do that rounding, and Legendre’s theorem ([Theorem 1.4.3](#)) is what guarantees the answer appears among the convergents. Ad-hoc rounding to “the nearest small fraction” has no uniqueness guarantee. And the convergents are computable in $O(\log^3 q)$ bit operations ([Algorithm 1.4.5](#)).

What if r is odd, or $a^{r/2} \equiv -1$?

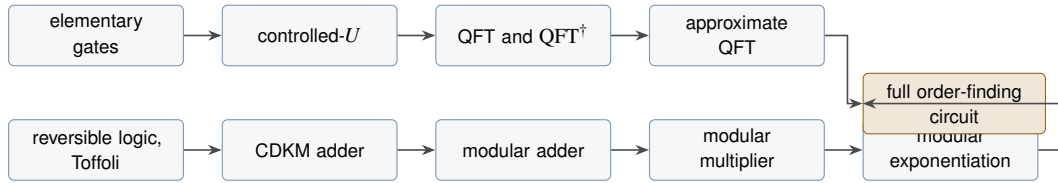
That base yields nothing; draw a new a . For odd composite N with $m \geq 2$ distinct prime factors the good-base probability is at least $1 - 2^{-(m-1)} \geq \frac{1}{2}$ ([Theorem 9.2.1](#)). This is a *different* failure mode from a useless measurement outcome, and the two must not be conflated.

Remark 11.4.1 (Cost of the classical stage). The whole classical post-processing — continued fractions, verification, gcd — is $O(n^3)$ bit operations, negligible beside the quantum circuit and negligible beside the number field sieve.

Chapter 12

Reversible modular arithmetic

This chapter discharges the debt incurred everywhere above. The statement “ $\Theta(n^3)$ gates, see [refs]” is replaced by an explicit construction and an explicit count. The conclusion is worth stating first, because it reorients one’s whole intuition: **Shor’s algorithm is 99.997% arithmetic.** The quantum Fourier transform, to which most expositions (including Chapters 6 and 8 of this one) devote the bulk of their pages, is a rounding error in the gate count.



A note on bit indexing in this chapter

Convention (C3) fixes the *register* convention: big-endian, $x = \sum_s x_s 2^{t-s}$, with x_1 on the top wire. That convention is about which *wire* carries which *weight*, and it is unchanged here.

Inside arithmetic circuits, however, it is far more natural to index bits by their *weight*, writing $x = \sum_i x_i 2^i$ so that x_0 is the least significant bit and carries propagate from x_0 upwards. This chapter and Chapter 17 do that. The two indexings are related by $x_i^{(\text{this chapter})} = x_{n-i}^{(\text{C3})}$; the physical wire order is identical, only the subscript is renumbered. Nothing about the circuits changes — but when transcribing a formula from Chapter 6 or Chapter 8 into an arithmetic block, check which indexing it uses.

Where the gates go

The Fourier transform is $\Theta(n^2)$ exact, or $O(n \log n)$ approximate. The modular exponentiation is $\Theta(n^3)$. The QFT is asymptotically the *cheapest* part of Shor’s algorithm; roughly 99.997% of the gates are arithmetic. This chapter is therefore the bulk of the engineering.

12.1 Addition: the carries are the cost

Schoolbook binary addition of a and b :

$$s_i = a_i \oplus b_i \oplus c_i, \quad c_{i+1} = \text{MAJ}(a_i, b_i, c_i) := a_i b_i \oplus a_i c_i \oplus b_i c_i, \quad c_0 = 0. \quad (12.1)$$

The sum bits are $\mathbb{F}_2$ -linear given the carries, so they are free (CNOTs). The carries are *quadratic*, so each costs at least one Toffoli-like gate. And the carries form a chain of length n , so the naive depth is $\Theta(n)$.

Definition 12.1.1 (The MAJ and UMA blocks). On an ordered triple of wires (c, b, a) define, in this order:

$$\begin{aligned} \text{MAJ} : & \quad \text{CNOT}(a \rightarrow b), \quad \text{CNOT}(a \rightarrow c), \quad \text{TOF}(c, b \rightarrow a); \\ \text{UMA} : & \quad \text{TOF}(c, b \rightarrow a), \quad \text{CNOT}(a \rightarrow c), \quad \text{CNOT}(c \rightarrow b). \end{aligned}$$

Each block contains *exactly one* Toffoli gate.

Lemma 12.1.2 (What the blocks do).

$$\text{MAJ} : (c, b, a) \mapsto (c \oplus a, b \oplus a, \text{MAJ}(a, b, c)), \quad (12.2)$$

$$\text{UMA} : (c \oplus a, b \oplus a, \text{MAJ}(a, b, c)) \mapsto (c, a \oplus b \oplus c, a). \quad (12.3)$$

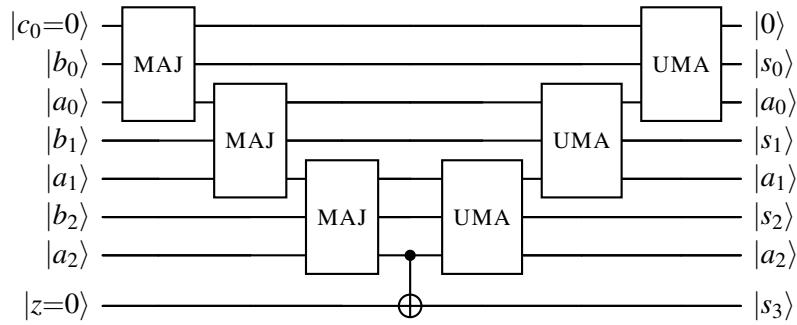
Proof. For MAJ, track the three assignments: $b \mapsto b \oplus a$, then $c \mapsto c \oplus a$, then

$$a \mapsto a \oplus (b \oplus a)(c \oplus a) = a \oplus (bc \oplus ab \oplus ac \oplus a) = ab \oplus ac \oplus bc = \text{MAJ}(a, b, c),$$

using $a \cdot a = a$ and $a \oplus a = 0$ over $\mathbb{F}_2$. For UMA, feed in $(C, B, A) = (c \oplus a, b \oplus a, \text{MAJ}(a, b, c))$: the Toffoli gives $A \oplus BC = \text{MAJ}(a, b, c) \oplus (b \oplus a)(c \oplus a) = a$ by the same computation read backwards; then $C \mapsto C \oplus a = c$; then $B \mapsto B \oplus c = a \oplus b \oplus c$. $\square$

Remark 12.1.3. The only step that is easy to get wrong is $a \oplus (b \oplus a)(c \oplus a) = ab \oplus ac \oplus bc$, where the two occurrences of a cancel because $a^2 = a$ and $a \oplus a = 0$. Everything in the CDKM adder follows from this single line, which is why the exhaustive check for $n = 6$ in [Appendix D](#) is worth quoting: it exercises the carry chain, the top-carry copy and the ancilla restoration simultaneously.

12.1.1 The CDKM ripple-carry adder



The CDKM ripple-carry adder for $n = 3$. Wires are interleaved $c_0, b_0, a_0, b_1, a_1, \dots, b_{n-1}, a_{n-1}, z$ so every block acts on three adjacent wires. The MAJ chain computes all carries; the single CNOT copies out the top carry; the UMA chain simultaneously uncomputes the carries and writes the sum bits — Bennett’s compute-copy-uncompute with the uncomputation doing useful work on the way back.

Theorem 12.1.4 (CDKM adder). *The circuit above, generalised to n bits, implements*

$$|c_0=0\rangle |b\rangle |a\rangle |z=0\rangle \mapsto |0\rangle |(a+b) \bmod 2^n\rangle |a\rangle |s_n\rangle,$$

an in-place $(n+1)$ -bit sum with the operand a preserved and no garbage. Its costs are

qubits	$2n + 2$	T -count (7 T per TOF)	$14n$
Toffoli gates	$2n$	T -count with temporary-AND	$4n$
CNOTs	$4n + 1$	T -depth	$6n$
garbage	0	KQ_T	$6n(2n + 2) = 12n^2 + 12n$

Proof. Correctness. Set $c_0 = 0$. By [Lemma 12.1.2](#), after the i -th MAJ block the a_i -wire holds $\text{MAJ}(a_i, b_i, c_i) = c_{i+1}$, so the carry propagates up the a -wires and after n blocks the top a -wire holds c_n ; the CNOT copies it to z , and $s_n = c_n$. Running the UMA blocks in reverse order, the i -th receives exactly the triple that MAJ_i produced, hence restores c_i on the carry wire and a_i on the a -wire, and writes $a_i \oplus b_i \oplus c_i = s_i$ on the b -wire, which is [Equation \(12.1\)](#). Since the UMA blocks run in reverse, each has its carry input already restored by the block above it. Finally c_0 is restored to 0.

Costs. There are n MAJ and n UMA blocks, each with one Toffoli and two CNOTs, plus one extra CNOT: $2n$ Toffolis and $4n + 1$ CNOTs. The blocks lie on a chain, so T -depths add: $2n \times 3 = 6n$. Qubits: $2n$ operand wires plus c_0 and z . Garbage: by [Definition 4.1.2](#), $(2n + 2) - (n + 1) - n - 1 = 0$.

The $4n$ figure. Each carry is computed by a MAJ block and uncomputed by the matching UMA block, so almost every Toffoli here is half of a compute/uncompute pair; by [Theorem 4.4.6](#) such a pair costs 4 T , not $2 \times 7 = 14$. $\square$

Remark 12.1.5 (Variants). Dropping the z wire gives an n -bit modulo- 2^n adder on $2n + 1$ qubits. Freeing c_0 gives an adder with carry-in. *Running the circuit backwards subtracts*, at the same cost; combined with two's complement this yields $|a\rangle |b\rangle \mapsto |a\rangle |b - a\rangle$. A *constant* addend deletes or fixes roughly half the Toffolis at compile time.

Remark 12.1.6 (Carry-lookahead: trading qubits for depth). The ripple-carry adder has depth $\Theta(n)$ because Equation (12.1) is a chain. Classically one breaks the chain with the generate/propagate algebra $g_i = a_i b_i$, $p_i = a_i \oplus b_i$, $c_{i+1} = g_i \oplus p_i c_i$, and observes that $(g, p) \circ (g', p') = (g \oplus p g', p p')$ is *associative*, so all carries follow from a parallel prefix computation of depth $O(\log n)$. The quantum carry-lookahead adder achieves depth $O(\log n)$, $O(n)$ Toffolis and $O(n)$ ancillas. Its Toffoli count is a constant factor worse than CDKM (roughly $4n$ versus $2n$); for Shor at cryptographic sizes the modular exponentiation is intrinsically sequential anyway, so the depth advantage is largely wasted and ripple-carry wins on volume.

12.2 Subtraction, comparison, and controlled addition

Proposition 12.2.1 (Comparator). $|a\rangle |b\rangle |0\rangle \mapsto |a\rangle |b\rangle |[a \geq b]\rangle$ using n Toffolis, $4n$ T gates, and no garbage: run the MAJ chain, copy the final carry to the output qubit with one CNOT, and run the MAJ chain backwards.

Proof. The MAJ chain leaves c_n on the top a -wire, and c_n is the carry-out of $a + \neg b + 1 = a - b + 2^n$, which is 1 iff $a - b \geq 0$, i.e. iff $a \geq b$. (Dropping the $+1$, i.e. computing $a + \neg b$, gives $[a > b]$ instead.) Since the chain is computed then uncomputed, each of its n Toffolis is half of a temporary-AND pair, so the T -count is $4n$. $\square$

Proposition 12.2.2 (Controlled adder). $|k\rangle |a\rangle |b\rangle \mapsto |k\rangle |a\rangle |b + ka\rangle$, $k \in \{0, 1\}$: control only the UMA half, giving $2n$ Toffolis for the carries plus n for the controlled sum bits. Note that the carry chain itself need not be controlled — it is computed and uncomputed either way. This is an instance of Lemma 13.1.1.

Constant operands halve the count

Every addition inside Shor's algorithm has a *compile-time classical operand*. A CNOT controlled on a constant bit $a_i = 0$ is deleted; one controlled on $a_i = 1$ becomes an X ; a Toffoli with a constant control collapses to a CNOT or vanishes. Roughly half the gates disappear before the circuit is emitted. *This is not a special case but the normal one.* Report gate counts after this folding, and always including uncomputation.

12.3 Modular addition

Fix an n -bit modulus $N < 2^n$ and use registers of $n + 1$ qubits, so intermediate values up to $2N$ fit.

Theorem 12.3.1 (Modular adder with a classical constant $a < N$). The following seven-step circuit implements $|x\rangle \mapsto |(x + a) \bmod N\rangle$ for all $0 \leq x < N$, using one ancilla which is returned to $|0\rangle$, and five additions of classical constants plus $O(1)$ Clifford gates:

Operation	Effect
1 ADD(a)	$ x\rangle \mapsto x + a\rangle$, with $x + a < 2N$
2 SUB(N)	$\mapsto x + a - N\rangle$, possibly negative
3 CNOT from the sign qubit to an ancilla t	$t = [x + a < N]$
4 ADD(N) controlled on t	$\mapsto (x + a) \bmod N\rangle$
5 SUB(a)	$\mapsto (x + a) \bmod N - a\rangle$
6 X on t , then CNOT from the sign qubit to t	$t \mapsto 0$
7 ADD(a)	$\mapsto (x + a) \bmod N\rangle$

Proof. Steps 1–2 give $x + a - N$, negative exactly when $x + a < N$; in two’s complement the sign qubit is $[x + a < N]$, so step 3 records it without disturbing the value, and step 4 restores $R := (x + a) \bmod N \in [0, N)$.

The step worth checking is clearing the ancilla. Now

$$t = 1 \iff x + a < N \iff R = x + a \geq a, \quad t = 0 \iff R = x + a - N < a,$$

the last because $x < N$. Hence $t = [R \geq a]$, *recomputable from the register alone*. Step 5 forms $R - a$, whose sign qubit is $[R < a] = \neg t$; applying X to t makes it $\neg t$, and the CNOT then sets $t \mapsto \neg t \oplus \neg t = 0$. Step 7 restores R . $\square$

The recurring pattern

Every block ends by recomputing its own flag from its output. That, and nothing else, is what makes the whole tower garbage-free — and by [Proposition 3.10.2](#), garbage-freeness is not optional. The same pattern reappears in the modular doubler below, and in the λ -uncomputation of the elliptic-curve point addition ([Algorithm 17.4.1](#)).

Remark 12.3.2 (Modular doubling, p odd). Two changes make it cheaper than the adder. It works in place on one n -qubit register, so it needs only $n + 2$ qubits; and the first integer addition is replaced by a multiplication by 2, implemented as a cyclic bit shift (a chain of swaps built from CNOTs). Because p is odd, the reduction flag is recoverable from the *least significant bit* of the result: $2x - p$ is odd while $2x$ is even. So a subtraction of the modulus has taken place if and only if that bit is 1, and no persistent ancilla is needed.

12.4 Modular multiplication and exponentiation

Theorem 12.4.1 (Out-of-place multiplication by a constant). *Let $\gcd(a, N) = 1$. The circuit*

$$\prod_{i=0}^{n-1} \left(\text{ADD mod } N \left(2^i a \bmod N \right) \text{ controlled on } x_i \right)$$

implements $|x\rangle |0\rangle \mapsto |x\rangle |ax \bmod N\rangle$ using n controlled modular additions, hence $\Theta(n^2)$ Toffoli gates.

Proof. $ax \bmod N = \left(\sum_i x_i (2^i a \bmod N) \right) \bmod N$, and the constants $2^i a \bmod N$ are computed classically at compile time. Each term is added exactly when $x_i = 1$. All partial sums are reduced mod N , so the accumulator never exceeds N . $\square$

Theorem 12.4.2 (In-place multiplication). *With $\bar{a} := a^{-1} \bmod N$ computed classically by the extended Euclidean algorithm,*

$$|x\rangle |0\rangle \xrightarrow{\text{out-of-place with } a} |x\rangle |ax\rangle \xrightarrow{n \text{ swaps}} |ax\rangle |x\rangle \xrightarrow{\text{inverse of out-of-place with } \bar{a}} |ax\rangle |0\rangle.$$

So $U_a : |x\rangle \mapsto |ax \bmod N\rangle$ costs two out-of-place modular multiplications, n swaps, and one n -qubit ancilla register.

Proof. The third step is the inverse of $|y\rangle |0\rangle \mapsto |y\rangle |\bar{a}y\rangle$ applied with $y = ax$: it maps $|ax\rangle |z\rangle \mapsto |ax\rangle |z - \bar{a}(ax)\rangle$, and $\bar{a}(ax) \equiv x$, so $|x\rangle \mapsto |0\rangle$. $\square$

Invertibility is exactly what makes an in-place circuit exist

Theorem 12.4.2 needs $a^{-1} \bmod N$, which exists iff $\gcd(a, N) = 1$. In Shor’s algorithm that is checked classically before the quantum part runs — and if it fails, you have already factored N . So the condition is free, but it is not vacuous: it is the precise reason an in-place modular *multiplier* exists, whereas an in-place modular *squarer* does not.

Theorem 12.4.3 (Modular exponentiation). *Let $\gcd(a, N) = 1$ and let t be the exponent-register size. Define the classical constants $a_\ell := a^{2^\ell} \bmod N$ for $\ell = 0, \dots, t-1$, computed by t classical modular squarings. Then*

$$\prod_{\ell=0}^{t-1} \left(U_{a_\ell} \text{ controlled on exponent qubit } \ell \right) : |k\rangle |1\rangle \mapsto |k\rangle |a^k \bmod N\rangle$$

for every $k \in \{0, \dots, 2^t - 1\}$, where U_{a_ℓ} is the in-place modular multiplier.

Proof. Writing $k = \sum_\ell k_\ell 2^\ell$, the circuit applies $\prod_\ell U_{a_\ell}^{k_\ell} = U_{a^k \bmod N}$, since $a \mapsto U_a$ is a group homomorphism (Proposition 5.5.2) and both sides are linear. Then $U_a^k |1\rangle = |a^k \bmod N\rangle$. $\square$

The schoolbook cost

$$\underbrace{\Theta(n)}_{\text{multiplications}} \times \underbrace{\Theta(n)}_{\text{modular additions each}} \times \underbrace{\Theta(n)}_{\text{Toffolis each}} = \Theta(n^3) \text{ Toffoli gates on } \Theta(n) \text{ qubits.}$$

The constant hidden in Θ is what the rest of this chapter attacks. It ranges from ≈ 600 for a minimal-width construction down to ≈ 0.3 for a windowed one — a spread of more than three orders of magnitude at fixed asymptotics. See Chapter 18 for the comparison.

12.5 The Fourier (Draper) adder, and why it loses

The identity $\text{QFT}_N^\dagger \text{diag}(\omega_N^{ak}) \text{QFT}_N = T_a$ of Theorem 6.1.4, read left to right, is an algorithm: to add, Fourier transform, apply phases, transform back.

Theorem 12.5.1 (Draper adder). *With $\Phi\text{ADD}(a) := \bigotimes_{j=1}^n P(2\pi a/2^j)$ acting on qubit j (MSB first),*

$$\text{QFT}_{2^n}^\dagger \Phi\text{ADD}(a) \text{QFT}_{2^n} |b\rangle = |(a+b) \bmod 2^n\rangle. \quad (12.4)$$

The circuit uses no ancilla qubits and, for classical a , exactly n single-qubit phase gates between the two transforms.

Proof. $\text{QFT} |b\rangle = 2^{-n/2} \sum_k \omega^{bk} |k\rangle$ with $\omega = e^{2\pi i/2^n}$. Now $\Phi\text{ADD}(a)$ multiplies $|k\rangle = |k_1 \dots k_n\rangle$ by

$$\prod_j e^{2\pi i a k_j / 2^j} = \exp\left(2\pi i a \sum_j k_j 2^{-j}\right) = e^{2\pi i a k / 2^n} = \omega^{ak},$$

using $k/2^n = \sum_j k_j 2^{-j}$. Hence $\Phi\text{ADD}(a) \text{QFT} |b\rangle = 2^{-n/2} \sum_k \omega^{(a+b)k} |k\rangle = \text{QFT} |a+b\rangle$, and applying $\text{QFT}^\dagger$ gives the claim. $\square$

n -bit adder	ancillas	$n = 32$	$n = 256$	$n = 2048$
CDKM ripple-carry, $7T$ per TOF ($14n$)	2	448	3584	28672
CDKM with temporary-AND ($4n$)	n	128	1024	8192
Draper, classical addend ($3n \log_2 \frac{1}{\epsilon}$)	0	3830	30600	245000
Draper, quantum addend ($3n^2 \log_2 \frac{1}{\epsilon}$)	0	126000	7.9×10^6	5.0×10^8

T-count of n -bit addition, with rotation synthesis at $\epsilon = 10^{-12}$ per rotation.

The moral of the table

The Draper adder is the direct algorithmic reading of the identity that makes the QFT interesting in the first place, and it is *the wrong circuit to build*. The reason is entirely a property of the fault-

tolerant cost model of [Section 4.5](#): rotations cost $3 \log_2(1/\epsilon)$ T gates and Toffolis cost 4.

- On a **NISQ** machine, where ancillas are the scarcest resource, the Draper adder can win — it needs zero.
- On a **fault-tolerant** machine ripple-carry wins by $\approx 30\times$.

Historically this reversed the field: Beauregard’s $2n + 3$ -qubit Fourier-based Shor circuit (2003) was the state of the art when qubits were the binding constraint; every modern estimate uses ripple-carry arithmetic instead.

Remark 12.5.2 (Where the Draper adder still pays). If a lives in a second register, each $P(2\pi a/2^j)$ becomes j controlled rotations, giving $n(n+1)/2$ of them. But *adjacent Draper adders share their QFTs, which cancel*: in a long chain of additions the transforms are paid for once. That is the design’s real selling point, and it is why Beauregard’s circuit is built from it.

12.6 Reducing the constant factor: windowed arithmetic

The idea

A modular multiplication is n controlled modular additions, one per bit. If instead c bits are consumed at a time and the appropriate multiple of a is *looked up* from a classically precomputed table of size 2^c , there is one addition per c bits. A lookup costs $\approx 2^c$ Toffolis while an addition costs $\approx n$, so windows of size $c \approx \log_2 n$ are effectively free and the number of additions drops by a factor $\log n$.

Proposition 12.6.1 (Table lookup / QROM by unary iteration). $|j\rangle |0^n\rangle \mapsto |j\rangle |T[j]\rangle$ for $j < 2^c$ costs $2^c - 1$ Toffoli gates and $c - 1$ ancillas: a binary tree of Toffolis computes the indicator of each j in turn, each indicator controlling a layer of CNOTs that xors $T[j]$ into the output. Uncomputing a lookup is cheaper still — measurement-based uncomputation costs $O(2^{c/2})$.

Proposition 12.6.2 (Windowed modular exponentiation). Let c_e be an exponent window and c_m a multiplication window. Precompute classically the table $T[j_e, j_m] = (a^{j_e \cdot 2^{\text{offset}}} \cdot j_m 2^{\text{offset}'}) \bmod N$. Then $|k\rangle |1\rangle \mapsto |k\rangle |a^k \bmod N\rangle$ needs

$$\frac{2n}{c_e} \cdot \frac{n}{c_m} \text{ modular additions} \quad \text{instead of} \quad 2n \cdot n,$$

each preceded by one $2^{c_e+c_m}$ -entry lookup. The windows must satisfy $c_e + c_m \lesssim \log_2 n$ so the lookup cost does not exceed the saved addition cost. At $n = 2048$ that permits windows of about 5 and 5, cutting the addition count by $\approx 25\times$.

Source of the constant	Mechanism	Rough gain
constant-operand folding	compile-time deletion of Toffolis	$\approx 2\times$
temporary logical-AND	$4T$ per pair instead of 14	$\approx 3.5\times$ in T
windowing, $c_e + c_m \lesssim \log_2 n$	fewer additions per multiplication	$\approx 25\times$ at $n=2048$
combined	$\approx 10n^3 \rightarrow \approx 0.3n^3$ Toffoli	$\approx 33\times$

12.7 Resource summary for the order-finding circuit

Theorem 12.7.1 (Cost of Shor’s order-finding circuit). For an n -bit modulus N and $t = 2n + O(1)$ exponent qubits, the quantum part of Shor’s algorithm can be implemented with

logical qubits	$\Theta(n)$
Toffoli gates	$\Theta(n^3)$; concretely $\approx 0.3n^3$ with windowed arithmetic
T gates	$\approx 4 \times$ the Toffoli count
QFT gates	$O(n \log n)$ using the approximate QFT

and the modular exponentiation accounts for essentially all of it.

Shor's algorithm is arithmetic, not Fourier analysis

Set $n = 2048$:

modular exponentiation	$\approx 2.6 \times 10^9$ Toffoli
approximate QFT on $t = 4097$ qubits	$\approx 1 \times 10^5$ gates
ratio	$\approx 3 \times 10^4$

The QFT — the part of the algorithm that carries all of the conceptual content, and to which this document devotes three chapters — is **0.003%** of the circuit. It can even be run semiclassically ([Section 8.6](#)) on a single qubit with classical feedback, at which point its cost is negligible in qubits too.

Everything that determines whether RSA falls is the cost of multiplying two 2048-bit numbers, reversibly, in superposition.

Chapter 13

The controlled- U tower

Phase estimation asks for the ladder $c-U_a^{2^0}, c-U_a^{2^1}, \dots, c-U_a^{2^{t-1}}$. Chapter 12 built U_a ; this chapter builds its *controlled* version — and shows that doing it the obvious way is wrong by an order of magnitude.

The two traps

Trap 1: computing $U_a^{2^\ell}$ by applying U_a 2^ℓ times. That costs $2^\ell - 1$ multiplications and destroys the speed-up entirely. Avoided by Proposition 5.5.2: $U_a^p = U_{a^p \bmod N}$, so the power is absorbed into a classically precomputed constant.

Trap 2: building $c-V$ by controlling every gate of V . That is what `V.control()` does in every circuit framework, and it is wrong by an order of magnitude — measured below at $12\times$ for $c-U_a$. Avoided by Lemma 13.1.1: control only the sub-blocks that must change, and *prove* the rest cancels.

Trap 1 was disposed of in Proposition 5.5.2. So there is only ever *one* circuit to build — $c-U_c$ for a classical constant c — and the ladder instantiates it t times with t different constants. The rest of this chapter is that one circuit.

13.1 The control-propagation principle

Lemma 13.1.1 (Which gates need the control). *Let $V = G_L \cdots G_1$ be a circuit and let $S \subseteq \{1, \dots, L\}$. Write V_S for the circuit obtained by replacing G_j with $c-G_j$ for every $j \in S$ and leaving the others alone. Then V_S is a correct implementation of $c-V$ on the subspace W of admissible inputs if and only if*

$$\underbrace{\prod_{j \notin S}^{\leftarrow} G_j}_{\text{the residue circuit } R_S} = \mathbb{I} \quad \text{on } W. \quad (13.1)$$

Proof. V_S acts on $|0\rangle \otimes |\psi\rangle$ by running every gate of S as the identity and every other gate as itself, i.e. by R_S ; and on $|1\rangle \otimes |\psi\rangle$ by running all of them, i.e. by V . So $V_S = |0\rangle\langle 0| \otimes R_S + |1\rangle\langle 1| \otimes V$, which equals $c-V = |0\rangle\langle 0| \otimes \mathbb{I} + |1\rangle\langle 1| \otimes V$ exactly when $R_S = \mathbb{I}$ on the relevant subspace. $\square$

How to use it

Lemma 13.1.1 turns circuit design into a search for the smallest S whose residue is the identity. Two patterns account for everything below.

- (a) **Conjugation.** If $V = ABA^\dagger$ then $S = \{B\}$ works, since the residue is $AA^\dagger = \mathbb{I}$. *Never control a conjugating layer.* Every QFT...QFT † sandwich in Shor's circuit falls under this, and the QFTs are the bulk of the gates.
- (b) **Compensating residue.** Sometimes the residue is not literally a cancelling pair but still computes the identity *on the admissible subspace*. Theorem 13.4.1 is exactly this: four of the seven blocks of the modular adder conspire to do nothing when the control is off — but *only because* $b < N$ is guaranteed. This is where “on W ” in Equation (13.1) earns its keep, and why the claim needs a proof rather than an inspection.

13.2 Level 4: $c\text{-}U_a$ from two multipliers and a swap

Theorem 13.2.1 (In-place controlled multiplication). *Let $\gcd(a, N) = 1$ and $\bar{a} := a^{-1} \bmod N$, computed classically. Let*

$$\text{CMULT}(a) : |c\rangle |x\rangle |b\rangle \mapsto |c\rangle |x\rangle |b + c \cdot ax \bmod N\rangle$$

be the controlled out-of-place multiplier of Theorem 13.3.1. Then

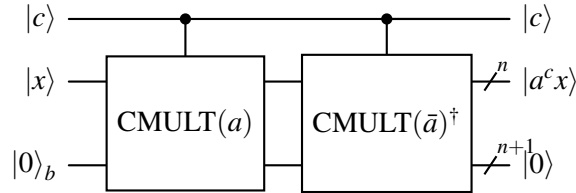
$$c\text{-}U_a = \text{CMULT}(\bar{a})^\dagger \cdot c\text{-SWAP} \cdot \text{CMULT}(a) \quad (13.2)$$

implements $|c\rangle |x\rangle |0\rangle \mapsto |c\rangle |a^c x \bmod N\rangle |0\rangle$ for all $0 \leq x < N$.

Proof. Take $c = 1$ and follow the register pair $(x\text{-reg}, b\text{-reg})$:

$$|x\rangle |0\rangle \xrightarrow{\text{CMULT}(a)} |x\rangle |ax\rangle \xrightarrow{\text{SWAP}} |ax\rangle |x\rangle \xrightarrow{\text{CMULT}(\bar{a})^\dagger} |ax\rangle |x - \bar{a}(ax)\rangle = |ax\rangle |0\rangle,$$

because $\bar{a}a \equiv 1 \pmod{N}$. For $c = 0$ every factor is the identity, so nothing happens. Both branches leave the b -register in $|0\rangle$, so the output is unentangled and $c\text{-}U_a$ is realised exactly. $\square$



Three controlled blocks and nothing else. Note what is not here: no controlled QFT, and no control on the $\bar{a}$ constant — the inverse is classical. The SWAP is n Fredkin gates, all sharing the single control c .

Why not control the SWAP only?

A natural-looking shortcut is to leave both CMULTs uncontrolled and control only the SWAP. It *fails*: with $c = 0$ the residue is $\text{CMULT}(\bar{a})^\dagger \text{CMULT}(a)$, which maps $|x\rangle |0\rangle \mapsto |x\rangle |ax - \bar{a}x\rangle \neq |x\rangle |0\rangle$. The residue condition Equation (13.1) is violated, and the b -register comes out *dirty* — which, by Proposition 3.10.2, silently destroys the phase-estimation interference. **Always check the residue.**

13.3 Level 3: the controlled multiplier

Theorem 13.3.1 (Controlled out-of-place multiplier). *Let $\text{ADD}_{c_1 c_2}(k)$ denote the **Fourier-basis** modular adder $|\phi(b)\rangle \mapsto |\phi(b + k \bmod N)\rangle$ of Theorem 13.4.1, doubly controlled by c_1, c_2 — i.e. the ϕADD blocks, which act on the transformed register. Then*

$$\text{CMULT}(a) = \text{QFT}^\dagger \left(\prod_{i=0}^{n-1} \text{ADD}_{c, x_i}(2^i a \bmod N) \right) \text{QFT}, \quad (13.3)$$

*where the QFTs act on the b -register only, implements $|c\rangle |x\rangle |b\rangle \mapsto |c\rangle |x\rangle |b + cax \bmod N\rangle$. **The QFTs are not controlled.***

Proof. Correctness for $c = 1$: $ax \bmod N = (\sum_i x_i (2^i a \bmod N)) \bmod N$, and the i -th adder fires exactly when $x_i = 1$; all partial sums stay below N because each adder reduces. The constants $2^i a \bmod N$ are classical.

That the QFTs need no control is Lemma 13.1.1(a): with $c = 0$ no adder fires, so the residue is $\text{QFT}^\dagger \text{QFT} = \mathbb{I}$. $\square$

The blocks must be Fourier-basis adders

Conjugating a *computational*-basis adder by the QFT does not give an adder. The interior blocks of Equation (13.3) are the ϕ ADD blocks of Theorem 13.4.1, which act on $|\phi(b)\rangle$; the outer $\text{QFT}^\dagger \cdots \text{QFT}$ is what converts the register into and out of that basis, and is exactly the sandwich Lemma 13.1.1(a) tells us not to control. If a ripple-carry adder is used instead, the QFTs simply disappear and the saving of this section is realised differently — by not controlling the carry chain (Proposition 12.2.1 and the controlled adder following it).

This is where the saving is

The QFT on $n + 1$ qubits contains $\Theta(n^2)$ two-qubit rotations, while the doubly-controlled adders contain $\Theta(n)$ gates each. Controlling the two QFTs would therefore roughly *double* the gate count of every multiplier — and by Theorem 13.2.1 there are 2 multipliers per $c\text{-}U_a$, each built from n doubly-controlled modular adders, with $t = 2n$ such $c\text{-}U_a$ blocks in the ladder. Lemma 13.1.1(a) deletes all of it, for free, with a one-line argument.

13.4 Level 2: the doubly-controlled modular adder

Now the interesting one. The modular adder of Theorem 12.3.1 is seven blocks; only *three* of them carry the control.

Theorem 13.4.1 (Beauregard’s controlled modular adder: three of seven). *Let $0 \leq a < N$ and let the value register hold $|\phi(b)\rangle$ with $0 \leq b < N$, plus one ancilla $|0\rangle_t$. Run*

<i>Block</i>	<i>Control</i>
(1) $\phi\text{ADD}(a)$	controlled by c_1, c_2
(2) $\phi\text{SUB}(N)$	<i>bare</i>
(3) $\text{QFT}^\dagger$; CNOT from the sign qubit to t ; QFT	<i>bare</i>
(4) $\phi\text{ADD}(N)$	<i>controlled by</i> t
(5) $\phi\text{SUB}(a)$	controlled by c_1, c_2
(6) $\text{QFT}^\dagger$; X ; CNOT to t ; X ; QFT	<i>bare</i>
(7) $\phi\text{ADD}(a)$	controlled by c_1, c_2

Then the circuit implements $|\phi(b)\rangle |0\rangle \mapsto |\phi(b + c_1 c_2 a \bmod N)\rangle |0\rangle$.

Proof. Case $c_1 c_2 = 1$. Steps 1–2 give $b + a - N$, negative exactly when $b + a < N$, so after step 3 the ancilla holds $t = [b + a < N]$ and step 4 restores the modulus in that case: the register holds $R := (b + a) \bmod N$. To clear t , note

$$t = 1 \iff b + a < N \iff R = b + a \geq a, \quad t = 0 \iff R = b + a - N < a \quad (\text{since } b < N),$$

so $t = [R \geq a]$ is recomputable from the register. Step 5 forms $R - a$, whose sign qubit is $[R < a] = \neg t$. The $X\text{--CNOT}\text{--}X$ of step 6 acts as follows: the two X gates conjugate the *sign qubit*, so what is copied into t is $\neg(\neg t) = t$ — and $t \oplus t = 0$. (This is the same clearing step as Theorem 12.3.1 step 6, written with the X s on the sign wire rather than on t ; the two forms are equivalent because conjugating either end of a CNOT by X complements the copied bit.) Step 7 restores R .

Case $c_1 c_2 = 0$ — the residue. Steps 1, 5, 7 do nothing, so the circuit runs $\phi\text{SUB}(N)$, detect, $c\text{-}\phi\text{ADD}(N)$, clear. Since $b < N$, the value $b - N$ is negative, so its sign qubit is 1 and step 3 sets $t = 1$; step 4 then adds N back, returning the register to b . Finally the sign qubit of b is 0 because $b < N < 2^n$, so the $X\text{--CNOT}\text{--}X$ of step 6 copies $\neg 0 = 1$ into t , giving $t \mapsto 1 \oplus 1 = 0$. The residue is the identity, and Lemma 13.1.1 applies with $W = \{b < N, t = 0\}$. $\square$

The hypothesis $b < N$ is load-bearing

Both halves of the proof use $b < N$: the $c = 1$ half to conclude $R - a < 0 \iff t = 0$, and the $c = 0$ half to conclude that the sign qubit of b is 0. Feed this circuit a value $\geq N$ and it *silently leaves t dirty*. Since the adders are chained inside [Theorem 13.3.1](#), the invariant $b < N$ has to hold at every step — which it does, because every adder reduces modulo N . This is exactly the kind of precondition that [Chapter 19](#) makes explicit.

13.5 Level 1: doubly-controlled to singly-controlled

Three doubly-controlled blocks remain per adder. The last reduction turns every double control into a single one, *once*, for four T gates.

Proposition 13.5.1 (Hoisting the AND). *Let V be any circuit and let d be a fresh ancilla in $|0\rangle$. Then*

$$c\text{-}c\text{-}V = \text{AND}(c_1, c_2 \rightarrow d)^\dagger \cdot c\text{-}V_d \cdot \text{AND}(c_1, c_2 \rightarrow d) \quad (13.4)$$

on the subspace $d = 0$, where AND is the temporary logical-and of [Theorem 4.4.6](#). The pair costs 4 T gates in total — 4 to compute, 0 to uncompute — independently of how many controlled blocks V contains.

Proof. AND maps $|c_1\rangle|c_2\rangle|0\rangle \mapsto |c_1\rangle|c_2\rangle|c_1c_2\rangle$, so the middle gate fires exactly when $c_1c_2 = 1$, and the final $\text{AND}^\dagger$ restores $|0\rangle$. Uncomputation is by X -basis measurement and a classically conditioned CZ, hence Clifford, by [Theorem 4.4.6](#). $\square$

The ancilla must be clean — and this is testable

[Proposition 13.5.1](#) is *false* as an identity of unitaries on the full space: the left-hand side acts as the identity on d , the right-hand side uses d as scratch. It is true only on $d = 0$. Numerically, on the $N = 15$ modular adder:

equal restricted to $d = 0$	true
equal as full unitaries	false

Any test that compares full unitaries will report a *false failure* here; any test that forgets to check d returns to $|0\rangle$ will *miss a real one*.

13.6 The complete tower, and what it costs

Level	Object	What carries the control
5	the ladder, $\ell = 0, \dots, t-1$	<i>nothing</i> — t separate circuits, one per classical a_ℓ
4	$c\text{-}U_{a_\ell}$	2 multipliers + n Fredkins (Theorem 13.2.1)
3	$\text{CMULT}(a)$	the n modular adders; not the two QFTs (Theorem 13.3.1)
2	$\text{ADD}_{c, x_i}(k) \bmod N$	3 of 7 blocks (Theorem 13.4.1)
1	$c\text{-}c\text{-}\phi\text{ADD}(k)$	one AND + $n+1$ singly-controlled rotations
0	$c\text{-}P(\theta)$, AND	Toffoli / $4T$

Reading down, the control is pushed from a single outer qubit to $\approx 12n^3$ leaf gates across the whole ladder; reading up, each level proves that the level above it need not be controlled.

Proposition 13.6.1 (Count). *Per $c\text{-}U_a$: $2n$ doubly-controlled modular adders and n Fredkin gates. Each modular adder has 3 controlled blocks of $n+1$ rotations, so*

$$\#\{\text{controlled rotations per } c\text{-}U_a\} = 2n \cdot 3 \cdot (n+1) \approx 6n^2,$$

and over the whole $t = 2n$ ladder, $\approx 12n^3$. The $4n$ QFTs per $c-U_a$, which contain $\Theta(n^2)$ rotations each, are uncontrolled.

Measured: $N = 15$, $a = 7$, transpiled to $\{\text{CNOT}, U\}$

Circuit	CNOTs	Depth	Ratio
modular adder, <code>.control(2)</code> on everything	2634	4836	
modular adder, 3-of-7 (Theorem 13.4.1)	206	354	12.8×
$c-U_a$, naive controls	20892	37727	
$c-U_a$, Theorem 13.2.1	1748	2957	12.0×

An order of magnitude, from one lemma and two pages of case analysis — *before* any T -count optimisation, windowing or approximation. This is the single highest-leverage decision in implementing Shor's algorithm.

13.7 Three things that bite

1. The padding values are not the identity

U_a is supposed to act as the identity on $y \geq N$, so that it is a permutation of all 2^n basis states. The circuit of Theorem 13.2.1 does *not* do this. Measured on $N = 15$, $n = 4$, $a = 7$, the only padding value $x = 15$ gives

$$|1\rangle|15\rangle|0\rangle_b|0\rangle_t \mapsto |1\rangle|0\rangle|28\rangle_b|0\rangle_t,$$

i.e. the b -register does not return to $|0\rangle$. The circuit is still unitary — it must be — but it is a *different* unitary from U_a , and on the padding subspace it entangles the work register.

This is harmless in Shor's algorithm, for a specific reason worth stating: the eigenvector register is initialised to $|1\rangle$ and every subsequent value lies in the orbit $\{1, a, a^2, \dots\} \subseteq (\mathbb{Z}/N\mathbb{Z})^\times$, all of which are $< N$. The admissible subspace W of Lemma 13.1.1 is preserved. But if you ever reuse this U_a outside that invariant — to test it against a matrix, say — it will not match.

2. Framework `.control()` is a trap

Qiskit's `QuantumCircuit.control(k)`, Cirq's `controlled_by`, and Q#'s `Controlled` all do the honest thing: they control *every* gate. That is correct, and it is exactly what Lemma 13.1.1 says you should avoid.

Use them to *define the specification*, then hand-build the optimised version and assert the two are equal *on the admissible subspace*. That is precisely how the 12× above was measured: the naive version is the oracle for the fast one.

3. $\bar{a}$ must exist

Theorem 13.2.1 needs $a^{-1} \bmod N$, which exists iff $\gcd(a, N) = 1$. In Shor's algorithm that is checked classically before the quantum part runs — and if it fails you have already factored N . So the condition is free, but it is not vacuous.

13.8 Two ways to make the control cheaper still

13.8.1 One control qubit instead of t

Every control qubit in the ladder is measured immediately afterwards and never reused. By Proposition 4.1.3 the whole exponent register can therefore be replaced by a *single* qubit, recycled t times with

classically conditioned phase corrections between rounds — the semiclassical QFT of Griffiths–Niu (Section 8.6). This is what makes Beauregard’s $2n + 3$ -qubit circuit possible, and it costs nothing: the same t controlled operations, on one wire instead of t .

13.8.2 Fold the control into the address

The modern construction does not control U_a at all. In windowed arithmetic (Section 12.6) one looks up a classically precomputed table $T[j] = a^j \bmod N$, j a window of c_e exponent bits, and the exponent qubits are the lookup *address*. There is no control qubit and no controlled operation: a table lookup indexed by j already does something different for each value of j , which is what “controlled” means. *Conditioning is free because it has been absorbed into data.*

The arc

naive	control every gate	$\Theta(n^3)$ controlled gates
Lemma 13.1.1	control the 3-of-7 core	12× cheaper
Proposition 13.5.1	one AND, then single controls	another $\sim 1.5\times$
windowing	<i>no controls at all</i> — control is an index	$\sim 25\times$

Each step replaces *conditioning* with *structure*. That is the whole engineering story of Shor’s algorithm in one column.

Chapter 14

Worked example: $N = 21, a = 2$

Every number in this chapter was computed exactly and verified numerically ([Appendix D](#)). $N = 21$ is chosen deliberately: it is the smallest instance that exercises *every* step of the algorithm. The ubiquitous $N = 15$ does not, for the reason given in [Section 14.7](#).

14.1 Setup

The instance. $1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 16 \rightarrow 11 \rightarrow 1$ (mod 21), so

$$\mathcal{O} = \{1, 2, 4, 8, 16, 11\}, \quad r = \text{ord}_{21}(2) = 6,$$

and by [Proposition 5.5.3](#) U_2 acts on $\mathcal{O}$ as the 6-cycle S . The eigenphases ([Theorem 5.5.4](#)) are

$$0, \frac{1}{6}, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, \frac{5}{6}.$$

Register size. $n = \lceil \log_2 21 \rceil = 5$; the bound $N^2 \leq Q < 2N^2$ of [Corollary 11.1.1](#) reads $441 \leq Q < 882$, so

$$Q = 512, \quad t = 9 \text{ counting qubits.}$$

Since $6 \nmid 512$, only $s = 0$ ($\varphi = 0$) and $s = 3$ ($\varphi = \frac{1}{2} = \frac{256}{512}$) are *dyadic*; the other four eigenphases are not representable as $j/2^t$, so their peaks must broaden.

14.2 One branch, by hand

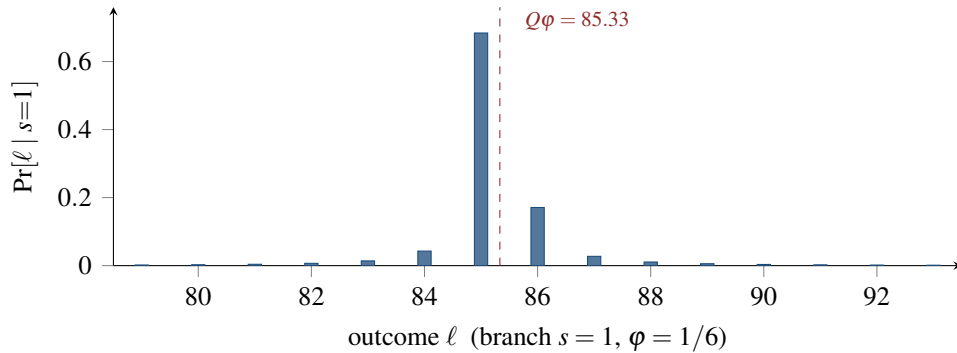
Take the branch $s = 1, \varphi = \frac{1}{6}$. Then $Q\varphi = 512/6 = 85.33\overline{3}$, not an integer; the nearest grid point is $y = 85$ with detuning

$$\delta_{85} = \frac{1}{6} - \frac{85}{512} = \frac{512 - 510}{3072} = \frac{1}{1536}, \quad Q\delta_{85} = \frac{512}{1536} = \frac{1}{3}.$$

Substituting into the Fejér kernel [Equation \(8.5\)](#),

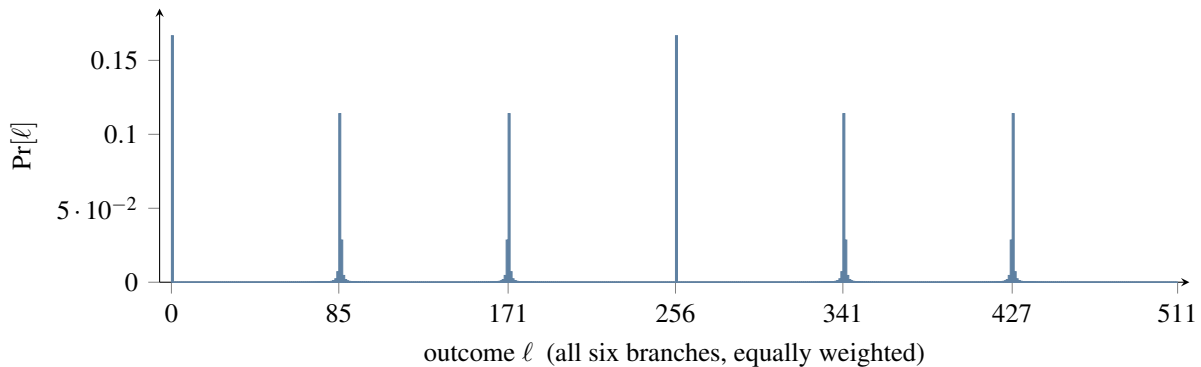
$$\Pr[85 \mid s=1] = \frac{\sin^2(\pi/3)}{512^2 \sin^2(\pi/1536)} = \frac{0.75}{1.0966} \approx \boxed{0.6839}.$$

For $y = 86$: $Q\delta_{86} = -\frac{2}{3}$ and $\Pr[86 \mid s=1] \approx 0.1710$. *The two adjacent outcomes hold 85.5% of the branch* — a concrete instance of [Proposition 8.3.6](#), whose guarantee was $8/\pi^2 = 0.8106$.



14.3 The full measured distribution

Averaging over the six branches, each occurring with probability $\frac{1}{6}$ by Theorem 8.5.1:



Reading the peaks

Six peaks at $\ell \approx Qs/6$ for $s = 0, \dots, 5$, namely $\ell \in \{0, 85, 171, 256, 341, 427\}$. The two *dyadic* branches ($s = 0$ at $\ell = 0$, $s = 3$ at $\ell = 256$) are single spikes of height $\frac{1}{6} = 0.1667$. The four non-dyadic branches spread: height 0.1140 at the nearest grid point plus 0.0285 at the neighbour, with a small tail beyond.

Where r is encoded

The *peak spacing* is Q/r . That — not any single amplitude — is where r is encoded, which is exactly why one measurement gives a *rational approximation* and needs continued fractions to be promoted to an exact fraction.

14.4 Reconstruction, branch by branch

s	$\phi = s/6$	nearest ℓ	continued fraction $\rightarrow$ convergent with denom. < 21	r'	verdict
0	0	0	$[0] \rightarrow 0/1$	1	$2^1 = 2 \neq 1$: retry
1	$1/6$	85	$[0; 6, 42, 2] \rightarrow \mathbf{1/6}$	6	$2^6 = 64 \equiv 1$: success
2	$1/3$	171	$[0; 2, 1, 170] \rightarrow 1/3$	3	$2^3 = 8 \neq 1$: divisor
3	$1/2$	256	$[0; 2] \rightarrow 1/2$	2	$2^2 = 4 \neq 1$: divisor
4	$2/3$	341	$[0; 1, 1, 1, 170] \rightarrow 2/3$	3	divisor
5	$5/6$	427	$[0; 1, 5, 42, 2] \rightarrow \mathbf{5/6}$	6	$2^6 \equiv 1$: success

14.4.1 Following $\ell = 171$ through the post-processing

Continued fraction of $171/512$.

$$512 = 2 \cdot 171 + 170,$$

$$171 = 1 \cdot 170 + 1, \quad \Rightarrow \quad \frac{171}{512} = [0; 2, 1, 170].$$

$$170 = 170 \cdot 1 + 0,$$

Convergents: $\frac{0}{1}, \frac{1}{2}, \frac{1}{3}, \frac{171}{512}$. The last with denominator $< N = 21$ is $\frac{1}{3}$, so $r' = 3$.

Verification and repair. $2^3 = 8 \not\equiv 1 \pmod{21}$ — fails. So 3 is only a *divisor* of r : this run came from branch $s = 2$, and $\frac{2}{6}$ reduced to $\frac{1}{3}$.

Test small multiples: $2 \cdot 3 = 6$, and $2^6 = 64 = 3 \cdot 21 + 1 \equiv 1 \pmod{21}$ ✓, so $r = 6$. Alternatively, combine with a run giving $r' = 2$: $\text{lcm}(3, 2) = 6$.

Why the denominator is never simply trusted

The measured ℓ carries s/r in *lowest terms*; the reduction is invisible from ℓ alone. The classical verification $a^{r'} \stackrel{?}{\equiv} 1$ is what makes the whole algorithm never return a wrong answer.

14.5 Success probability, exactly

Branches $s = 1, 5$ have $\gcd(s, 6) = 1$ and recover $r = 6$ directly: $\phi(6)/6 = \frac{1}{3}$ of the branches. Summing the *exact* Fejér weights of all ℓ whose convergent has denominator 6 gives

Pr[one run yields $r = 6$ directly]	0.320762
Pr[one run yields $r = 6$, allowing the multiple-testing repair]	0.814927
Pr[two runs, lcm of denominators = r]	0.529028
mass at $\ell = 0$ (the $s = 0$ branch, uninformative)	0.166672
ceiling $1 - \frac{1}{6}$	0.833333

Reading these numbers

The direct figure 0.3208 is close to, but not equal to, $\phi(6)/6 = 0.3333$: the shortfall is the probability mass in the tails of the four broadened peaks that lands on outcomes whose convergent is *not* 6. The repaired figure 0.8149 is capped at $5/6 = 0.8333$, because the entire $s = 0$ branch — a full $\frac{1}{6}$ of the probability — lands exactly on $\ell = 0$, which carries no information about r whatsoever and must be discarded. The repair recovers essentially everything else.

14.6 Finishing the factorisation

$r = 6$ is even and $2^3 = 8 \not\equiv -1 \equiv 20 \pmod{21}$, so both conditions of [Lemma 9.1.1](#) hold. Therefore

$$\gcd(2^3 - 1, 21) = \gcd(7, 21) = 7, \quad \gcd(2^3 + 1, 21) = \gcd(9, 21) = 3,$$

giving $\boxed{21 = 3 \times 7}$.

14.7 Contrast with $N = 15$

$N = 15, a = 7, t = 3, Q = 8$	Why it misleads
$7^0, 7^1, 7^2, 7^3 \equiv 1, 7, 4, 13$ and $7^4 \equiv 1$, so $r = 4$. Since $4 \mid 8$, every eigenphase $s/4 = 2s/8$ is an <i>exact</i> grid point: by Theorem 8.2.1 phase estimation is exact — four perfectly sharp peaks at $\ell \in \{0, 2, 4, 6\}$, each of probability $\frac{1}{4}$.	$(\mathbb{Z}/15\mathbb{Z})^\times \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$, so <i>every</i> order is a power of two, no matter which base is chosen. Continued fractions therefore do literally no work. It is exactly the case in which the hardest step disappears.

ℓ	ℓ/Q	continued fraction	r'	verdict
0	0	$[0] \rightarrow 0/1$	1	$7^1 = 7 \neq 1$: fail, retry
2	1/4	$[0; 4] \rightarrow 1/4$	4	$7^4 \equiv 1$: success
4	1/2	$[0; 2] \rightarrow 1/2$	2	$7^2 = 4 \neq 1$: fail, retry
6	3/4	$[0; 1, 3] \rightarrow 3/4$	4	$7^4 \equiv 1$: success

A single run succeeds with probability exactly $\frac{1}{2} = \phi(4)/4$, as [Proposition 11.3.1](#) predicts (the successes are $s = 1, 3$, the two values coprime to 4). Then $r = 4$ is even and $7^2 = 4 \not\equiv -1 \equiv 14$, so $\gcd(3, 15) = 3$ and $\gcd(5, 15) = 5$.

Test your implementation on a non-dyadic period

If you build the $N = 21$ circuit, the constants are $b_k = 2^k \bmod 21$ for $k = 0, \dots, 8$, giving 2, 4, 16, 4, 16, 4, 16, 4, 16 — it becomes periodic but *never collapses to 1*, because $r = 6$ is not a power of two. **Every one of the nine controlled multipliers must be built.**

For $N = 15$, by contrast, $2^k \bmod 15$ collapses to 1 from $k = 2$ onwards, so six of the eight controlled multipliers are the identity and are simply omitted. That is not an optimisation — it is the dyadic structure of $N = 15$ appearing in the circuit, and it is why that instance is easy.

Also: $M_2 \bmod 21$ is a permutation with cycle structure (1 2 4 8 16 11)(3 6 12)(5 10 20 19 17 13)(7 14)(9 18 15)(0) — not a single rotation, so no swap-only shortcut exists.

Sizes to expect: $n = 5$ target qubits, $t = 9$ control qubits, so 14 qubits before ancillas. If your circuit is much smaller than that, check what you assumed.

14.8 What the published demonstrations show

Work	Platform	N	Technique
Vandersypen et al. (2001)	7-qubit NMR	15	first demonstration
Martín-López et al. (2012)	photonic	15	qubit recycling; one control qubit reused
Monz et al. (2016)	ion trap	15	Kitaev iterative phase estimation; 7 control + 4 cache qubits
Amico, Saleem, Kumph (2019)	superconducting	15, 21, 35	semiclassical (Kitaev) QFT to minimise qubits and gates
Skosana & Tame (2021)	superconducting	21	compiled QPE, approximate Toffolis

The question to ask of any claimed record

Was the modular exponentiation circuit compiled using prior knowledge of r ?

Nearly every demonstration above uses a compiled circuit — multipliers simplified using the known order, or the known factorisation. That removes precisely the part that does the work. No cryptographically relevant number has been factored by Shor's algorithm, and none is close.

The demonstrations are valuable as *engineering milestones*, not as evidence about scaling.

Two legitimate savings do recur, and both are in [Chapter 13](#): the semiclassical/iterative QFT (exact, legitimate at any scale) and compiling classical constants at synthesis time (legitimate when the constant is genuinely public). Using knowledge of r is neither.

Part IV

Discrete Logarithms and Elliptic Curves

The discrete logarithm as a hidden subgroup problem

The same eigenvalue-to-phase mechanism solves the discrete logarithm, but on a *two*-dimensional hidden subgroup. This chapter derives, from character orthogonality alone, that measuring two index registers returns a pair (y_1, y_2) obeying the linear relation $y_2 \equiv m y_1 \pmod{r}$, from which the logarithm m follows by one modular inversion.

What changes, and what does not

Links 5–7 of the chain (kickback, ladder, interference) are **identical**. Three things change:

1. The hiding function becomes $f(k, \ell) = [k]P + [\ell]Q = [k + m\ell]P$, hiding the subgroup $L = \langle (-m, 1) \rangle \leq \mathbb{Z}/r\mathbb{Z} \times \mathbb{Z}/r\mathbb{Z}$.
2. The transform becomes $\text{QFT} \otimes \text{QFT}$: the group is two-dimensional and its dual factors.
3. Post-processing becomes $m \equiv y_2 y_1^{-1} \pmod{r}$ — **no continued fractions**, because r is public.

One sentence: factoring is the one-dimensional hidden subgroup problem, where the unknown is the *size* of the hidden subgroup; the DLP is the two-dimensional one, where the size is known and the unknown is the *slope*.

15.1 Setup

Let $G = \langle P \rangle$ be a finite cyclic group of *known* order r , written additively. Fix a generator P and an element $Q = [m]P$ for an unknown $m \in \mathbb{Z}/r\mathbb{Z}$; the problem is to recover m . (Multiplicatively: $h = g^m$ in $\mathbb{F}_p^\times$.)

Definition 15.1.1 (Hidden subgroup problem). Given $f : A \rightarrow X$ on a finite abelian group A that is *constant on cosets* of an unknown subgroup $L \leq A$ and *distinct on distinct cosets*, find L . We say f *hides* L .

Order finding was the one-dimensional case: $A = \mathbb{Z}$ truncated to $\mathbb{Z}/Q\mathbb{Z}$, $f(x) = a^x \bmod N$, and $L = r\mathbb{Z}$.

Proposition 15.1.2 (The hidden subgroup). Take $A = \mathbb{Z}/r\mathbb{Z} \times \mathbb{Z}/r\mathbb{Z}$ and $f(k, \ell) := [k]P + [\ell]Q$. Since $Q = [m]P$ and scalar multiplication is a group homomorphism $\mathbb{Z} \rightarrow G$,

$$f(k, \ell) = [k]P + [\ell m]P = [k + m\ell]P, \quad (15.1)$$

so f depends only on $k + m\ell \bmod r$, and f hides

$$L = \{(k, \ell) \in A : k + m\ell \equiv 0 \pmod{r}\} = \langle (-m, 1) \rangle, \quad (15.2)$$

of order r and index r . Knowing L is equivalent to knowing m .

Proof. $f(k, \ell) = f(k', \ell') \iff (k - k') + m(\ell - \ell') \equiv 0 \iff (k - k', \ell - \ell') \in L$, so f is constant on cosets of L and injective across them. Every element of L has the form $(-m\ell, \ell) = \ell \cdot (-m, 1)$, so $L = \langle (-m, 1) \rangle$; it has r elements, whence index r . Reading m off the generator $(-m, 1)$ recovers m . $\square$

Geometric reading

The unknown is now the *slope of a line* in $\mathbb{Z}/r\mathbb{Z} \times \mathbb{Z}/r\mathbb{Z}$. The quantum machine will hand over one point of the *dual* line, which is enough — a line is determined by one generic point of its annihilator.

15.2 The measured relation, derived

Assume for the derivation that the index registers have dimension exactly r ; the finite-register version follows in [Section 15.5](#).

Theorem 15.2.1 (The linear relation). *Prepare the uniform superposition over A , apply f into a point register, apply QFT_r to each index register, and measure. (Using $\text{QFT}_r^\dagger$ instead gives the same conclusion: the residual phase changes from $\omega_r^{cy_1}$ to $\omega_r^{-cy_1}$, which a computational-basis measurement cannot see. [Algorithm 16.5.1](#) is written with $\text{QFT}^\dagger$ to match [Algorithm 8.1.3](#).) The outcome (y_1, y_2) satisfies*

$$y_2 \equiv my_1 \pmod{r} \quad (15.3)$$

with each admissible y_1 occurring with probability $1/r$.

Proof. (1) *Superpose and apply the oracle.*

$$\frac{1}{r} \sum_{k, \ell \in \mathbb{Z}/r\mathbb{Z}} |k\rangle |\ell\rangle |\mathcal{O}\rangle \xrightarrow{f} \frac{1}{r} \sum_{k, \ell} |k\rangle |\ell\rangle |[k + m\ell]P\rangle.$$

(2) *Measure (or, by [Proposition 4.1.3](#), ignore) the point register.* It holds $[c]P$ for a uniformly random coset label $c \in \mathbb{Z}/r\mathbb{Z}$; by [Proposition 3.6.1](#) the index registers collapse to the uniform superposition over the coset $\{(k, \ell) : k + m\ell \equiv c\}$. Parametrising that coset by the free variable ℓ (so $k = c - m\ell$):

$$\frac{1}{\sqrt{r}} \sum_{\ell \in \mathbb{Z}/r\mathbb{Z}} |c - m\ell\rangle |\ell\rangle.$$

(3) *Apply $\text{QFT}_r \otimes \text{QFT}_r$.*

$$\begin{aligned} & \frac{1}{\sqrt{r}} \sum_{\ell} \left(\frac{1}{\sqrt{r}} \sum_{y_1} \omega_r^{(c-m\ell)y_1} |y_1\rangle \right) \left(\frac{1}{\sqrt{r}} \sum_{y_2} \omega_r^{\ell y_2} |y_2\rangle \right) \\ &= \frac{1}{r^{3/2}} \sum_{y_1, y_2} \omega_r^{cy_1} \underbrace{\left(\sum_{\ell} \omega_r^{\ell(y_2 - my_1)} \right)}_{= r[y_2 \equiv my_1] \text{ by Lemma A.0.1}} |y_1\rangle |y_2\rangle = \frac{1}{\sqrt{r}} \sum_{y_1} \omega_r^{cy_1} |y_1\rangle |my_1 \bmod r\rangle. \end{aligned}$$

(4) *Measure.* The outcome is $(y_1, my_1 \bmod r)$, uniform over the r values of y_1 . □

The leftover phase, and why it is harmless

$\omega_r^{cy_1}$ is *not* a global phase — it depends on y_1 . But it is a phase attached to each basis state of the register that is measured next, and a *computational-basis* measurement is insensitive to phases on its own basis states ([Definition 3.4.2](#): only $|\langle m|\psi\rangle|^2$ enters). So $\Pr[(y_1, y_2)] = 1/r$ regardless of c . That is why the unknown coset label does no harm, and why the point register need not be measured at all.

Corollary 15.2.2 (Recovering m). *If $\gcd(y_1, r) = 1$ then $m \equiv y_2 y_1^{-1} \pmod{r}$, computed by one extended-Euclid inversion. When r is prime (the NIST curves have cofactor 1), every $y_1 \neq 0$ is invertible, so a single run succeeds with probability $(r-1)/r \approx 1$.*

Proof. $y_2 \equiv my_1$ with y_1 invertible gives $m \equiv y_2 y_1^{-1}$. For prime r every $y_1 \in \{1, \dots, r-1\}$ is invertible; only $y_1 = 0$ (probability $1/r$) is uninformative — and it is uninformative for *every* m , because $\chi_{0,0} \equiv 1$ is trivial on every subgroup. □

15.3 Why two QFTs

The dual group factors

The translations $(k, \ell) \mapsto (k + \alpha, \ell + \beta)$ of $A = \mathbb{Z}/r\mathbb{Z} \times \mathbb{Z}/r\mathbb{Z}$ commute, so by [Theorem 5.1.2](#) they are simultaneously diagonalised — by the characters of A ,

$$\chi_{y_1, y_2}(k, \ell) = \omega_r^{ky_1 + \ell y_2}.$$

$\text{QFT}_r \otimes \text{QFT}_r$ is precisely the change to that eigenbasis, and it *factors* over the two coordinates because $\widehat{A} \cong \widehat{\mathbb{Z}/r\mathbb{Z}} \times \widehat{\mathbb{Z}/r\mathbb{Z}}$.

What measurement gives. It samples a character that is *trivial on the hidden subgroup*:

$$\chi_{y_1, y_2}|_L \equiv 1 \iff \omega_r^{\ell(y_2 - my_1)} = 1 \quad \forall \ell \iff y_2 \equiv my_1.$$

So the measured pair names a point of $L^\perp$, the annihilator of L . Since L is a line, one generic point of $L^\perp$ determines it, hence determines m .

15.4 Factoring and DLP side by side

	Factoring	DLP / ECDLP
A	$\mathbb{Z}/Q\mathbb{Z}$	$\mathbb{Z}/r\mathbb{Z} \times \mathbb{Z}/r\mathbb{Z}$
L	$r'\mathbb{Z}/Q\mathbb{Z}$	$\langle (-m, 1) \rangle$
rank A	1	2
QFTs	1	2
unknown	the <i>size</i> of L	the <i>slope</i> of L
output	$\ell \approx Qs/r$	(y_1, y_2) with $y_2 \equiv my_1$
post-processing	continued fractions	one modular inverse
success per run	$\phi(r)/r$	$(r-1)/r$

15.5 The finite-register version and post-processing

In practice r is not a power of two, so each index register uses $n+1$ qubits with $2^{n+1} > r$ and one applies $\text{QFT}_{2^{n+1}}$. (That $n+1$ qubits suffice follows from Hasse's theorem; see [Proposition 16.3.1](#).) The exact delta function of [Theorem 15.2.1](#) broadens into a Fejér peak — the *same* kernel as [Theorem 8.3.2](#) — and the measured pair satisfies

$$\frac{y_1}{2^{n+1}} \approx \frac{j_1}{r}, \quad \frac{y_2}{2^{n+1}} \approx \frac{j_2}{r}, \quad j_2 \equiv m j_1 \pmod{r}. \quad (15.4)$$

Rounding $(y_1/2^{n+1}, y_2/2^{n+1})$ to the nearest rationals with the *known* denominator r — a two-dimensional rounding, *not* a search over unknown denominators — recovers (j_1, j_2) and hence $m \equiv j_2 j_1^{-1}$.

A real and easily made implementation error

The relation obeyed by the *post*-transform outcomes is $y_2 \equiv my_1$, **not** the pre-transform relation $k + m\ell \equiv c$. Applying the pre-transform relation to measured data — e.g. $\hat{m} = (c - k)\ell^{-1}$ — gives a recovery rate at the $1/r$ random baseline *even on a perfect simulator*, because the pre-transform registers were never measured.

Always verify $[\hat{m}]P = Q$ before believing an answer. (This costs one scalar multiplication and makes the ECDLP algorithm Las Vegas too.)

15.6 Is the finite-field discrete logarithm the same algorithm?

Yes — identical structure, different oracle

The derivation of Equation (15.3) used only that the group is cyclic of known order with an efficiently computable operation. *Nothing about elliptic curves entered.*

Setting	Oracle $f(k, \ell)$	Cost of the oracle
finite-field DLP ($\mathbb{F}_p^\times, r = p - 1$)	$g^k h^\ell \bmod p$	modular exponentiation
ECDLP ($Q = [m]P$)	$[k]P + [\ell]Q$	elliptic scalar multiplication

Practical reading. RSA, finite-field Diffie–Hellman, DSA, ECDH and ECDSA are all broken by the *same machine*. That is one mechanism, four deployed cryptosystems — and it is the reason the migration to post-quantum algorithms is a single migration rather than several.

Chapter 16

Elliptic curves and their group law

The oracle of Chapter 15 needs the group operation $+$ on $E(\mathbb{F}_p)$ in *explicit coordinate formulas*, because those formulas are what the reversible circuit of Chapter 17 must implement. We derive them from the chord–tangent construction and read off exactly which field operations appear — crucially, a *modular inversion*, which is what makes an elliptic point addition so much dearer than a modular multiplication.

16.1 The curve and its group

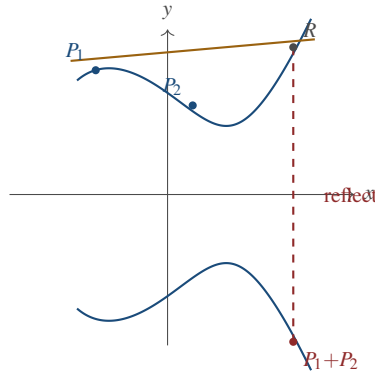
Definition 16.1.1 (Short Weierstrass form). Let $p > 3$ be prime. An *elliptic curve* in short Weierstrass form is

$$E : y^2 = x^3 + ax + b, \quad a, b \in \mathbb{F}_p, \quad \Delta = -16(4a^3 + 27b^2) \neq 0, \quad (16.1)$$

the non-vanishing discriminant ensuring no repeated roots (a smooth curve). Its set of points is $E(\mathbb{F}_p) = \{(x, y) \in \mathbb{F}_p^2 : y^2 = x^3 + ax + b\} \cup \{\mathcal{O}\}$, where $\mathcal{O}$ is the *point at infinity*. Under the chord–tangent addition below, $E(\mathbb{F}_p)$ is a finite abelian group with identity $\mathcal{O}$ and inverse $-(x, y) = (x, -y)$.

16.2 Deriving the group law

Three points of E on a common line sum to $\mathcal{O}$; this *defines* $+$. Concretely, to add P_1 and P_2 , draw the line through them and reflect its third intersection with E .



Theorem 16.2.1 (Affine addition formulas). For $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$ with $P_1 \neq \pm P_2$ set

$$\lambda = \frac{y_2 - y_1}{x_2 - x_1}; \quad \text{for doubling } (P_1 = P_2, y_1 \neq 0) \text{ set } \lambda = \frac{3x_1^2 + a}{2y_1}. \quad (16.2)$$

Then $P_3 = P_1 + P_2 = (x_3, y_3)$ with

$$x_3 = \lambda^2 - x_1 - x_2, \quad y_3 = \lambda(x_1 - x_3) - y_1. \quad (16.3)$$

Proof. The line through P_1, P_2 is $y = \lambda x + v$ with $v = y_1 - \lambda x_1$. Substituting into $y^2 = x^3 + ax + b$,

$$(\lambda x + v)^2 = x^3 + ax + b \implies x^3 - \lambda^2 x^2 + (a - 2\lambda v)x + (b - v^2) = 0,$$

a cubic whose roots are the three intersection abscissae x_1, x_2, x_3 . By Viète, the sum of the roots equals the negated x^2 -coefficient:

$$x_1 + x_2 + x_3 = \lambda^2 \implies x_3 = \lambda^2 - x_1 - x_2.$$

The third intersection is $R = (x_3, \lambda x_3 + v)$; reflecting in the x -axis (the group inverse) gives $y_3 = -(\lambda x_3 + v) = -(\lambda x_3 + y_1 - \lambda x_1) = \lambda(x_1 - x_3) - y_1$.

For doubling, differentiate $y^2 = x^3 + ax + b$ implicitly: $2yy' = 3x^2 + a$, so the tangent slope is $\lambda = (3x_1^2 + a)/(2y_1)$, and the same Viète argument applies with a double root at x_1 . $\square$

The operation that dominates the cost

Both slopes are a *quotient* in $\mathbb{F}_p$: computing λ requires a modular inversion, $(x_2 - x_1)^{-1}$ or $(2y_1)^{-1}$. Modular inversion is by far the most expensive field operation — worth $\Theta(n)$ multiplications — and it appears in every point addition.

This is why a 256-bit ECDLP is a comparable quantum target to a 2048–3072-bit RSA modulus despite the far smaller key: an elliptic point addition is individually much dearer than a modular multiplication. The smaller n still wins — but by less than the key sizes alone would suggest.

16.3 Scalar multiplication, order, and register size

For $m \in \mathbb{Z}$, $[m]P = \underbrace{P + \dots + P}_m$ (with $[0]P = \mathcal{O}$, $[-m]P = [m](-P)$) is a group homomorphism $\mathbb{Z} \rightarrow E(\mathbb{F}_p)$;

its image $\langle P \rangle$ is cyclic of order $r = \text{ord}(P)$. Scalar multiplication is computed by *double-and-add*: writing $m = \sum_j m_j 2^j$,

$$[m]P = \sum_j m_j [2^j]P, \quad [2^j]P \text{ by } j \text{ repeated doublings}, \quad (16.4)$$

which is the exact analogue of repeated squaring (Proposition 10.2.2) and the structure the quantum circuit mirrors.

Proposition 16.3.1 (Register size from Hasse's bound). *By Hasse's theorem $|\#E(\mathbb{F}_p) - (p+1)| \leq 2\sqrt{p}$, so*

$$r \leq \#E(\mathbb{F}_p) \leq p+1+2\sqrt{p} < 2^{n+1}, \quad n = \lceil \log_2 p \rceil.$$

Hence $n+1$ qubits suffice to hold any scalar or coordinate.

Proof. $p+1+2\sqrt{p} = (\sqrt{p}+1)^2 < (2^{n/2}+1)^2 \leq 2^{n+1}$, the last inequality holding whenever $2^{n/2} \geq 1+\sqrt{2}$, i.e. $n \geq 3$ — which is automatic since $p > 3$ forces $n \geq 3$. $\square$

16.4 The running curve

We fix a toy curve, carried end to end in Section 17.6:

$$E : y^2 = x^3 + 3x + 2 \quad \text{over } \mathbb{F}_5.$$

Finding the points. Squares in $\mathbb{F}_5$ are $\{0, 1, 4\}$.

Test each x :

- $x = 0$: 2 — not a square;
- $x = 1$: $1 + 3 + 2 = 6 \equiv 1$ — square, $y = \pm 1$;
- $x = 2$: $8 + 6 + 2 = 16 \equiv 1$ — square, $y \in \{1, 4\}$;
- $x = 3$: $27 + 9 + 2 = 38 \equiv 3$ — not a square;
- $x = 4$: $64 + 12 + 2 = 78 \equiv 3$ — not a square.

So $E(\mathbb{F}_5) = \{\mathcal{O}, (1, 1), (1, 4), (2, 1), (2, 4)\}$ and $\#E = 5$, *prime*, so every non-identity point generates. Discriminant check:

$$-16(4 \cdot 27 + 27 \cdot 4) = -16 \cdot 216 \equiv 4 \neq 0 \checkmark$$

The cyclic subgroup. Take $P = (2, 1)$. Doubling

with $a = 3$: $\lambda = \frac{3 \cdot 4 + 3}{2 \cdot 1} = \frac{15}{2} \equiv \frac{0}{2} = 0$, so

$$x_3 = 0 - 2 - 2 = -4 \equiv 1 \text{ and}$$

$$y_3 = 0 \cdot (2 - 1) - 1 = -1 \equiv 4: [2]P = (1, 4).$$

Continuing,

$$[1]P = (2, 1), \quad [2]P = (1, 4), \quad [3]P = (1, 1),$$

$$[4]P = (2, 4), \quad [5]P = \mathcal{O}.$$

The instance to be solved. $P = (2, 1)$, $Q = (1, 1)$.

Then $Q = [3]P$, so the answer is $m = 3$ — which the quantum algorithm will produce without knowing it in advance. Inverses mod 5: $1^{-1} = 1$, $2^{-1} = 3$, $3^{-1} = 2$, $4^{-1} = 4$.

A caveat on small demonstrations

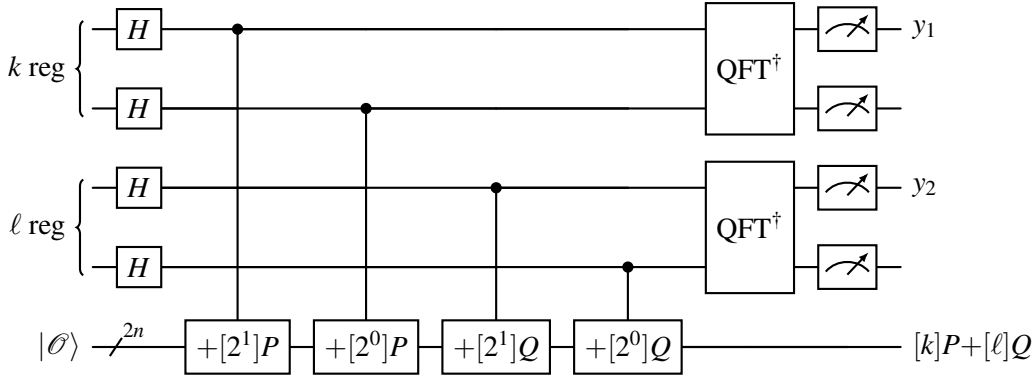
Encoding points by their *group index* requires enumerating $\langle P \rangle$, which is $O(r)$ classical work — and the index of Q is m . A demonstration built that way has already solved the ECDLP classically before a single qubit is allocated.

The path to real sizes is the *coordinate* oracle built in Chapter 17, where the circuit implements the group law on actual field elements.

16.5 The algorithm and its circuit

Algorithm 16.5.1 (Shor for the ECDLP). **Input:** $E(\mathbb{F}_p)$, a point P of order r , and $Q = [m]P$. Set $n = \lceil \log_2 p \rceil$.

1. Prepare two index registers $|0\rangle_{n+1} |0\rangle_{n+1}$ and a point register in $|\mathcal{O}\rangle$.
2. Apply $H^{\otimes(n+1)}$ to each index register.
3. **Double scalar multiplication (the oracle):** controlled on bit k_j , add the classical point $[2^j]P$; controlled on bit ℓ_j , add $[2^j]Q$. This maps the point register to $[k]P + [\ell]Q$.
4. Apply $\text{QFT}_{2^{n+1}}^\dagger$ to each index register.
5. Measure both index registers, obtaining (y_1, y_2) .
6. Classically: round to (j_1, j_2) with $j_2 \equiv m j_1 \pmod{r}$ and output $m \equiv j_2 j_1^{-1}$. **Verify** $[m]P = Q$.



Shown with $n + 1 = 2$ bits per register; in general the k -register contributes the ladder $+ [2^n]P, \dots, + [2^0]P$ and the ℓ -register $+ [2^n]Q, \dots, + [2^0]Q$.

Proposition 16.5.2 (Double scalar multiplication). *For a classically known point R and an $(n+1)$ -qubit scalar register $|s\rangle$,*

$$\prod_{j=0}^n \left(\text{controlled-} + [2^j]R \text{ on qubit } s_j \right) : |s\rangle |\mathcal{O}\rangle \mapsto |s\rangle |[s]R\rangle,$$

since $[s]R = \sum_j s_j [2^j]R$ and the constants $[2^j]R = [2]([2^{j-1}]R)$ are computed by j classical doublings. Running this with $R = P$ on the k -register and $R = Q$ on the ℓ -register realises step 3.

Why every addition is by a constant point

The points $[2^j]P$ and $[2^j]Q$ are precomputed *classically*, so every box is “add a fixed, known point to a quantum point” — **never** a quantum–quantum point addition.

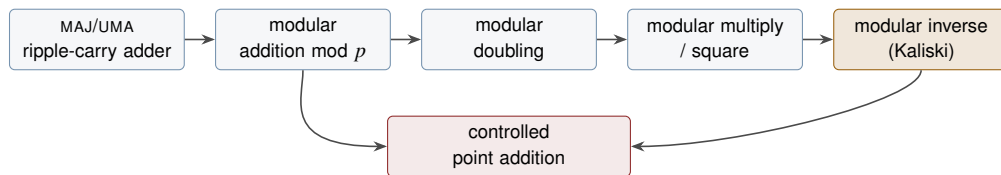
Three consequences: the constant’s coordinates are folded into the gate pattern at compile time (a

constant control deletes or fixes a Toffoli, roughly halving the gates); unlike RSA, no quantum–quantum multiplication ever appears; and additions commute, so their order is free. The one irreducible cost inside each box is the modular inversion for λ .

Remark 16.5.3 (The semiclassical variant). Every control of the two $\text{QFT}^\dagger$ s is a qubit measured at the end and never reused, so by [Proposition 4.1.3](#) each index register collapses to a *single reused ancilla* with classical feedback ([Section 8.6](#)). This saves $2n$ qubits at no gate cost.

The reversible elliptic-curve oracle

This chapter builds the oracle of [Algorithm 16.5.1](#) all the way down to Toffoli gates. The tower is five levels deep, and every level must be *garbage-free* — not for tidiness, but because by [Proposition 3.10.2](#) leftover scratch would record which-path information and destroy the interference that produces $y_2 \equiv my_1$.



Conventions

All registers hold n -bit integers with $p < 2^n$; operations are in place unless stated. Every constant $(p, 2^i a \bmod p, \text{the coordinates of } [2^i]P)$ is folded in at compile time, which roughly halves the gate count. Modular multiplication, squaring and inversion operate *out of place* and XOR the result into a separate output register — which is what allows them to be run a second time to clear that register.

17.1 The cost of each field operation

Operation	Reversible cost
modular add / subtract	$O(n \log n)$ Toffoli
modular double	$O(n \log n)$ Toffoli
modular multiply / square	$O(n^2 \log n)$ Toffoli
modular inverse	$O(n^2 \log n)$ Toffoli, $O(n)$ ancillas

The $\log n$ factors come from the constant-incrementer construction; exact leading terms are tabulated in [Section 17.7](#). Textbook treatments quote $\Theta(n) / \Theta(n^2)$, which is the count without that construction.

Why affine coordinates, despite the inversion

Projective coordinates avoid almost all inversions classically, by multiplying through with denominators. Two obstacles make that hard here: each point is an *equivalence class* of coordinate vectors, and the extra temporary variables must all be uncomputed.

Since the binding constraint in the published estimates is *qubits*, and the inversion circuit already supplies the ancillas that the multiplier needs, affine coordinates win. Whether projective coordinates help remains open.

17.2 Modular addition and doubling

Modular addition, $|x\rangle|y\rangle \mapsto |x\rangle|(x+y) \bmod p\rangle$. Add; subtract the constant p unconditionally; copy the resulting sign bit to a reduction flag with a CNOT; add p back conditioned on that flag; finally uncompute the flag by a comparison of the result with x , *since a reduction happened if and only if $(x+y) \bmod p < x$* . Two auxiliary qubits: one flag, and one *dirty* ancilla used by the constant adders and returned unchanged. Modular subtraction is the same circuit reversed. To control it, control every operation.

Modular doubling, p odd, $|x\rangle \mapsto |2x \bmod p\rangle$.

Two changes make it cheaper. It works in place on one n -qubit register, so it needs only $n+2$ qubits; and the first integer addition is replaced by a multiplication by 2, implemented as a cyclic bit shift (a chain of swaps built from CNOTs). Because p is odd, the reduction flag is recoverable from the *least significant bit* of the result: $2x - p$ is odd while $2x$ is even. So a subtraction of the modulus has taken place if and only if that bit is 1, and *no persistent ancilla is needed*.

The recurring pattern, once more

Every block ends by recomputing its own flag from its output. That, and nothing else, is what makes the whole tower garbage-free.

17.3 Modular multiplication, squaring, and inversion

Proposition 17.3.1 (Multiplication by doubling and adding). *By Horner's rule,*

$$x \cdot y = \sum_{i=0}^{n-1} x_i 2^i y = \left(\cdots ((x_{n-1}y) \cdot 2 + x_{n-2}y) \cdot 2 + \cdots \right) + x_0 y,$$

so with an accumulator z starting at 0, looping $i = n-1, \dots, 0$ with

$$z \leftarrow 2z \bmod p \text{ (modular doubling),} \quad z \leftarrow (z + y) \bmod p \text{ controlled on } x_i,$$

computes $|x\rangle|y\rangle|0\rangle \mapsto |x\rangle|y\rangle|xy \bmod p\rangle$ in n modular doublings and n controlled modular additions. Squaring saves the y -register by copying bit x_i into one ancilla each round. Montgomery multiplication replaces the conditional subtraction by a truncation plus one addition, roughly halving the Toffoli count at the cost of more ancillas — which are free here, because the inversion circuit already supplies them.

17.3.1 Kaliski's binary extended GCD, made reversible

Algorithm 17.3.2 (Kaliski's almost-inverse). Maintain (u, v, r, s) with the loop invariant $p = rv + su$. Start $u = p, v = x, r = 0, s = 1, k = 0$, and iterate

$$\begin{cases} u \text{ even :} & u \leftarrow u/2, \quad s \leftarrow 2s; \\ v \text{ even :} & v \leftarrow v/2, \quad r \leftarrow 2r; \\ u > v : & u \leftarrow (u-v)/2, \quad r \leftarrow r+s, \quad s \leftarrow 2s; \\ \text{else :} & v \leftarrow (v-u)/2, \quad s \leftarrow s+r, \quad r \leftarrow 2r; \end{cases}$$

incrementing k each round, until $v = 0$. Then $r \equiv x^{-1}2^k \bmod p$ (the *almost inverse*), and $n-k$ controlled modular doublings plus a sign flip convert it to $x^{-1}2^n$, the Montgomery form.

The obstacle: the loop length depends on the input

k lies between n and $2n$ and *depends on x* , so the naive loop is irreversible — different inputs would halt at different times, and a circuit has a fixed shape.

Two devices that make it reversible

1. **Run a fixed $2n$ rounds.** A flag f , initially 1, flips to 0 once $v = 0$ is reached, after which each round merely increments a counter k instead of touching (u, v, r, s) . The circuit shape is therefore *independent of x* . Since $n \leq k \leq 2n$, the counter needs only $\lceil \log_2 n \rceil + 1$ qubits.
2. **Record the branch.** Each round stores which of the four branches was taken in *one* qubit m_i , so the whole computation can be uncomputed after the result is copied out.

Proposition 17.3.3 (Why one qubit per round suffices). *Naively the four branches need two bits, which would cost $4n$ ancillas rather than $2n$. But in each step of Algorithm 17.3.2 exactly one of r and s is even and the other odd — an inductive consequence of the invariant $p = rv + su$. If the updated r is even the branch was “ v even” or “both odd, $u \leq v$ ”; if the updated s is even it was “ u even” or “both odd, $u > v$ ”. So the second encoding bit is recomputable from the register contents and can be uncomputed immediately by a single CNOT from the least significant bit of r .*

Cost, and where the qubits go

Those $2n$ round-record qubits **dominate the qubit count of the whole algorithm**. The inversion needs

$$7n + 2\lceil \log_2 n \rceil + 9 \text{ qubits,} \quad 32n^2 \log_2 n \text{ Toffoli,} \quad (17.1)$$

which is $\Theta(n)$ multiplications’ worth of work — and it is why the point addition is expensive.

Round-function structure. Each round: a CNOT testing whether u is even; another testing v ; a subtractor whose most significant qubit decides $u > v$; a five-gate encoder producing the two branch bits ab (‘10’ for u even, ‘01’ for u odd and v even, ‘11’ for both odd with $u > v$, ‘00’ for both odd with $u \leq v$), where $b = m_i$; a six-gate dispatcher; and a final CNOT from the low bit of r that uncomputes a . After $2n$ rounds the result is copied out and the entire circuit is run backwards.

17.4 Reversible controlled point addition

$P_2 = (x_2, y_2) = [2^j]P$ is a compile-time constant, so every operation involving it is a *constant* operation.

Algorithm 17.4.1 (Controlled addition of a constant point). Registers hold the quantum point (x_1, y_1) , the control qubit, and auxiliaries λ, t_0 . The notation $[\cdot]_1$ shows the value *when* $\text{ctrl} = 1$.

Which steps are controlled. Exactly the five steps whose name begins `ctrl_` (steps 2, 8, 9, 15, 16) are conditioned on the control qubit; every other step — including the inversions, squarings and multiplications that build and clear λ and t_0 — runs *unconditionally*. On the $\text{ctrl} = 0$ branch those steps still compute λ and t_0 and still uncompute them, so all auxiliaries return to $|0\rangle$ and (x_1, y_1) is returned unchanged by steps 1 and 17, which cancel. This is Lemma 13.1.1 again: the residue circuit is the identity on the admissible subspace.

Operation	Effect
1 sub_const $x_1 - x_2$	$x_1 \leftarrow x_1 - x_2$
2 ctrl_sub_const $y_1 - y_2$	$y_1 \leftarrow [y_1 - y_2]_1$
3 inv $x_1 \rightarrow t_0$	$t_0 \leftarrow 1/(x_1 - x_2)$
4 mul $y_1 t_0 \rightarrow \lambda$	$\lambda \leftarrow [y_1 - y_2]_1$
5 mul $\lambda x_1 \rightarrow y_1$	$y_1 \leftarrow 0$
6 inv $x_1 \rightarrow t_0$	$t_0 \leftarrow 0$
7 squ $\lambda \rightarrow t_0$	$t_0 \leftarrow \lambda^2$
8 ctrl_sub $x_1 - t_0$	$x_1 \leftarrow [x_1 - x_2 - \lambda^2]_1$
9 ctrl_add_const $3x_2$	$x_1 \leftarrow [x_2 - x_3]_1$
10 squ $\lambda \rightarrow t_0$	$t_0 \leftarrow 0$
11 mul $\lambda x_1 \rightarrow y_1$	$y_1 \leftarrow [y_3 + y_2]_1$
12 inv $x_1 \rightarrow t_0$	$t_0 \leftarrow [1/(x_2 - x_3)]_1$
13 mul $t_0 y_1 \rightarrow \lambda$	$\lambda \leftarrow 0$
14 inv $x_1 \rightarrow t_0$	$t_0 \leftarrow 0$
15 ctrl_neg x_1	$x_1 \leftarrow [x_3 - x_2]_1$
16 ctrl_sub_const $y_1 - y_2$	$y_1 \leftarrow [y_3]_1$
17 add_const $x_1 + x_2$	$x_1 \leftarrow [x_3]_1$

Checking two steps. Step 9. After step 8 the register holds $x_1 - x_2 - \lambda^2$. Since $x_3 = \lambda^2 - x_1 - x_2$ we have $x_1 - x_2 - \lambda^2 = -x_3 - 2x_2$; adding the constant $3x_2$ gives $x_2 - x_3$.

Step 11. y_1 is 0 after step 5, so it receives $\lambda \cdot (x_2 - x_3)$. By the identity opposite this equals $y_3 + y_2$, which step 16 turns into y_3 .

Why λ can be uncomputed. The slope is recomputable from the output and the constant:

$$\lambda = \frac{y_1 - y_2}{x_1 - x_2} = -\frac{y_3 + y_2}{x_3 - x_2}.$$

Reason: the three collinear points on the chord are P_1, P_2 and $R = (x_3, -y_3)$, so the slope through P_2 and R is $\frac{-y_3 - y_2}{x_3 - x_2}$, and it equals λ .

The λ -uncomputation is the point of the whole design

The same inversion-and-multiply that *computed* λ (steps 3–4) *recomputes* it from the output (steps 12–13) and clears the ancilla. This is Bennett’s compute–copy–uncompute (Theorem 4.3.3) specialised to the group law.

It is **mandatory**: by Proposition 3.10.2, leftover garbage would tag the branch and destroy the interference that produces $y_2 \equiv my_1$. A circuit that skips it does not compute a worse answer — it computes the $1/r$ random baseline.

Note that this is why steps 3–6 and 10–14 are *uncontrolled*. Skipping them when $\text{ctrl} = 0$ would leave step 14 unable to clear t_0 , which would end holding $-(x_1 - x_2)^{-1} \neq 0$ — precisely the dirty ancilla the design exists to avoid.

17.5 Exceptional cases

The three cases the affine formulas do not cover.

- *Doubling* ($P_1 = P_2$) needs the tangent slope $\lambda = (3x_1^2 + a)/(2y_1)$. It does not arise inside a single controlled addition, because the added point is a *constant* generically different from the quantum point — the doublings all happen classically, before the circuit is compiled.
- *The point at infinity* is carried by a flag qubit (a distinguished bit pattern), with the addition conditioned to pass the other summand through when either point is $\mathcal{O}$.
- *Collisions* $x_1 = x_2$ (so $\Delta x = 0$ and no inverse exists) occur on a measure-zero fraction of inputs; the published analysis argues they can be avoided by the choice of curve and base point. Complete Edwards or projective formulas remove the exceptions entirely, at extra cost.

Where approximation is legitimate, and where it is not. The correctness of the measured relation $y_2 \equiv my_1$ requires the oracle to compute $[k]P + [\ell]Q$ on *every branch* of the superposition. A branch on which the affine formulas fail would contribute a wrong or garbage-tagged term. The literature's position is that such branches carry negligible amplitude for well-chosen parameters — and **this is the least rigorous part of the published analysis.**

Garbage-freeness is not in that category. The interference genuinely depends on it, it must be exactly right, and the λ -uncomputation is how it is achieved.

17.6 Worked example over $\mathbb{F}_5$, end to end

Continuing Section 16.4: $E : y^2 = x^3 + 3x + 2$ over $\mathbb{F}_5$, $P = (2, 1)$ of order $r = 5$, $Q = (1, 1)$, true answer $m = 3$. We use the *exact* form (index registers of dimension $r = 5$, QFT₅), which shows the mechanism cleanly.

Oracle values and hidden subgroup. $[0]P = \mathcal{O}$,
 $[1]P = (2, 1)$, $[2]P = (1, 4)$, $[3]P = (1, 1) = Q$,
 $[4]P = (2, 4)$, so

$$f(k, \ell) = [k]P + [\ell]Q = [k + 3\ell \bmod 5]P.$$

For instance

$$f(2, 4) = [2 + 12]P = [14 \bmod 5]P = [4]P = (2, 4)$$

✓. And

$$L = \langle (-3, 1) \rangle = \langle (2, 1) \rangle \leq \mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z},$$

so f is constant exactly when $k + 3\ell$ is constant mod 5.

State evolution.

$$\begin{aligned} \frac{1}{5} \sum_{k, \ell} |k\rangle |\ell\rangle |\mathcal{O}\rangle &\xrightarrow{f} \frac{1}{5} \sum_{k, \ell} |k\rangle |\ell\rangle |[k + 3\ell]P\rangle \\ &\xrightarrow{\text{meas. pt}} \frac{1}{\sqrt{5}} \sum_{\ell} |c - 3\ell\rangle |\ell\rangle \\ &\xrightarrow{\text{QFT}_5^{\otimes 2}} \frac{1}{\sqrt{5}} \sum_{y_1} \omega_5^{cy_1} |y_1\rangle |3y_1\rangle \end{aligned}$$

outcome (y_1, y_2)	check $y_2 \equiv 3y_1$	$m = y_2 y_1^{-1} \bmod 5$	result
(1, 3)	$3 \equiv 3$	$3 \cdot 1 = 3$	$m = 3$ ✓
(2, 1)	$1 \equiv 6$	$1 \cdot 3 = 3$	$m = 3$ ✓
(3, 4)	$4 \equiv 9$	$4 \cdot 2 = 8 \equiv 3$	$m = 3$ ✓
(4, 2)	$2 \equiv 12$	$2 \cdot 4 = 8 \equiv 3$	$m = 3$ ✓
(0, 0)	—	(undefined)	retry

Reading the table, and the contrast with factoring

Since $r = 5$ is prime, every $y_1 \neq 0$ is invertible, so a single run returns $m = 3$ unless $y_1 = 0$ (probability $1/5$). Verification: $[3]P = (1, 1) = Q \checkmark$.

No continued fractions, no divisor ambiguity, no lcm of several runs. For prime-order curves the post-processing is one modular inversion and the success probability is $(r-1)/r \approx 1$. The quantum part, however, is much more expensive per run.

17.7 Resources

Circuit	qubits	ancillas	Toffoli gates
add_const_modp	$2n$	n	$16n \log_2 n - 26.9n$
ctrl_add_const_modp	$2n+1$	n	$16n \log_2 n - 26.9n$
ctrl_sub_modp	$2n+4$	3	$16n \log_2 n - 23.8n$
ctrl_neg_modp	$n+3$	2	$8n \log_2 n - 14.5n$
mul_modp (dbl/add)	$3n+2$	2	$32n^2 \log_2 n - 59.4n^2$
mul_modp (Montgomery)	$5n+4$	$2n+4$	$16n^2 \log_2 n - 26.3n^2$
squ_modp (dbl/add)	$2n+3$	3	$32n^2 \log_2 n - 59.4n^2$
squ_modp (Montgomery)	$4n+5$	$2n+5$	$16n^2 \log_2 n - 26.3n^2$
inv_modp	$7n+2\lceil\log_2 n\rceil+9$	$5n+2\lceil\log_2 n\rceil+9$	$32n^2 \log_2 n$

Per point addition, and for the whole run

The maximum qubit usage occurs during an inversion, so

$$9n + 2\lceil\log_2 n\rceil + 10 \quad \text{logical qubits} \quad (17.2)$$

— the $7n + 2\lceil\log_2 n\rceil + 9$ of the inversion, plus one control qubit, plus $2n$ for the two registers holding intermediate results.

Algorithm 17.4.1 contains **4 inversions, 2 squarings and 4 multiplications**, so the leading coefficient is $4 \cdot 32 + 2 \cdot 16 + 4 \cdot 16 = 224$:

$$\lesssim 224n^2 \log_2 n + 2045n^2 \quad \text{Toffoli per addition,} \quad (17.3)$$

and for the whole ECDLP run ($2n$ additions, sequential),

$$(448 \log_2 n + 4090) n^3 \quad \text{Toffoli.} \quad (17.4)$$

Where the cost is, precisely. The inversion supplies the $2n$ round-record ancillas that dominate the *qubit* count; the modular multiplications inside the point additions dominate the *Toffoli* count. Not any Fourier transform: the two QFTs are $\Theta(n^2)$ exact, or $O(n \log n)$ approximate — negligible.

Part V

Cost, Specification, and Context

Resource estimates and cryptographic impact

The asymptotics — $\Theta(n^3)$ Toffoli gates on $\Theta(n)$ qubits — have not moved since 1994. The constants have moved by orders of magnitude, and it is the constants that decide whether RSA falls in this century. This chapter collects the numbers.

18.1 Logical resources for factoring

Stage	Cost
$H^{\otimes t}$	$t = 2n + O(1)$ gates
modular exponentiation	$\Theta(n^3)$ Toffoli
QFT [†] exact / approximate	$\Theta(n^2) / O(n \log n)$ gates
continued fractions	$O(n^3)$ bit operations (classical)
total quantum	$\Theta(n^3)$ Toffoli, $\Theta(n)$ qubits

Concretely $\approx 0.3n^3$ Toffoli with windowed arithmetic; $T \approx 4 \times$ the Toffoli count.

Two Toffoli counts for RSA-2048, and why they differ by $\sim 2000 \times$

Two figures circulate for the same problem, and conflating them is a common error.

Construction	RSA-2048	What it is
Gidney–Ekerå, windowed	$\approx 0.3n^3 = 2.6 \times 10^9$	Windowed arithmetic (Section 12.6), $\approx 3n$ logical qubits, optimised for <i>spacetime volume</i> . This is the figure used for the “QFT is 0.003%” comparison in Chapter 12.
Häner–Roetteler–Svore, $2n + 2$ qubits	5.2×10^{12}	A <i>minimal-width</i> construction: only $2n + 2$ qubits, achieved by giving up windowing and paying $\approx 605n^3$ Toffoli. The table below is this one.

The two optimise different things — gates versus qubits — and both are correct for their own design point. Always state which construction a count belongs to. The “ $\approx 5 \times 10^{12}$ ” quoted later in this chapter is the Häner–Roetteler–Svore figure.

18.1.1 Simulated counts (Häner–Roetteler–Svore, minimal width)

One modular constant multiplication costs $32n^2(\log_2 n - 2) + 14.73n^2$ Toffoli, and the whole factoring run — $2n$ of them — costs $(64(\log_2 n - 2) + 29.46)n^3$.

$\lceil \log_2 N \rceil$	qubits	Toffoli gates
512	1 026	6.41×10^{10}
1 024	2 050	5.81×10^{11}
2 048	4 098	5.20×10^{12}
3 072	6 146	1.86×10^{13}
7 680	15 362	3.30×10^{14}
15 360	30 722	2.87×10^{15}

18.2 Twenty years of constant factors

	Beauregard 2003	Gidney–Ekerå 2021	Gidney 2025
arithmetic	Fourier (Draper)	ripple-carry, windowed	approximate residues
logical qubits	$2n + 3$	$\approx 3n$	$\approx 1.4n$
Toffoli count	$O(n^3 \log n)$	$\approx 0.3n^3$	$> 100\times$ fewer than 2024 alternatives
key technique	QFT adder	temporary AND; windowing	Chevignard–Fouque–Schrottenloher residue arithmetic
<i>Physical estimate for RSA-2048; gate error 10^{-3}, cycle time $1\ \mu\text{s}$:</i>			
physical qubits	—	20 million	< 1 million
runtime	—	8 hours	< 1 week
enabling ideas	—	lattice surgery; magic-state distillation	yoked surface codes; magic state cultivation

What moved, and what did not

The asymptotics ($\Theta(n^3)$ Toffoli, $\Theta(n)$ qubits) have not changed since 1994. The *qubit count for RSA-2048 has fallen by more than an order of magnitude in six years* — from 20 million to under 1 million — entirely through better constants: approximate residue arithmetic, storing idle logical qubits in yoked surface codes, and allocating less space to magic state production by using cultivation rather than distillation. The longer runtime in the 2025 estimate is the trade: more Toffoli gates, fewer magic state factories.

What these numbers are, and are not

The physical estimates assume a *specific, uniformly good* hardware model: a square grid of qubits with nearest-neighbour connectivity, a uniform 10^{-3} physical gate error rate (below the surface-code threshold), a $1\ \mu\text{s}$ surface-code cycle time, a $10\ \mu\text{s}$ control-system reaction time, and fast classical decoding.

No existing device meets all of these at the required scale. The estimates answer “how much hardware of this quality would be needed”, *not* “when will it exist”. They are nevertheless the right quantity for planning a migration to post-quantum cryptography, because they bound from below what an adversary must build.

18.3 Elliptic curves versus RSA

	ECDLP, P-256	RSA-3072	ratio
classical security	≈ 128 bit	≈ 128 bit	matched
field / modulus size n	256	3072	$12\times$
logical qubits	2330	6146	$0.38\times$
Toffoli gates	1.26×10^{11}	1.86×10^{13}	$0.007\times$

ECC bits	qubits	Toffoli	RSA bits	qubits
110	1014	9.4×10^9	512	1026
160	1466	3.0×10^{10}	1024	2050
224	2042	8.4×10^{10}	2048	4098
256	2330	1.3×10^{11}	3072	6146
384	3484	4.5×10^{11}	7680	15362
521	4719	1.1×10^{12}	15360	30722

The argument in one paragraph

At comparable classical security a 256-bit elliptic curve matches a 3072-bit RSA modulus, but both quantum attacks scale as $\Theta(n^3)$ in the *operand* size n . The $12\times$ smaller n therefore makes the ECDLP dramatically cheaper — in qubits (2330 versus 6146) and by two orders of magnitude in Toffoli gates — *despite* each elliptic point addition being individually far harder than a modular multiplication, because of the inversion.

In short: the smaller operand size that makes ECC attractive classically is what makes it the cheaper quantum target.

Caveats on the comparison

These are *logical*-resource estimates for a uniformly good hardware model; physical overheads (error correction, magic-state production) multiply both columns similarly. The Ekerå–Håstad argument divides the RSA Toffoli column by 4 but does not apply to the ECC column. And a hybrid quantum-classical number field sieve using $n^{2/3+o(1)}$ qubits would shift the security alignment in a way that complicates the analysis further.

18.4 Variants that change the picture

Variant	What it changes
Ekerå–Håstad	Reduces the phase-estimation rounds from $2n$ to $n/2$ at the cost of a lattice-enumeration post-processing step, dividing every Toffoli count by 4 <i>without saving qubits</i> . Applies to RSA (where the exponent structure is known) but not to a general ECDLP.
Regev	Replaces the one-dimensional order-finding structure with a d -dimensional one, reducing the per-run circuit to $\tilde{O}(n^{3/2})$ gates at the price of $\sqrt{n} + 4$ runs and a lattice-reduction post-processing step. Total work is still $\tilde{O}(n^2)$, so the gain is in <i>circuit depth</i> — which is what coherence budgets care about. Regev’s original construction needs $O(n^{3/2})$ qubits, worse than Shor.
Ragavan–Vaikuntanathan	Brings Regev’s qubit count back to $O(n \log n)$ using Fibonacci-based reversible exponentiation — a pure Chapter 12 technique.
Chevignard–Fouque–Schrottenloher	Approximate residue arithmetic, which is the main source of Gidney’s 2025 qubit reduction.

18.5 Cryptographic impact and migration

Scheme family	Status under a large fault-tolerant quantum computer
RSA, finite-field Diffie–Hellman, DSA, ECDH, ECDSA	Broken. All are instances of the abelian hidden subgroup problem and fall to the machine built in this document. One mechanism, four deployed cryptosystems.
Symmetric ciphers (AES), hash functions	Weakened, not broken. Only Grover’s algorithm applies — a <i>quadratic</i> speed-up by a different mechanism (amplitude amplification, not eigenvalue estimation). Doubling the key length restores the margin.
Lattice-, code-, isogeny- and hash-based schemes	No known abelian-HSP attack. These are the basis of the post-quantum standards.

On timelines

Two things are true at once, and conflating them produces bad planning.

(1) No cryptographically relevant number has been factored by Shor’s algorithm, and none is close. The gap between a 12-qubit, 260-two-qubit-gate demonstration and RSA-2048 — which needs $\approx 5 \times 10^{12}$ Toffoli gates and, with error correction, on the order of 10^6 physical qubits at circuit depth $\sim 10^9$ — is nine to ten orders of magnitude in both dimensions.

(2) Anything encrypted today that must stay secret for a decade should *already* be migrating, because of **harvest-now-decrypt-later**: an adversary can record ciphertext today and decrypt it whenever the machine arrives. The relevant question is not “when will the machine exist” but “how long must this data stay secret, and how long will migration take”.

A note on the Las Vegas property under noise

Because every candidate order is verified classically ($a' \stackrel{?}{\equiv} 1$) and every candidate factor by division, *noise can only waste runs, never produce a wrong factorisation*. The Las Vegas property of

Theorem 10.5.1 holds on hardware as well as in theory.

On a current device the practical consequence is visible: for a 12-qubit $N = 15$ circuit with 260 two-qubit gates, out of 1024 shots only about 181 land on the four correct bitstrings — the rest leak across the other 252 outcomes. *The signal survives; the contrast does not.* A threshold filter recovers the answer, and the classical verification guarantees it is right.

Chapter 19

Formal specifications of every subroutine

Textbook presentations of Shor’s algorithm are written as a *procedure*: a numbered sequence of steps, each annotated with the state of the system at that point. That style — the Nielsen–Chuang presentation is the canonical example — is already halfway to a formal specification. This chapter completes the passage, writing each subroutine as a Hoare triple

$$\{\text{precondition}\} \quad \text{Circuit} \quad \{\text{postcondition}\}$$

together with its resource bound. The point is not formality for its own sake. It is that [Chapters 13 and 17](#) produced several circuits whose correctness depends on a precondition that is *invisible in the diagram* — most notably $b < N$ in [Theorem 13.4.1](#), whose violation silently leaves an ancilla dirty and silently destroys the interference. Writing the preconditions down is how such bugs stop being silent.

The anatomy of a specification

A specification has three parts, following the classical style:

1. **Inputs** — register sizes, initial states, oracle assumptions, and any *classical* preconditions on the parameters.
2. **Functional correctness** — the input/output relation, stated as a postcondition on the joint state, together with the probability with which it holds.
3. **Complexity** — a runtime and width bound.

A *procedure* then decomposes the circuit into blocks, and each block carries intermediate assertions describing the state of the system at that step. Functions decorated with pre- and postconditions in this way are exactly what an annotated-programming environment (Why3, Qbricks, and similar) checks mechanically.

19.1 The top-level specification

SHOR-FACTOR

Inputs. A composite integer N , odd and not a prime power, with $n = \lceil \log_2 N \rceil$.

Outputs. A non-trivial factor d of N (so $1 < d < N$ and $d \mid N$).

Runtime. $\tilde{O}(n^3)$ quantum operations on $\Theta(n)$ qubits; succeeds with probability $\Omega(1)$ per attempt, expected $O(1)$ attempts.

Procedure.

- | | |
|---|-----------------------------|
| 1. N even $\Rightarrow$ return 2 | trivial case |
| 2. $N = b^c \Rightarrow$ return b | prime-power case |
| 3. $a \leftarrow \mathcal{U}\{2, \dots, N-1\}$ | random base |
| 4. $d \leftarrow \text{gcd}(a, N)$; $d > 1 \Rightarrow$ return d | lucky guess |
| 5. $r \leftarrow \text{ORDER-FIND}(N, a)$ | quantum |
| 6. r odd $\vee a^{r/2} \equiv -1 \Rightarrow$ goto 3 | unlucky base |
| 7. return $\text{gcd}(a^{r/2} \mp 1, N)$ | Lemma 9.1.1 |

Key invariant. Steps 1–2 establish N odd, composite, with $m \geq 2$ distinct prime factors — *the hypothesis of Theorem 9.2.1*, without which the $\frac{1}{2}$ bound fails. Step 4 establishes $\text{gcd}(a, N) = 1$, the hypothesis of [Proposition 5.5.2](#) and of [Theorem 13.2.1](#) (existence of $\bar{a}$).

Las Vegas. Steps 6–7 verify; a wrong value is rejected, never returned.

19.2 The quantum core

ORDER-FIND

Inputs. (1) An oracle U_a performing $|j\rangle|k\rangle \mapsto |j\rangle|a^j k \bmod N\rangle$, for a coprime to the L -bit number N ; (2) $t = 2L + 1 + \lceil \log(2 + \frac{1}{2\varepsilon}) \rceil$ qubits initialised to $|0\rangle$; and (3) L qubits initialised to the state $|1\rangle$. (With the minimal choice $t = \lceil 2\log_2 N \rceil$ the same specification holds with $1 - \varepsilon$ replaced by $4/\pi^2$; see [Section 11.1](#).)

Outputs. The least integer $r > 0$ such that $a^r \equiv 1 \pmod{N}$.

Runtime. $O(L^3)$ operations. Succeeds with probability $O(1)$.

Procedure.

1. $|0\rangle|u\rangle$ initial state
2. $\rightarrow \frac{1}{\sqrt{2^t}} \sum_{j=0}^{2^t-1} |j\rangle|1\rangle$ create superposition
3. $\rightarrow \frac{1}{\sqrt{2^t}} \sum_{j=0}^{2^t-1} |j\rangle|a^j \bmod N\rangle$ apply U_a
 $\approx \frac{1}{\sqrt{r2^t}} \sum_{s=0}^{r-1} \sum_{j=0}^{2^t-1} e^{2\pi i s j / r} |j\rangle|u_s\rangle$ [Theorem 10.3.1](#)
4. $\rightarrow \frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |\widetilde{s/r}\rangle|u_s\rangle$ apply $\text{QFT}^\dagger$ to register 1
5. $\rightarrow \widetilde{s/r}$ measure register 1
6. $\rightarrow r$ continued-fraction algorithm

Preconditions. $\gcd(a, N) = 1$; $2^t \geq N^2$ (else [Corollary 11.1.1](#) fails and step 6 is not guaranteed to return s/r); the work register is exactly $|1\rangle$, so that [Theorem 10.1.1](#) applies.

Postcondition. $\Pr[\text{output} = \text{ord}_N(a)] \geq (1 - \varepsilon) \cdot \phi(r)/r$, and whenever a value is output it has been verified to satisfy $a^r \equiv 1$.

19.2.1 Block-level assertions

Each block of [Algorithm 10.2.1](#) carries a pre/post pair. These are the assertions a verification tool discharges.

create_superposition

pre: $\text{state} = |0\rangle^{\otimes t} \otimes |u\rangle$

apply $H^{\otimes t}$ to register 1

post: $\text{state} = \frac{1}{\sqrt{2^t}} \sum_{j=0}^{2^t-1} |j\rangle \otimes |u\rangle$

Cost: t Clifford gates. *Proof:* [Proposition 6.1.3](#).

modular_exponentiation

$$\text{pre: state} = \frac{1}{\sqrt{Q}} \sum_j |j\rangle \otimes |1\rangle, \quad \gcd(a, N) = 1$$

apply $\prod_{\ell} c\text{-}U_{a_{\ell}}$ with $a_{\ell} = a^{2^{\ell}} \bmod N$ (classical)

$$\text{post: state} = \frac{1}{\sqrt{Q}} \sum_j |j\rangle \otimes |a^j \bmod N\rangle, \quad \text{all ancillas} = |0\rangle$$

Cost: $\Theta(n^3)$ Toffoli, or $\approx 0.3n^3$ windowed. *Proof:* [Theorem 12.4.3](#). *Critical side condition:* **all ancillas restored** — by [Proposition 3.10.2](#) a violation destroys the interference in the next block, silently.

inverse_qft_and_measure

$$\text{pre: state} = \frac{1}{\sqrt{r}} \sum_s \left(\frac{1}{\sqrt{Q}} \sum_j e^{2\pi i s j / r} |j\rangle \right) \otimes |u_s\rangle$$

apply $\text{QFT}_Q^{\dagger}$ to register 1; measure register 1

$$\text{post: } \Pr[\ell] = \frac{1}{r} \sum_s \frac{1}{Q^2} \frac{\sin^2(\pi Q \delta_{s,\ell})}{\sin^2(\pi \delta_{s,\ell})}, \quad \delta_{s,\ell} = \frac{s}{r} - \frac{\ell}{Q}$$

Cost: $\Theta(t^2)$ exact, $O(t \log t)$ approximate. *Proof:* [Theorems 8.3.2](#) and [8.5.1](#). *Guarantee:* $\Pr[\text{nearest grid point}] \geq 4/\pi^2$ per branch ([Theorem 8.3.4](#)).

19.3 Arithmetic blocks

cdkm_adder

$$\{ c_0 = 0 \wedge z = 0 \}$$

MAJ chain; CNOT; UMA chain

$$\{ \text{reg}_b = (a + b) \bmod 2^n \wedge z = s_n \wedge c_0 = 0 \wedge \text{reg}_a = a \}$$

Cost: $2n$ Toffoli, $4n + 1$ CNOT, $2n + 2$ qubits, $4n$ T with temporary-AND, **zero garbage**. *Proof:* [Theorem 12.1.4](#).

modular_adder (constant $a < N$)

$$\{ 0 \leq x < N \wedge t = 0 \}$$

7-block circuit of [Theorem 12.3.1](#)

$$\{ \text{reg} = (x + a) \bmod N \wedge t = 0 \}$$

Cost: 5 constant additions + $O(1)$ Clifford; **one** ancilla, cleared.

Proof: [Theorem 12.3.1](#). *Clearing argument:* $t = [R \geq a]$ is recomputable from the output alone.

doubly_controlled_modular_adder — the load-bearing precondition

$$\{ 0 \leq b < N \wedge t = 0 \}$$

3-of-7 controlled circuit of [Theorem 13.4.1](#)

$$\{ \text{reg} = (b + c_1 c_2 a) \bmod N \wedge t = 0 \}$$

Cost: 3 of 7 blocks controlled; $12.8\times$ cheaper than controlling every gate.

Why this box is different

The precondition $b < N$ is used in *both* branches of the proof — the $c_1 c_2 = 1$ branch to conclude $R - a < 0 \iff t = 0$, and the $c_1 c_2 = 0$ branch to conclude that the sign qubit of b is 0.

Feed this circuit a value $\geq N$ and it *silently leaves t dirty*. The circuit still runs. It is still unitary. It computes the wrong distribution. **This is precisely the class of bug that a written precondition catches and a circuit diagram does not.**

in_place_multiplier $c \cdot U_a$

$$\{ 0 \leq x < N \wedge \gcd(a, N) = 1 \wedge \text{reg}_b = 0 \}$$

$$\text{CMULT}(\bar{a})^\dagger \cdot c\text{-SWAP} \cdot \text{CMULT}(a)$$

$$\{ \text{reg}_x = a^c x \bmod N \wedge \text{reg}_b = 0 \}$$

Cost: 2 out-of-place multipliers + n Fredkin gates; $\Theta(n^2)$ Toffoli.

Proof: [Theorem 13.2.1](#). *Preconditions:* $\gcd(a, N) = 1$ is what makes $\bar{a}$ exist and therefore what makes an in-place circuit *possible at all*.

Known limitation: the postcondition holds only on the admissible subspace $\{x < N\}$; on padding values $x \geq N$ the circuit is a different unitary ([Chapter 13](#), Warning 1). Harmless in Shor because the orbit of $|1\rangle$ never leaves $(\mathbb{Z}/N\mathbb{Z})^\times$.

19.4 Elliptic-curve blocks

controlled_point_addition

$$\{ (x_1, y_1) \in E(\mathbb{F}_p) \setminus \{\mathcal{O}\} \wedge (x_1, y_1) \neq \pm P_2 \wedge \lambda = t_0 = 0 \}$$

17-step circuit of [Algorithm 17.4.1](#)

$$\{ (x_1, y_1) = [(x_1, y_1) + c P_2] \wedge \lambda = t_0 = 0 \}$$

Cost: $9n + 2\lceil \log_2 n \rceil + 10$ qubits; $\lesssim 224n^2 \log_2 n + 2045n^2$ Toffoli; 4 inversions, 2 squarings, 4 multiplications.

Ancilla-clearing argument: $\lambda = -\frac{y_3 + y_2}{x_3 - x_2}$ is recomputable from the *output* and the constant, because P_1, P_2 and $(x_3, -y_3)$ are collinear.

Unmet obligations: the precondition $(x_1, y_1) \neq \pm P_2$ (no collision, so $\Delta x \neq 0$ and the inverse exists) is argued in the literature to hold on all but a measure-zero fraction of branches. **This is the least rigorous part of the published analysis** and is flagged as such in [Chapter 17](#).

modular_inverse (Kaliski)

$$\{ \gcd(x, p) = 1 \wedge \text{round records} = 0 \wedge k = 0 \}$$

$2n$ fixed rounds; copy out; run backwards

$$\{ \text{out} = x^{-1} 2^n \bmod p \wedge \text{all scratch} = 0 \}$$

Cost: $7n + 2\lceil \log_2 n \rceil + 9$ qubits, $32n^2 \log_2 n$ Toffoli.

Reversibility argument: the loop length is data-dependent ($n \leq k \leq 2n$), so the circuit runs a *fixed* $2n$ rounds with a flag that freezes the state once $v = 0$; and one qubit per round records the branch, which suffices because exactly one of r, s is even at each step ([Proposition 17.3.3](#)).

19.5 The whole algorithm as one judgement

The following is the specification of the *factoring* circuit as it appears in the formal-verification literature (the Qbricks development), reproduced here because it shows what a fully discharged, machine-checked statement of Shor’s algorithm actually looks like — functional correctness, probabilistic correctness and all three resource bounds in a single judgement.

SHOR-CIRC — the mechanised factoring judgement

$$\begin{aligned} & \Gamma(a, b, n : \text{int}), (j : \text{ghost int}) \vdash \\ & \left\{ \text{co_prime}(a, b) \wedge 1 \leq b < 2^n \wedge 1 \leq j < b \wedge a^j \bmod b = 1 \right\} \\ & \text{SHOR-CIRC}(a, b, n) \\ & \left\{ \begin{array}{l} \text{proba_partial_measure_p}(|1\rangle_n, \text{error} \leq \frac{1}{2^{2n+1}}) \geq \frac{4}{\pi^2} \wedge \\ \text{size}(\text{result}) = \text{SHOR-POLY}(n) \wedge \\ \text{ancillas}(\text{result}) = n + 2 \wedge \text{width}(\text{result}) = 3n \end{array} \right\} \end{aligned}$$

Here $\text{proba_partial_measure_p}(|1\rangle_n, \text{error} \leq \frac{1}{x})$ denotes the probability that the measured outcome ℓ approximates the target ratio to within $1/x$, i.e. $|\ell/2^t - j/r| < 1/x$ with $x = 2^{2n+1}$ — the hypothesis of [Corollary 11.1.1](#); size, width and ancillas are the gate count, qubit count and ancilla count; and $\text{SHOR-POLY}(n)$ is polynomial in n (degree 4 for the schoolbook construction of [Chapter 12](#)).

The ECDLP specification has the same shape, with the postcondition replaced by the two-dimensional relation of [Theorem 15.2.1](#) and the resource bounds by [Equations \(17.2\) and \(17.4\)](#):

SHOR-ECDLP

Inputs. $E(\mathbb{F}_p)$ with $p > 3$ prime, a point P of prime order r , and $Q = [m]P$; two index registers of $n + 1$ qubits ($n = \lceil \log_2 p \rceil$, sufficient by [Proposition 16.3.1](#)) and a point register in $|\mathcal{O}\rangle$.

Postcondition. The measured pair (y_1, y_2) satisfies $y_2 \equiv m y_1 \pmod{r}$ after rounding, and $m \equiv y_2 y_1^{-1}$; the returned $\hat{m}$ has been *verified* to satisfy $[\hat{m}]P = Q$. Success probability $(r - 1)/r$ per run for prime r .

Resources. $9n + 2\lceil \log_2 n \rceil + 10$ logical qubits; $(448 \log_2 n + 4090)n^3$ Toffoli gates.

Preconditions carried through every point addition. $(x_1, y_1) \neq \pm P_2$ (so $\Delta x \neq 0$); all auxiliaries $\lambda = t_0 = 0$ on entry and exit.

What a specification buys

1. **It separates the three claims.** *Functional correctness* (the state relation), *probabilistic correctness* (the $4/\pi^2$ bound), and *resource bounds* (size, width, ancillas) are logically independent, and a circuit can satisfy any two while failing the third. Most informal presentations blur them.
2. **It surfaces preconditions that diagrams hide.** $b < N$ in [Theorem 13.4.1](#); $d = 0$ in [Proposition 13.5.1](#); $\gcd(a, N) = 1$ in [Theorem 13.2.1](#); $\Delta x \neq 0$ in [Algorithm 17.4.1](#). Every one of these has been the source of a real implementation error.
3. **It makes the ancilla obligation explicit.** “All ancillas returned to $|0\rangle$ ” is a postcondition like any other, and by [Proposition 3.10.2](#) it is not a matter of hygiene but of correctness.
4. **It is mechanically checkable.** Embedding these triples in a proof environment such as Why3 — the approach taken by the Qbricks project — turns “the ancilla is restored” from a claim in prose into a discharged proof obligation.

Testing discipline that follows from the specifications

1. Verify $\text{QFT}^\dagger \circ \text{QFT} = \mathbb{I}$ numerically (catches the swap layer and bit order).
2. Then the adder *exhaustively* for $n \leq 6$ — all 2^{2n} operand pairs, checking ancilla restoration.
3. Then `cc_add_mod` over all (x, k, ctrl) for small N , checking ancilla cleanliness.
4. Then the bit order *end to end* on a known basis state.
5. Compare the hand-optimised controlled circuit against the naive `.control()` version *restricted to the admissible subspace* — comparing full unitaries reports false failures ([Proposition 13.5.1](#)), and forgetting to check the ancilla misses real ones.

The abelian hidden subgroup problem, and where it stops

20.1 One mechanism, several broken cryptosystems

The unifying statement

Factoring (hidden group $\mathbb{Z}$, via $\mathbb{Z}/r\mathbb{Z}$), the finite-field discrete logarithm, and the ECDLP (both on $\mathbb{Z}/r\mathbb{Z} \times \mathbb{Z}/r\mathbb{Z}$) are *all* instances of the abelian hidden subgroup problem.

In every case the group is **abelian**, so its irreducible representations are **one-dimensional**, so each eigenvalue is a **single phase** — the fact underlying every line of [Chapter 7](#) through [Chapter 17](#).

Problem	Group A , hidden subgroup L	Oracle
order finding / factoring	$\mathbb{Z}/Q\mathbb{Z}$, $L = r\mathbb{Z}$	$x \mapsto a^x \bmod N$
finite-field DLP	$\mathbb{Z}/r\mathbb{Z} \times \mathbb{Z}/r\mathbb{Z}$, $L = \langle (-m, 1) \rangle$	$(k, \ell) \mapsto g^k h^\ell \bmod p$
ECDLP	$\mathbb{Z}/r\mathbb{Z} \times \mathbb{Z}/r\mathbb{Z}$, $L = \langle (-m, 1) \rangle$	$(k, \ell) \mapsto [k]P + [\ell]Q$
Simon's problem	$\mathbb{F}_2^n$, $L = \{0, s\}$	$x \mapsto f(x)$ with $f(x) = f(x \oplus s)$
Deutsch–Jozsa	$\mathbb{F}_2^n$	a balanced/constant function

Where it fails

For **non-abelian** groups the irreducible representations have dimension > 1 : an eigenvalue becomes a *matrix*, and there is no single phase to kick back. [Lemma 7.2.1](#) has no analogue, and the entire chain breaks at link 5.

No efficient quantum algorithm is known for the non-abelian HSP in general — which is why graph isomorphism (symmetric group) and lattice problems (dihedral group) remain open, and why post-quantum cryptography is built on them.

This is not a footnote. It is the reason the post-quantum standards are believed to work.

20.2 What each ingredient contributes — and what it does not

Ingredient	Contribution, and its limit
Superposition	Exposes the whole orbit at once as the state $ 1\rangle$. It does <i>not</i> , by itself, “try all answers” — an unmeasured uniform superposition is information-free (Section 3.8).
Linearity	Lets phase estimation act on every eigenbranch of $ 1\rangle$ simultaneously; it is why a non-eigenstate input works at all (Theorem 8.5.1).
Complex relative phase	The <i>medium</i> carrying the eigenvalue. A real-amplitude theory could not encode s/r in a phase, since eigenvalues of unitaries are points on the unit circle (Corollary 2.5.2(b)).
Tensor products	Give the exponentially large index space and the big-endian ladder $e^{2\pi ix\varphi}$. They do not, alone, create speed-up.
Controlled unitaries	Convert an eigenvalue into a relative phase. They do <i>not</i> force the target into an eigenstate (Chapter 7).
Entanglement	Tags each ladder with its branch; it suppresses cross-branch interference and is necessary <i>bookkeeping</i> , not the engine. Gottesman–Knill (Theorem 4.4.2) shows it cannot be sufficient.
Interference	Coherent addition of the x -indexed amplitudes — what actually concentrates the distribution. <i>Name the two contributions</i> ; do not say “the phases interfere”.
Measurement	Returns one sample ℓ . It does not read the whole spectrum and does not uncover a pre-existing classical value.
Reversible modular arithmetic	Makes the controlled powers cheap, via classical repeated squaring or double-and-add. This, not the QFT’s gate count, is where the algorithm spends its time.

Three principles

- P1.** Garbage is which-path information, so it kills interference — hence every arithmetic block must be uncomputed, and the elliptic point addition must recompute λ from its own output.
- P2.** The cost is arithmetic $\Theta(n^3)$, not Fourier $\Theta(n^2)$ (or $O(n \log n)$ approximate).
- P3.** The speed-up needs *all three* of: a structured input, cheap controlled powers, and interference — and it needs the group to be **abelian**.

20.3 Six common misstatements, and precise replacements

Imprecise	Precise
<i>It tries all possible answers at once.</i>	It computes on all inputs at once, but a measurement returns a <i>single</i> bitstring — so the algorithm has to arrange interference first. An unmeasured uniform superposition carries zero information.
<i>The QFT computes the whole Fourier spectrum.</i>	It applies the Fourier <i>change of basis</i> to the amplitudes. The Q coefficients are never made classically accessible; one outcome is sampled. That is also why it is not a fast FFT.
<i>Entanglement is the source of the speed-up.</i>	Entanglement appears when the oracle correlates the registers, and its role is to keep the r eigenphase branches from mixing. The advantage comes from interference <i>within</i> each branch, plus the fact that controlled powers are cheap. Gottesman–Knill settles this.
<i>Measurement reveals the value the qubit already had.</i>	There was no such value. The relative phases were physically real and had to be converted into magnitudes first — that conversion is what the inverse QFT does.
<i>The phases interfere.</i>	The contributions $V_{yx}\alpha_x$ and $V_{yx'}\alpha_{x'}$ to the outcome $ y\rangle$ add with relative phase $\theta - \varphi$. If the two contributions cannot be named, there is no interference to discuss.
<i>A qubit is 0 and 1 at the same time.</i>	A qubit's state is a unit vector in $\mathbb{C}^2$. The two coefficients are complex amplitudes; only their squared moduli are probabilities, and the <i>relative phase</i> between them is physical — that is the part with no classical analogue.

20.4 Analogies: which to use, and exactly where each breaks

Analogy	What it gets right	Where it breaks
Diffraction grating	Exact: $\Pr[y]$ is the Q -slit intensity pattern, same formula (Equation (8.5))	The slits are basis states, not apertures; nothing propagates
Arrows added head to tail	Exact: β_y is a sum of complex numbers	Nothing — use it freely; no significant failure mode
Tuning a radio dial	Fourier is a test for frequency match	You get one <i>sample</i> , and only after all the tests happened coherently; there is no spectrum to look at
Musical beats / resonance	Constructive vs. destructive addition	The index x is not time; nothing oscillates physically
Combination lock whose wrong turns cancel	Conveys interference as <i>active</i> cancellation	Suggests checking candidates one at a time — wrong candidates don't get checked and rejected, their amplitudes destroy each other

20.5 Fourteen review questions

1. Why does factoring reduce to order finding?
2. Why is modular multiplication a cyclic translation?
3. Why do roots of unity appear — and why are they *forced*?
4. Why is a unitary's eigenvalue always a pure phase?
5. Why is the Fourier basis a change of basis, not a trick?
6. Why are all probabilities uniform *before* the inverse QFT?
7. Why does the free state $|1\rangle$ work as input?
8. Why does $c-U$ *not* project the target onto an eigenstate?
9. Why does the inverse QFT produce a peak, and how wide is it?
10. Why does $N = 15$ hide the continued-fraction step, and $N = 21$ not?
11. Why does the ECDLP need two QFTs and no continued fractions?
12. Why is ECC-256 a cheaper quantum target than RSA-3072 despite the harder oracle?
13. Why must the oracle be garbage-free, and what exactly breaks otherwise?
14. Why is the QFT the cheapest part of the algorithm?

Where each answer lives. 1–2: Chapter 9. 3–5: Chapter 5. 6–9: Chapters 7 and 8. 10: Chapter 14. 11–12: Chapters 15 and 18. 13: Chapters 4 and 17. 14: Chapter 12.

Self-assessment

Answering a question never seen before by *re-deriving* rather than searching memory; being able to say, without preparation, what each ingredient contributes *and what it does not*; and being able to catch an error in someone else's explanation and *name which link of the chain it broke*.

20.6 The mechanism in one line

modular periodicity $\rightarrow$ cyclic shift $\rightarrow$ character eigenvectors
 $\rightarrow$ roots-of-unity eigenvalues $\rightarrow$ phase kickback $\rightarrow$ binary phase ladder
 $\rightarrow$ inverse-QFT unwinding $\rightarrow \frac{\ell}{Q} \approx \frac{s}{r} \rightarrow$ continued fractions $\rightarrow r \rightarrow$ factors

Multiplication by a modulo N restricts, on the orbit of 1, to the cyclic shift S ; S is diagonalised by the characters of $\mathbb{Z}/r\mathbb{Z}$, whose eigenvalues are the r -th roots of unity $e^{2\pi is/r}$; controlled powers of U_a kick these eigenphases onto the counting register as a binary phase ladder $e^{2\pi ixs/r}$; crucially $|1\rangle$ is the uniform superposition of all r eigenvectors, so one runs phase estimation *without knowing* r ; the inverse QFT unwinds each ladder into a peak at $\ell \approx Qs/r$; continued fractions promote ℓ/Q to s/r ; and r yields the factors.

And the two-dimensional version:

$$f(k, \ell) = [k]P + [\ell]Q = [k + m\ell]P \text{ hides } L = \langle (-m, 1) \rangle$$

$$\xrightarrow{\text{QFT}_r \otimes \text{QFT}_r} \text{ samples a character trivial on } L$$

$$\longrightarrow y_2 \equiv my_1 \pmod{r} \longrightarrow m \equiv y_2 y_1^{-1}, \text{ one run for prime } r.$$

Appendices

Appendix A

Sums of roots of unity and elementary inequalities

Every derivation in this document rests on the two lemmas below. They are collected here so that each can be cited by name rather than re-derived.

Lemma A.0.1 (Geometric sum / character orthogonality). *For $\ell \in \mathbb{Z}$ and $M \geq 1$, with $\omega_M = e^{2\pi i/M}$,*

$$\sum_{k=0}^{M-1} \omega_M^{k\ell} = \begin{cases} M, & \ell \equiv 0 \pmod{M}, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{A.1})$$

Proof. If $\ell \equiv 0$ every term is 1. Otherwise $z := \omega_M^\ell \neq 1$ and $\sum_{k=0}^{M-1} z^k = \frac{1-z^M}{1-z} = 0$, since $z^M = \omega_M^{\ell M} = 1$. $\square$

What generates

Unitarity of the QFT ([Proposition 6.1.2](#)); orthonormality of the shift eigenvectors ([Theorem 5.3.1](#)); the identity $|1\rangle = \frac{1}{\sqrt{r}} \sum_s |u_s\rangle$ ([Theorem 10.1.1](#)); exactness of QPE on dyadic phases ([Theorem 8.2.1](#)); the exact zeros of the Fejér kernel ([Theorem 8.3.2](#)); and the entire ECDLP measurement relation ([Theorem 15.2.1](#)).

It is by a wide margin the most frequently used identity in this document. *Geometrically:* the M -th roots of unity taken ℓ at a time form a balanced closed polygon, and a closed polygon of vectors sums to zero.

Lemma A.0.2 (Partial geometric sum). *For $z \neq 1$, $\sum_{k=0}^{M-1} z^k = \frac{1-z^M}{1-z}$.*

Proof. Multiply the sum by $(1-z)$ and telescope. $\square$

Lemma A.0.3 (Chord length). *For real θ ,*

$$|1 - e^{i\theta}| = 2 \left| \sin \frac{\theta}{2} \right|, \quad \text{and} \quad |1 - e^{i\theta}| \geq \frac{2|\theta|}{\pi} \text{ for } |\theta| \leq \pi. \quad (\text{A.2})$$

Proof. $1 - e^{i\theta} = e^{i\theta/2} (e^{-i\theta/2} - e^{i\theta/2}) = -2ie^{i\theta/2} \sin(\theta/2)$, whose modulus is $2|\sin(\theta/2)|$. For the inequality, apply Jordan's inequality ([Lemma A.0.4\(b\)](#)) to $\sin(|\theta|/2) \geq \frac{2}{\pi} \cdot \frac{|\theta|}{2}$ on $|\theta|/2 \in [0, \pi/2]$, then double. $\square$

Lemma A.0.4 (Two sine inequalities). (a) $|\sin x| \leq |x|$ for all real x .

(b) (Jordan) $\sin \theta \geq \frac{2}{\pi} \theta$ for $\theta \in [0, \pi/2]$.

Proof. (a) For $x \geq 0$, $f(x) = x - \sin x$ has $f(0) = 0$ and $f'(x) = 1 - \cos x \geq 0$; extend by oddness, and note $|\sin x| \leq 1 \leq |x|$ for $|x| \geq 1$. (b) $g(\theta) = \sin \theta / \theta$ is decreasing on $(0, \pi/2]$, since $g'(\theta) = (\theta \cos \theta - \sin \theta) / \theta^2$ and $\theta \cos \theta - \sin \theta$ vanishes at 0 with derivative $-\theta \sin \theta \leq 0$. Hence $g(\theta) \geq g(\pi/2) = 2/\pi$. $\square$

Where the inequalities are used

[Lemma A.0.4](#) is exactly what produces the constant $4/\pi^2$ in [Theorem 8.3.4](#): (a) bounds the denominator of the Fejér kernel from above, (b) bounds the numerator from below, and $(\frac{2}{\pi})^2 = 4/\pi^2$. [Lemma A.0.3](#) is what produces the tail bound [Equation \(8.9\)](#) in [Theorem 8.4.1](#).

Appendix B

The continued-fraction algorithm

Section 1.4 proved what is needed; this appendix records the algorithm concretely and works an example, because the procedure is easy to state and easy to get wrong.

B.1 The algorithm

The expansion is a sequence of *invert-and-split* steps. Given x , take the integer part; invert the remaining fraction; take the integer part of that; and continue.

Example B.1.1 (3.1415).

$$\begin{aligned} 3.1415 &= \frac{6283}{2000} = 3 + \frac{283}{2000} = 3 + \frac{1}{\frac{2000}{283}} = 3 + \frac{1}{7 + \frac{19}{283}} = 3 + \frac{1}{7 + \frac{1}{\frac{283}{19}}} \\ &= 3 + \frac{1}{7 + \frac{1}{14 + \frac{17}{19}}} = \dots = [3; 7, 14, 1, 8, 2]. \end{aligned}$$

Why continued fractions are the natural representation

The continued-fraction expansion of a rational number always *terminates*: $x \in \mathbb{Q}$ iff $x = [a_0; a_1, \dots, a_n]$ for a finite sequence of integer coefficients. Decimal representations do not — $2/3 = 0.666\dots$ has infinitely many digits, whereas $2/3 = [0; 1, 2]$ has two.

Requiring integer coefficients makes the expansion of an irrational number *unique*, and that of a rational number *semi-unique*: since $a_n = (a_n - 1) + \frac{1}{1}$, every rational has exactly two expansions, $[a_0; \dots, a_n]$ and $[a_0; \dots, a_n - 1, 1]$. Without loss of generality one may therefore choose the expansion to have an even or an odd number of terms — a freedom used in the proof of [Theorem 1.4.3](#).

B.2 The convergent recursion

By [Proposition 1.4.2](#), with $p_{-1} = 1, q_{-1} = 0, p_0 = a_0, q_0 = 1$:

$$p_k = a_k p_{k-1} + p_{k-2}, \quad q_k = a_k q_{k-1} + q_{k-2}, \quad p_k q_{k-1} - p_{k-1} q_k = (-1)^{k-1}.$$

The last identity has two consequences used throughout: every convergent is automatically in lowest terms, and the convergents alternate around x .

B.3 Worked example: 171/512

This is the case followed in [Chapter 14](#).

$$\begin{aligned} 512 &= 2 \cdot 171 + 170, \\ 171 &= 1 \cdot 170 + 1, \\ 170 &= 170 \cdot 1 + 0, \end{aligned} \quad \implies \quad \frac{171}{512} = [0; 2, 1, 170].$$

k	a_k	p_k	q_k	$q_k < N = 21?$
0	0	0	1	yes
1	2	1	2	yes
2	1	1	3	yes $\leftarrow$ last such
3	170	171	512	no

The last convergent with denominator < 21 is $\frac{1}{3}$, so $r' = 3$. Verification: $2^3 = 8 \not\equiv 1 \pmod{21}$, so 3 is only a divisor of r . Testing $2 \cdot 3 = 6$: $2^6 = 64 \equiv 1 \pmod{21}$, so $r = 6$.

B.4 Cost

By [Proposition 1.4.2](#) the denominators satisfy $q_k \geq q_{k-1} + q_{k-2}$, the Fibonacci recursion, so $q_k \geq F_{k+1}$ grows exponentially and there are $O(\log q)$ convergents. Each step is one division and two multiply-adds on $O(\log q)$ -bit integers, giving $O(\log^3 q)$ bit operations in total with schoolbook arithmetic — *utterly negligible* beside the quantum circuit.

Appendix C

Gate and cost reference

C.1 Everything, in one place

Object	Action	Cost
CNOT, H , S , Paulis	Clifford	free (fault-tolerantly)
$T = P(\pi/4)$	$ 1\rangle \mapsto e^{i\pi/4} 1\rangle$	the unit of cost
TOF standalone	$(x, y, z) \mapsto (x, y, z \oplus xy)$	$7T$, T -depth 3
TOF as compute/uncompute pair	Theorem 4.4.6	$4T$ for the pair
FREDKIN	controlled swap	$7T$
$R_z(\theta)$ to precision ε	arbitrary rotation	$3 \log_2(1/\varepsilon) T$
$C^k(X)$, $k \geq 3$	k -controlled NOT	$2k - 3$ TOF, $k - 2$ ancillas
QFT $_{2^n}$ exact	Algorithm 6.3.2	$\Theta(n^2)$ rotations
AQFT to error ε	Theorem 6.4.2	$O(n \log(n/\varepsilon))$ gates
n -bit adder (CDKM)	$ a\rangle b\rangle \mapsto a\rangle a+b\rangle$	$2n$ TOF, $2n+2$ qubits
n -bit adder (temporary-AND)	same	$4nT$, n ancillas
n -bit comparator	$\mapsto [a \geq b]\rangle$	n TOF, $4nT$
modular adder	$ x\rangle \mapsto x+a \bmod N\rangle$	5 additions
modular multiplier (in place)	$ x\rangle \mapsto ax \bmod N\rangle$	$2n$ modular additions
modular exponentiation	$ k\rangle 1\rangle \mapsto k\rangle a^k\rangle$	$\Theta(n^3)$ TOF; $\approx 0.3n^3$

C.2 Gate usage in Shor's algorithm, layer by layer

Layer	Gates used	Built from	Cost
counting-register preparation	$H^{\otimes t}$	H	t Clifford gates
work-register preparation	X	X	1 Clifford gate
MAJ/UMA block	CNOT $\times 2$, TOF $\times 1$	Clifford + TOF	1 TOF; $4T$ per pair
CDKM adder (n bits)	$2n$ TOF, $(4n+1)$ CNOT	MAJ/UMA	$2n$ TOF; $4nT$
constant adder	as above, $\approx$ half deleted	compile-time folding	$\approx n$ TOF
comparator	MAJ chain, CNOT, MAJ $^{-1}$	MAJ	n TOF, $4nT$
modular adder	5 constant additions, X , CNOT	adder	$\Theta(n)$ TOF, 1 ancilla, cleared
modular doubler (p odd)	swap chain, $\pm p$	swap, adder	$\Theta(n)$ TOF, no persistent ancilla
modular multiplier	$n \times$ controlled modular add	modular adder	$\Theta(n^2)$ TOF
in-place multiplier U_a	2 multipliers, n swaps	above + swap	$\Theta(n^2)$ TOF
controlled U_a	doubly-controlled adds, FREDKIN	$C^2(X)$, FREDKIN	same order, $\approx 2 \times$ constant
modular exponentiation	t controlled U_a	above	$\Theta(n^3)$ TOF; $\approx 0.3n^3$ windowed
QFT factor	H , then $c\text{-}R_k$, $k = 2..n-s+1$	H , $P(\theta)$, CNOT	$1H + (n-s)$ rotations
$c\text{-}R_k$	$P(\pm\theta/2) \times 3$, CNOT $\times 2$	Clifford + P	$3 \log_2(1/\varepsilon) T$ if $k > 3$
bit reversal	$\lfloor n/2 \rfloor$ swaps	3 CNOT each	Clifford
QFT †	same, reversed, $R_k \mapsto R_k^\dagger$	—	same
approximate QFT	drop $c\text{-}R_k$ for $k > m$	—	$O(nm)$, $m \approx \log_2 n + 2$
measurement	computational basis	—	—
AND uncomputation	H , measure, CZ	Clifford	zero T

Bit order

Mismatching the bit order between the controlled-power ladder and the QFT is the most common implementation bug. Fix the convention once (here: big-endian, x_1 on the top wire) and check it end to end on a known basis state.

C.3 Reference numbers

$N = 15, a = 2$	orbit $\{1, 2, 4, 8\}$, $r = 4$; outcomes $\{0, \frac{Q}{4}, \frac{Q}{2}, \frac{3Q}{4}\}$ each with probability $\frac{1}{4}$ — exact spikes; success outright $\phi(4)/4 = \frac{1}{2}$; $\gcd(3, 15) = 3$, $\gcd(5, 15) = 5$. <i>Dyadic, and misleadingly easy.</i>
$N = 21, a = 2$	orbit $\{1, 2, 4, 8, 16, 11\}$, $r = 6$; $n = 5$, $Q = 512$, $t = 9$ (from $441 \leq Q < 882$); eigenphases $0, \frac{1}{6}, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, \frac{5}{6}$ — only $s = 0, 3$ dyadic; peaks at $\ell = 0, 85, 171, 256, 341, 427$; branch $s=1$: $\Pr[85] = 0.6839$, $\Pr[86] = 0.1710$; per-run success 0.3208 direct, 0.8149 with repair; $\gcd(7, 21) = 7$, $\gcd(9, 21) = 3$.
Toy curve	$E : y^2 = x^3 + 3x + 2$ over $\mathbb{F}_5$; $E(\mathbb{F}_5) = \{\mathcal{O}, (1, 1), (1, 4), (2, 1), (2, 4)\}$, $\#E = 5$ prime; $P = (2, 1)$ gives $[1..5]P = (2, 1), (1, 4), (1, 1), (2, 4), \mathcal{O}$; with $Q = (1, 1)$ the answer is $m = 3$; inverses mod 5: 1, 3, 2, 4.
Complexity	modular exponentiation / scalar multiplication $\Theta(n^3)$; QFT $\Theta(n^2)$ exact, $O(n \log n)$ approximate; continued fractions $O(\log^3 q)$; qubits $\Theta(n)$.
Peak bounds	nearest grid point $\Pr \geq 4/\pi^2 \approx 0.4053$; two straddling values $\geq 8/\pi^2 \approx 0.8106$; exact zeros at $\delta_y = k/Q$, $k \not\equiv 0$; Legendre threshold $1/(2q^2)$; register $t = 2n + O(1)$.
Gate accounting	1 TOF = $7T$ (T -depth 3), or $4T$ per compute/uncompute pair; rotation to precision $\epsilon = 3 \log_2(1/\epsilon) T$; CDKM adder $2n$ TOF, $2n+2$ qubits.
ECDLP resources	per point addition $9n + 2\lceil \log_2 n \rceil + 10$ qubits, $\lesssim 224n^2 \log_2 n + 2045n^2$ TOF; full run $(448 \log_2 n + 4090)n^3$ TOF.
Head-to-head	P-256: 2330 logical qubits, 1.26×10^{11} TOF. RSA-3072: 6146 qubits, 1.86×10^{13} TOF. Matched security ≈ 128 bits; operand ratio $12\times$.

Appendix D

Verification log

Every numerical claim in this document was checked before being written down. This appendix records what was checked and how, so that the claims can be *reproduced* rather than trusted.

D.1 What was checked

Claim	Where	Check performed
$\text{ord}_{21}(2) = 6$, orbit $\{1, 2, 4, 8, 16, 11\}$	Chapter 14	direct enumeration
$n = 5$, $441 \leq Q < 882 \Rightarrow Q = 512$, $t = 9$	Chapter 14	bit-length computation
$\Pr[85 \mid s=1] = 0.683919$	Chapter 14	Fejér kernel Equation (8.5) evaluated exactly
$\Pr[86 \mid s=1] = 0.170980$; pair sum 0.854899	Chapter 14	same
$\delta_{85} = 1/1536$, $Q\delta_{85} = 1/3$	Chapter 14	exact rational arithmetic
full distribution sums to 1.0000000	Chapter 14	$\sum_{\ell=0}^{511} \Pr[\ell]$, all six branches
peaks at $\ell \in \{0, 85, 171, 256, 341, 427\}$	Chapter 14	argmax over the distribution
dyadic branch height $= 1/6 = 0.16667$	Chapter 14	direct
per-run success (direct) $= 0.320762$	Chapter 14	summed Fejér weights over all ℓ whose convergent has denominator 6
per-run success (repaired) $= 0.814927$	Chapter 14	same, with the multiple-testing rule of Algorithm 11.2.1, $\ell = 0$ excluded
two-run lcm success $= 0.529028$	Chapter 14	double sum over independent outcome pairs
$171/512 = [0; 2, 1, 170]$, convergents $\frac{0}{1}, \frac{1}{2}, \frac{1}{3}, \frac{171}{512}$	Chapter 14, Appendix B	Proposition 1.4.2 recursion
$2^3 = 8$, $2^6 \equiv 1$, $\gcd(7, 21) = 7$, $\gcd(9, 21) = 3$	Chapter 14	direct
$N = 15$, $a = 7$: $r = 4$, four spikes of $\frac{1}{4}$	Section 14.7	exact distribution at $Q = 8$
$(\mathbb{Z}/15\mathbb{Z})^\times$ orders $\subseteq \{1, 2, 4\}$	Section 14.7	exhaustive over the 8 units
$4/\pi^2 = 0.4052847$, $8/\pi^2 = 0.8105695$	Chapter 8	direct
tightness: 0.41053 ($t=3$), 0.40537 ($t=6$), 0.405285 ($t=10$)	Chapter 8	$(Q \sin(\pi/2Q))^{-2}$
$E(\mathbb{F}_5)$ has 5 points; $\Delta \equiv 4 \neq 0$	Section 16.4	exhaustive over $\mathbb{F}_5^2$
$[1..5]P = (2, 1), (1, 4), (1, 1), (2, 4), \emptyset$	Section 16.4	Theorem 16.2.1 formulas
$Q = (1, 1) = [3]P$, so $m = 3$	Section 16.4	direct
$f(2, 4) = [14 \bmod 5]P = [4]P$	Section 17.6	Equation (15.1)
(y_1, y_2) table all yield $m = 3$	Section 17.6	$y_2 y_1^{-1} \bmod 5$ for $y_1 = 1..4$
Roetteler table (qubits, Toffoli)	Chapter 18	$9n + 2 \lceil \log_2 n \rceil + 10$ and Equation (17.3)
Häner factoring table	Chapter 18	$(64(\log_2 n - 2) + 29.46)n^3$
$0.3n^3$ at $n=2048 = 2.58 \times 10^9$	Chapter 12	direct

D.2 Circuit identities

The following were verified symbolically or by exact matrix computation. They are quoted from the literature and re-checked here.

Claim	Where	Check
TOF = the Clifford+ T circuit, T -count 7	Proposition 4.4.4	exact 8×8 matrix product
every permutation of $\{0, 1\}^2$ is $\mathbb{F}_2$ -affine	Theorem 4.3.1	exhaustive over all 24 permutations
MAJ and UMA block identities	Lemma 12.1.2	symbolic over $\mathbb{F}_2$, all 8 inputs
CDKM adder, ancillas restored	Theorem 12.1.4	exhaustive for $n = 6$: all 2^{12} operand pairs
$\text{QFT}^\dagger \text{diag}(\omega^{ak}) \text{QFT} = T_a$	Theorem 12.5.1	all a , for $n = 3, 4, 5$
$\text{QFT} T_a \text{QFT}^\dagger = \text{diag}(\omega^{+ak})$ — sign +, not —	Theorem 6.1.4	all a , $n = 3, 4, 5$; both signs tested, only + holds
$F_r = \sum_s s\rangle\langle u_s $ has $(F_r)_{sk} = \omega_r^{+sk}/\sqrt{r}$	Corollary 5.3.4	$S = F_r^\dagger D F_r$ and $C_a = F_r^\dagger \text{diag}(\hat{a}) F_r$ checked at $r = 6$; the opposite sign fails
AQFT error $\leq 2\pi n/2^m$	Theorem 6.4.2	operator norm at $n = 5$, $m = 2, 3, 4$: $1.9996 \leq 7.854$, $0.9428 \leq 3.927$, $0.1960 \leq 1.964$
group axioms on $E(\mathbb{F}_5)$	Theorem 16.2.1	commutativity and associativity, all 5^3 triples
$\Pr[\text{nearest}] \geq 4/\pi^2$, $\Pr[b_0] + \Pr[b_1] \geq 8/\pi^2$	Theorem 8.3.4, Proposition 8.3.6	1600 phases $\times t \in \{4, 6, 8, 10\}$; empirical minima 0.40659 and 0.81126
$t = n + \lceil \log_2(2 + \frac{1}{2\varepsilon}) \rceil$ suffices	Theorem 8.4.1	300 random phases at three (n, ε) pairs; failure probability never exceeded ε
in-place multiplier identity	Theorem 12.4.2	all coprime a , all $x < N$, $N \in \{15, 21, 33\}$
$ x - p_k/q_k \geq \frac{1}{q_k(q_{k+1} + q_k)}$	Corollary 11.1.1 proof	156 000 convergents of 4000 random reals
last convergent with denominator $< N$ is r	Corollary 11.1.1	122 712 instances, all $N < 80$, all admissible ℓ
nearest grid point is the <i>unique</i> mode	Theorem 8.3.4 proof	3596 phases $\times t \in \{3, 4, 5, 6\}$; no failures, no ties. Also $\sin^2(\pi Q \delta_\ell)$ constant in ℓ to machine precision — it is exactly $\sin^2(\pi Q \varphi)$, since $Q \delta_\ell$ differs from $Q \varphi$ by an integer
point addition, ctrl = 0 branch	Algorithm 17.4.1	over $\mathbb{F}_{101}$: skipping only the ctrl_* steps returns (x_1, y_1, λ, t_0) exactly unchanged. Skipping <i>all</i> [.] ₁ -marked steps instead leaves $t_0 = -(x_1 - x_2)^{-1} \neq 0$ — a dirty ancilla
AND uncomputation by X -basis measurement	Theorem 4.4.6	200 random input states, both outcomes
QFT product form and circuit	Theorem 6.2.1	$n = 4$: circuit matrix = DFT matrix
$\text{QFT}^\dagger \circ \text{QFT} = \mathbb{I}$	Section 6.3	numerical, $n \leq 5$
3-of-7 control correct, both control values, ancilla clean	Theorem 13.4.1	exhaustive: all $a < N$, all $b < N$, all (c_1, c_2) — 900 cases at $N = 15$
c- U_a acts as $ x\rangle \mapsto a^c x\rangle$, b and t restored	Theorem 13.2.1	all x coprime to N , both control values
AND hoisting valid on $d = 0$, <i>invalid</i> in general	Proposition 13.5.1	full $2^q \times 2^q$ unitaries, then restricted to $d = 0$
12 $\times$ saving from 3-of-7 control	Chapter 13 126	transpiled to $\{\text{CNOT}, U\}$, optimisation level 1
padding failure at $x = 15$	Chapter 13, Warning 1	basis-state simulation of every $\geq N$

Two sign conventions the verification caught

Two statements in early drafts of this document carried the wrong sign, and the numerical checks above are what surfaced them. Both are recorded here because they are exactly the errors a reader re-deriving the material is most likely to make independently.

(1) In [Corollary 5.3.4](#), the eigenvector $|u_s\rangle$ carries ω_r^{-sk} but the *matrix* F_r carries ω_r^{+sk} , because $F_r = \sum_s |s\rangle\langle u_s|$ is built from the *bras*, and passing from ket to bra conjugates. Using ω_r^{-sk} for both silently replaces S by $S^\dagger$.

(2) In [Theorem 6.1.4](#), $\text{QFT}_Q T_a \text{QFT}_Q^\dagger = \text{diag}(\omega_Q^{+ak})$. The exponent is $+ak$. The Draper adder of [Section 12.5](#) uses this sign, and a mismatch there produces a circuit that subtracts instead of adding.

D.3 Reproducing the numbers

The distribution used for [Chapter 14](#) is

$$\Pr[\ell] = \frac{1}{r} \sum_{s=0}^{r-1} \frac{1}{Q^2} \frac{\sin^2(\pi Q(\frac{s}{r} - \frac{\ell}{Q}))}{\sin^2(\pi(\frac{s}{r} - \frac{\ell}{Q}))}, \quad r=6, Q=512,$$

with the removable singularity handled as the limit 1. The success probabilities are obtained by running [Algorithm 11.2.1](#) on every ℓ and summing $\Pr[\ell]$ over the accepting set. Two conventions matter and change the answer:

Two conventions that change the repaired figure

1. $\ell = 0$ **must be rejected outright**. It carries no information about r — and because the entire $s = 0$ branch lands exactly there, it accounts for $\frac{1}{6}$ of the probability. Including it and then “repairing” by searching multiples of $r' = 1$ is not a repair; it is brute-force order finding, and it inflates the figure to 0.98.
2. **The multiple search must be bounded** by $k \leq \lfloor N/r' \rfloor$, so that the candidate stays below N . This is what keeps the repair polynomial and honest.

With both conventions, the repaired figure is 0.814927, capped by $1 - \frac{1}{6} = 0.8333$.

References

Primary

1. P. W. Shor, *Algorithms for quantum computation: discrete logarithms and factoring*, Proc. 35th FOCS, 124–134 (1994); SIAM J. Comput. **26**(5), 1484–1509 (1997).
2. M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, 10th Anniversary Edition, Cambridge University Press (2010).
3. A. Yu. Kitaev, *Quantum measurements and the Abelian stabilizer problem*, arXiv:quant-ph/9511026 (1995).

Interferometry, phase estimation, and the unifying picture

4. R. Cleve, A. Ekert, C. Macchiavello and M. Mosca, *Quantum algorithms revisited*, Proc. R. Soc. Lond. A **454**, 339–354 (1998); arXiv:quant-ph/9708016. (*Source of the multi-particle interferometry framing of Chapter 7.*)
5. A. Ekert, P. Hayden and H. Inamori, *Basic concepts in quantum computation*, arXiv:quant-ph/0011013 (2000). (*Mach–Zehnder picture, optimal phase estimation.*)
6. R. B. Griffiths and C.-S. Niu, *Semiclassical Fourier transform for quantum computation*, Phys. Rev. Lett. **76**, 3228–3231 (1996).
7. R. L. Singleton Jr, *Shor’s factoring algorithm and modular exponentiation operators: a pedagogical presentation with examples*, arXiv:2306.09122 (2023).

The Fourier basis, circulants, and simultaneous diagonalisation

8. B. Bamieh, *Discovering transforms: a tutorial on circulant matrices, circular convolution, and the discrete Fourier transform*, arXiv:1805.05533 (2018–2022). (*The “DFT is discovered, not postulated” derivation of Chapter 5.*)
9. D. Coppersmith, *An approximate Fourier transform useful in quantum factoring*, IBM Research Report RC19642 (1994); arXiv:quant-ph/0201067.

Gates, universality, and cost

10. A. Barenco, C. H. Bennett, R. Cleve, D. P. DiVincenzo, N. Margolus, P. Shor, T. Sleator, J. A. Smolin and H. Weinfurter, *Elementary gates for quantum computation*, Phys. Rev. A **52**, 3457–3467 (1995); arXiv:quant-ph/9503016.
11. C. H. Bennett, *Logical reversibility of computation*, IBM J. Res. Dev. **17**(6), 525–532 (1973).
12. R. Landauer, *Irreversibility and heat generation in the computing process*, IBM J. Res. Dev. **5**(3), 183–191 (1961).
13. D. Gottesman, *The Heisenberg representation of quantum computers*, arXiv:quant-ph/9807006 (1998); S. Aaronson and D. Gottesman, *Improved simulation of stabilizer circuits*, Phys. Rev. A **70**, 052328 (2004).
14. M. Amy, D. Maslov, M. Mosca and M. Roetteler, *A meet-in-the-middle algorithm for fast synthesis of depth-optimal quantum circuits*, IEEE TCAD **32**(6), 818–830 (2013).

15. P. Selinger, *Quantum circuits of T -depth one*, Phys. Rev. A **87**, 042302 (2013).
16. N. J. Ross and P. Selinger, *Optimal ancilla-free Clifford+ T approximation of z -rotations*, Quantum Inf. Comput. **16**(11–12), 901–953 (2016).
17. E. Muñoz-Coreas and H. Thapliyal, *Everything you always wanted to know about quantum circuits*, arXiv:2208.11725 (2022). (Resource-cost taxonomy of [Section 4.5](#).)

Reversible arithmetic

19. S. A. Cuccaro, T. G. Draper, S. A. Kutin and D. P. Moulton, *A new quantum ripple-carry addition circuit*, arXiv:quant-ph/0410184 (2004).
20. T. G. Draper, *Addition on a quantum computer*, arXiv:quant-ph/0008033 (2000).
21. T. G. Draper, S. A. Kutin, E. M. Rains and K. M. Svore, *A logarithmic-depth quantum carry-lookahead adder*, Quantum Inf. Comput. **6**(4–5), 351–369 (2006).
22. V. Vedral, A. Barenco and A. Ekert, *Quantum networks for elementary arithmetic operations*, Phys. Rev. A **54**, 147–153 (1996). (The in-place modular multiplier by swap-and-uncompute, [Theorem 12.4.2](#).)
23. D. Beckman, A. N. Chari, S. Devabhaktuni and J. Preskill, *Efficient networks for quantum factoring*, Phys. Rev. A **54**, 1034–1063 (1996).
24. S. Beauregard, *Circuit for Shor’s algorithm using $2n + 3$ qubits*, Quantum Inf. Comput. **3**(2), 175–185 (2003); arXiv:quant-ph/0205095.
25. C. Gidney, *Halving the cost of quantum addition*, Quantum **2**, 74 (2018); arXiv:1709.06648. (The temporary logical-and of [Theorem 4.4.6](#).)
26. T. Häner, M. Roetteler and K. M. Svore, *Factoring using $2n + 2$ qubits with Toffoli based modular multiplication*, arXiv:1611.07995 (2016).
27. B. S. Kaliski, *The Montgomery inverse and its applications*, IEEE Trans. Computers **44**(8), 1064–1065 (1995).

Resource estimates, elliptic curves, and variants

28. M. Roetteler, M. Naehrig, K. M. Svore and K. Lauter, *Quantum resource estimates for computing elliptic curve discrete logarithms*, ASIACRYPT 2017; arXiv:1706.06752. (Primary source for [Chapter 17’s circuits and counts](#).)
29. J. Proos and C. Zalka, *Shor’s discrete logarithm quantum algorithm for elliptic curves*, Quantum Inf. Comput. **3**(4), 317–344 (2003).
30. T. Häner, S. Jaques, M. Naehrig, M. Roetteler and M. Soeken, *Improved quantum circuits for elliptic curve discrete logarithms*, PQCrypto 2020; arXiv:2001.09580.
31. C. Gidney and M. Ekerå, *How to factor 2048 bit RSA integers in 8 hours using 20 million noisy qubits*, Quantum **5**, 433 (2021); arXiv:1905.09749.
32. C. Gidney, *How to factor 2048 bit RSA integers with less than a million noisy qubits*, arXiv:2505.15917 (2025).
33. C. Chevalignard, P.-A. Fouque and A. Schrottenloher, *Reducing the number of qubits in quantum factoring*, IACR ePrint 2024/222 (2024).
34. C. Gidney, M. Newman, P. Brooks and C. Jones, *Yoked surface codes*, arXiv:2312.04522 (2023).
35. C. Gidney, N. Shuitty and C. Jones, *Magic state cultivation: growing T states as cheap as CNOT gates*, arXiv:2409.17595 (2024).
36. M. Ekerå and J. Håstad, *Quantum algorithms for computing short discrete logarithms and factoring*

RSA integers, PQCrypto 2017.

- 37. O. Regev, *An efficient quantum factoring algorithm*, arXiv:2308.06572 (2023).
- 38. S. Ragavan and V. Vaikuntanathan, *Space-efficient and noise-robust quantum factoring*, arXiv:2310.00899 (2023–2025).
- 39. D. Litinski, *How to compute a 256-bit elliptic curve private key with only 50 million Toffoli gates*, arXiv:2306.08585 (2023).

Experimental demonstrations

- 40. L. M. K. Vandersypen et al., *Experimental realization of Shor’s quantum factoring algorithm using nuclear magnetic resonance*, Nature **414**, 883–887 (2001).
- 41. E. Martín-López et al., *Experimental realization of Shor’s quantum factoring algorithm using qubit recycling*, Nature Photonics **6**, 773–776 (2012).
- 42. T. Monz et al., *Realization of a scalable Shor algorithm*, Science **351**, 1068–1070 (2016).
- 43. M. Amico, Z. H. Saleem and M. Kumph, *Experimental study of Shor’s factoring algorithm using the IBM Q Experience*, Phys. Rev. A **100**, 012305 (2019).
- 44. U. Skosana and M. Tame, *Demonstration of Shor’s factoring algorithm for $N = 21$ on IBM quantum processors*, Sci. Rep. **11**, 16599 (2021).

Formal verification

- 45. C. Chareton, S. Bardin, F. Bobot, V. Perrelle and B. Valiron, *An automated deductive verification framework for circuit-building quantum programs (Qbricks)*, ESOP 2021; arXiv:2003.05841. (*The pre/post-condition style of Chapter 19.*)
- 46. IBM Quantum Documentation, *Shor’s algorithm*; J. Watrous, *Fundamentals of quantum algorithms*.